

KAFRIADA NET

(KOWAGURU AFRICAN SPORT NETWORK PLATFORM)
PROJECT GENERAL DOCUMENTATION

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KAFRIADA NET is a unified digital sports ecosystem developed to manage athlete identity, club operations, competitions, scouting, education, and sports intelligence across Africa and the Middle East.

This document serves as the single source of truth for:

- Software Developers
- Quality Assurance Teams
- UI/UX Designers
- DevOps Engineers
- Investors
- Government Partners
- Sports Federations

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CHAPTER 1: INTRODUCTION

1.1 Purpose

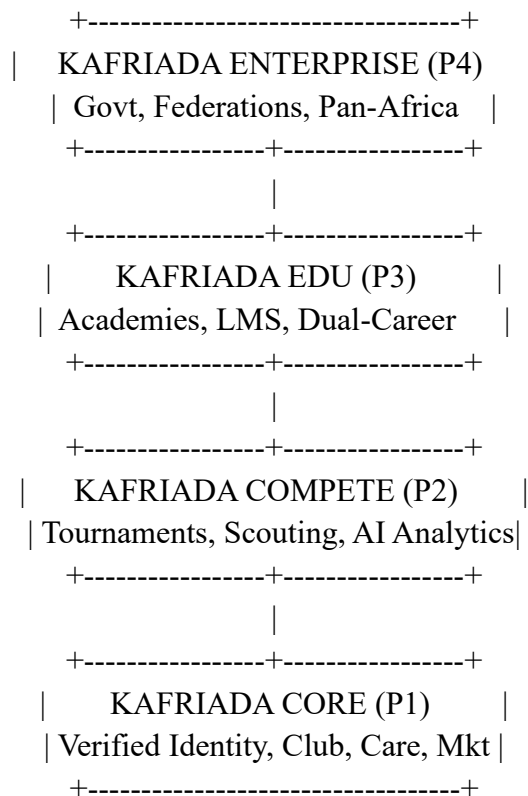
This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for KAFRIADA NET.

KAFRIADA NET is a unified digital sports ecosystem to manage athlete identity, club operations, competitions, scouting, education, and sports intelligence across Africa and the Middle East.

This document is the single source of truth for: Developers, QA, UI/UX, DevOps, Investors, Government Partners, and Federations.

Executive Strategy & Ecosystem Mapping

The **KAFRIADA NET** ecosystem, developed by KowaGuru Technology Limited, operates on a unified architecture where each phase expands upon the data layer established by the previous software release.



See: [2.0 Business Overview](#)

[KAFRIADA COMPETE](#) | [KAFRIADA EDU \(PHASE 3\)](#) | [KAFRIADA ENTERPRISE \(PHASE 4\)](#)

1.2 Document Conventions

- FR-X.Y.Z: Functional Requirement. X=Group, Y=Module, Z=ID
- NFR-X.Y: Non-Functional Requirement
- SHALL: Mandatory requirement
- SHOULD: Recommended requirement
- MAY: Optional requirement

1.3 Intended Audience

1. Software Developers and Engineers
2. Product Managers and Business Analysts
3. Investors and Strategic Partners
4. Government Agencies and Sports Federations
5. QA and Security Teams

1.4 Project Scope

database, and a single KAFRIADA ID system.

In Scope (Version 1.0 – KAFRIADA CORE)

- Athlete Identity
- Club Management
- Athlete Care
- Marketplace

Out of Scope (Version 1.0)

The following modules are scheduled for future phases:

- Full Tournament Automation
- Artificial Intelligence Performance Analytics
- Learning Management System (LMS)
- Enterprise Dashboards

Supported Sports (Version 1.0)

- Football
- Basketball
- Athletics
- Volleyball
- Swimming
- Cycling
- Handball
- Combat Sports

The platform is designed to remain sport-agnostic, allowing additional sports to be introduced without major architectural changes.

1.5 References

1. IEEE Std 29148-2018: Requirements Engineering

2. NDPR 2019: Nigeria Data Protection Regulation
3. GDPR: General Data Protection Regulation
4. Paystack API, Flutterwave API, Stripe API Documentation

CHAPTER 2: BUSINESS OVERVIEW & STAKEHOLDER ANALYSIS

2.0 Business Overview

KAFRIADA NET is a flagship SportTech platform developed by KowaGuru Technology Limited to provide a unified digital ecosystem for sports development across Africa and the Middle East.

The platform SHALL connect every stakeholder within the sports value chain through a single digital infrastructure. The core principle is "One Account • One Athlete Identity • One Ecosystem" powered by the KAFRIADA Unique ID.

Unlike traditional single-function systems, KAFRIADA NET integrates: Athlete Identity, Club Operations, Competition Management, Scouting, Education, Healthcare, and Government Intelligence into one ecosystem.

Brand: African Sport Network Platform

Background & Problem Domain

Across Africa and the Middle East, millions of talented athletes remain undiscovered due to:

1. Fragmented records and paper-based registration systems
2. Inconsistent player verification and identity fraud
3. Limited access to scouts and international opportunities
4. Isolated databases with no standardized data-sharing mechanism

Resulting Impacts:

1. Duplicate athlete registrations
2. Age and identity fraud
3. Poor player tracking and career history
4. Inefficient tournament administration
5. Weak sports policy planning and inadequate athlete welfare

Solution: KAFRIADA NET establishes a trusted digital infrastructure with verified athlete identities to enable management from grassroots to elite levels.

Problem Statement

Domain	Problem
Athlete Identity	No universal digital ID. Duplicate registrations. Difficulty tracking careers
Talent Discovery	Talented athletes are invisible to scouts. No centralized verified profile database
Club Management	Manual record keeping. Poor contract management. Limited digital communication
Tournament Administration	Paper-based registration. Manual fixtures. Inaccurate statistics. Low transparency
Athlete Welfare	Scattered medical records. Poor injury and rehabilitation monitoring

Education	Limited access to structured sports education for athletes and coaches
Government & Federations	Insufficient data for planning. No grassroots monitoring. Limited performance analytics

Solution Overview

KAFRIADA NET delivers 4 integrated flagship products under one account and one KAFRIADA ID:

Product	Phase	Purpose
KAFRIADA CORE	Phase 1	Verified Athlete ID, Club Management, Athlete Care, Marketplace
KAFRIADA COMPETE	Phase 2	Competition Management, Scouting, Performance Analytics & AI
KAFRIADA EDU	Phase 3	School/Academy Management, LMS, Coach/Athlete Education
KAFRIADA ENTERPRISE	Phase 4	Govt/Federation Dashboards, National Intelligence, APIs, Pan-Africa

2.1 Product Vision

"One Account. One Athlete Identity. One Ecosystem."

KAFRIADA NET aims to become Africa's leading digital sports infrastructure, providing trusted athlete identity, intelligent sports management, and data-driven talent development across Africa and the Middle East.

The platform is designed to unify athletes, clubs, federations, governments, scouts, educational institutions, healthcare providers, and commercial partners into one secure digital ecosystem..

2.2 Mission Statement

To transform sports development through digital technology that provides identity, verification, education, competition management, talent discovery, and intelligence while promoting integrity, inclusion, and equal access.

2.3 Business Goals & Objectives - 5 Year

ID	Goal	Success Metric	Target Year 5
BG-1	Digitize grassroots sports across Africa	5,000 clubs onboarded	10,000+
BG-2	Establish unified athlete digital identity	250,000 verified athletes users	500,000+
BG-3	Digitize competitions	5,000 tournaments managed	5,000+
BG-4	Increase athlete visibility to scouts	10,000 verified scouts active	25,000+
BG-5	Modernize club administration and operations	70% reduction in admin time	by Year 5

BG-6	Improve athlete healthcare tracking	100% digital medical records for clubs	by Year 5
BG-7	Strengthen scouting and recruitment	10,000 trial invitations/year	25,000+
BG-8	Expand access to education	10,000 course enrollments	25,000+
BG-9	Provide reliable sports intelligence	1000 Enterprise Govt/Federation clients	1000+
BG-10	Support evidence-based sports policies	National KPI dashboards deployed	by Year 5
BG-11	Build sustainable digital sports economy (Revenue Generation)	₦1,000,000,000 - ₦3,000,000,000 ARR	by Year 5

2.4 Stakeholder Analysis

ID	Stakeholder	Classification	Primary Need	Key Interaction with System
STK-01	Athlete	Primary User	Digital identity, career development, opportunities	Register, Verify, Profile, Apply
STK-02	Coach	Primary User	Player management, training records	Manage team, Attendance
STK-03	Club	Organizational User	Membership, contracts, competitions	Club dashboard, Player reg
STK-04	Club Management Team (Club Manager & Other Assistants)	Primary User	Club page management, activities scheduling, transfers & contracts overseeing, logistics planning and others.	Manage club non-player activities
STK-05	School	Organizational User	Student-athlete management	Academic + Sports data
STK-06	Academy	Organizational User	Talent development and scouting	Athlete pipeline
STK-07	Tournament Organizer	Organizational User	Registration, fixtures, live results	Create/manage tournaments
STK-08	Scout	Paying User	Verified athlete profiles, search	Advanced search, Reports
STK-09	Scout Agency	Organizational User	Talent pipeline management	Team of scouts, Analytics
STK-10	Sports Agent	Paying User	Player discovery and transfers	Transfer requests, Contracts
STK-11	Medical Team	Primary User	Medical records and injury tracking	Medical records, Clearance
STK-12	Parent/Guardian	Primary User	Minor protection, consent	Consent, Monitor minor
STK-13	Enterprise/Sponsor	Partner	Access to verified communities	Branding, Campaign data

STK-14	Sports Federation	Enterprise User	National databases and analytics	Governance dashboard
STK-15	Ministry of Sports	Enterprise User	Sports intelligence and reporting	Policy KPI dashboard
STK-16	University/Institutions	Partner	Sports data and research	Data access, Research
STK-17	NGO	Partner	Monitoring and evaluation	Program tracking
STK-18	Media	Partner	Competition updates and statistics	Public API, Live data
STK-19	KowaGuru Tech Ltd	Platform Owner	Platform governance and sustainability	Super Admin, Billing
STK-20	Support Officer	Admin	Customer Support	Resolve tickets <24hrs
STK-21	Guest/Public User	Viewer	View Public Profiles	Search athletes

(Note: We have total of 28 Roles, See [3.5 User Roles & Access Control Overview \[SYS-5.0\]](#))

2.5 Business Model

KAFRIADA NET operates as a SaaS + Marketplace + Data platform with over 10 revenue streams:

1. Verification Fees: Athlete/Coach/Club Manager/Team Assistant
 - Stage 2 ₦5,000,
 - Stage 3 ₦25,000
2. Subscriptions:
 - Club ₦50,000/yr,
 - Scout ₦100,000/yr,
 - Academy ₦200,000/yr,
 - Enterprise ₦10M-₦50M/yr
3. Tournament Hosting Fees: 5% transaction fee
4. Premium Scout Subscriptions: Advanced search & reports
5. Online Learning & Certification: Course ₦10K-₦50K
6. . Marketplace Commissions:
 1. Event Hosting 10%
 2. Transfer 3%,
 3. Marketplace Premium 7%
7. . Enterprise Licensing: Govt/Federation ₦10M-₦50M/yr
8. API Access Fees: Tiered, agreement-based
9. Sponsorship & Advertising: Branded pages, ads
10. Data Analytics & Consulting: Custom reports for stakeholders

2.6 Product Roadmap

Phase	Product	Timeline	Key Deliverable
1	KAFRIADA CORE	3-6 Months	Identity, Club, Care, Marketplace. 10K Athletes, 200 Clubs
2	KAFRIADA COMPETE	3-6 Months	Tournaments, Scouting, AI Analytics. Digitize 50+ competitions
3	KAFRIADA EDU	3-6 Months	School/Academy, LMS, Certifications. 10 Schools
4	KAFRIADA ENTERPRISE	3-6 Months	Govt Dashboards, APIs, Pan-Africa. 1-3 Countries

2.7 Value Proposition

Stakeholder Group	Value Delivered
Athletes	Permanent digital ID, verified profile, Career timeline, Scout exposure, Scholarships, Medical records, Digital portfolio
Clubs	Digital player registration, Team/Contract management, Attendance, Competition tools, Reports & Analytics
Schools & Academies	Student-athlete management, Parent portal, Training records, Performance monitoring, Learning resources
Scouts & Agents	Advanced athlete search, Verified performance data, Trial invitations, Recruitment workflow, Transfer tools
Governments & Federations	National athlete database, Participation statistics, Development KPIs, Policy support tools, Compliance reporting
Sponsors	Access to verified talent, Brand visibility, Partnership opportunities, Campaign analytics

2.8 Success Metrics / KPIs

Phase	KPI	Target
Phase 1: CORE	Athletes Registered	10,000+
	Clubs Onboarded	1000+
	Medical Partners	250+
	MVP Deployment	Live
Phase 2: COMPETE	Competitions Digitized	500+
	Verified Scouts	500+
	Performance Records	100,000+
Phase 3: EDU	Schools Onboarded	100+
	Academies Onboarded	100+
	Course Enrollments	1,000+
Phase 4: ENTERPRISE	Countries Deployed	3+
	Govt Partnerships	5+
	Athlete Records Supported	10,000,000+

2.9 Critical Success Factors

The success of KAFRIADA NET SHALL depend on:

1. CSF-1: High-quality, scalable software development
2. CSF-2: Secure infrastructure compliant with NDPR & GDPR
3. CSF-3: Strong partnerships with Enterprises, Organizations, Federations and Govt
4. CSF-4: Grassroots adoption and user training
5. CSF-5: Reliable 3-stage verification process
6. CSF-6: Sustainable revenue generation to hit ~~₦500M-₦3B~~
7. CSF-7: Continuous innovation and product iteration
8. CSF-8: Excellent customer support <24hr SLA

CHAPTER 3: PRODUCT OVERVIEW & SYSTEM ARCHITECTURE

3.1 Product Overview

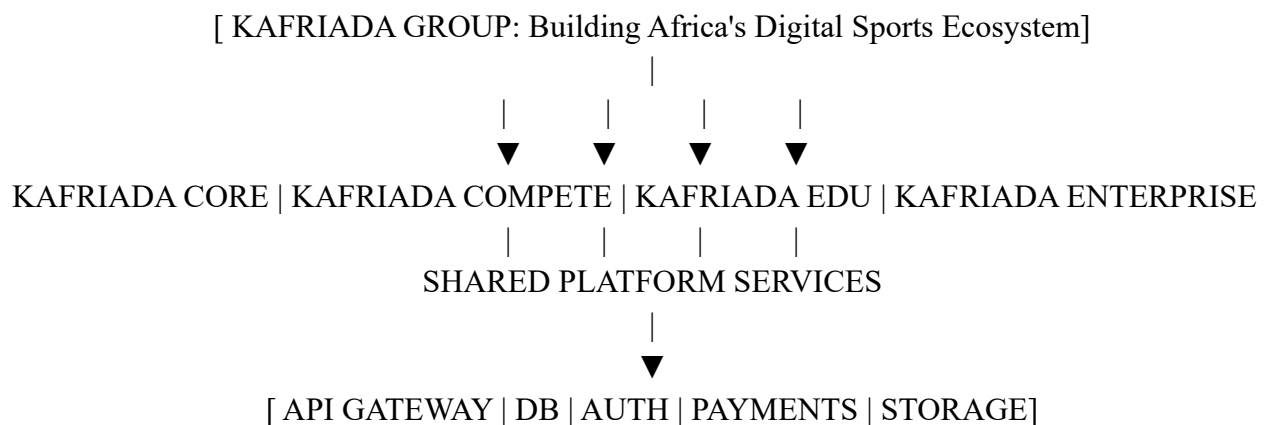
KAFRIADA NET is a cloud-based, multi-tenant Sports Technology (SportTech) platform that provides a unified digital ecosystem for sports development across Africa and the Middle East.

Core Principle: `One Account • One Athlete Identity • One Ecosystem`

Every user SHALL create one KAFRIADA Account. Eligible users SHALL be assigned one permanent KAFRIADA ID in format `KA-[COUNTRY]-[STATE]-[LGA]-[YEAR]-[6 DIGIT]`. The KAFRIADA ID serves as the unique foreign key across all products and services, enabling seamless movement between Identity, Club, Competition, Education, Healthcare, Scouting, and Enterprise without duplicate accounts.

3.2 Product Architecture & Context

KAFRIADA NET consists of 4 integrated flagship products sharing a common authentication system, centralized database, API gateway, and digital identity framework.



Architectural Principle: Each product SHALL operate independently but SHALL exchange data securely through shared platform services.

3.3 Product Components & Modules

ID	Product	Phase	Purpose	Core Modules
P-1	KAFRIADA CORE	Phase 1	Digital Foundation	Athlete Identity, Club Management, Athlete Care, Marketplace
P-2	KAFRIADA COMPETE	Phase 2	Competition & Talent	Tournament Management, League Management, Scout Network,

				Performance Intelligence, AI Analytics
P-3	KAFRIADA EDU	Phase 3	Education & Development	School Mgmt, Academy Mgmt, LMS, Coach Education, Athlete Development, Certifications
P-4	KAFRIADA ENTERPRISE	Phase 4	Intelligence & Scale	National DB, Federation/Govt Dashboard, KPIs, API Services, AI Decision Support, Research

3.4 Shared Platform Services

All 4 products SHALL rely on the following mandatory shared services:

Service ID	Service	Description
SPS-1	Authentication & RBAC	JWT, OAuth 2.0, 2FA, Role-Based Access Control
SPS-2	KAFRIADA ID Management	Generation, QR Code, Verification, Digital Passport
SPS-3	Notifications	Push via FCM, Email, SMS, WhatsApp API
SPS-4	Messaging	In-app messaging between roles
SPS-5	File & Media Storage	AWS S3 for photos, videos, documents
SPS-6	Payment Processing	Paystack, Flutterwave, Stripe integration
SPS-7	Search Engine	Elasticsearch for athlete, club, scout search
SPS-8	Reporting & Analytics	BI dashboards for all roles
SPS-9	API Gateway	REST + GraphQL. Rate limiting. Versioned
SPS-10	Audit & Compliance	Audit logs, Data retention, NDPR compliance
SPS-11	Backup & DR	Daily backups, Multi-AZ deployment
SPS-12	Localization	Multi-language, Multi-currency support

3.5 User Roles & Access Control Overview

The system SHALL implement Role-Based Access Control (RBAC). Future versions MAY support Attribute-Based Access Control (ABAC). (Note: For [2.4 Stakeholder Analysis](#) and [4.2 Core Access Principles](#))

Total Roles: 28

A. Core Operational Roles: 16

Athlete, Coach, Club Administrator, Club Manager, Team Assistant, Academy Administrator, School Administrator, Tournament Organizer/organization, Scout, Scout Manager, Sports Agent, Medical Personnel, Referee, Parent/Guardian, Sponsor, Volunteer, Media Representative.

B. Administrative & Governance Roles: 11

Super Administrator, Country Administrator, State Administrator, LGA Administrator, Federation Administrator, Ministry Administrator, Finance Officer, Support Officer, System Auditor, Guest/Public User.

(Note: Detailed permissions matrix defined in Chapter 4.)

3.6 Supported Platforms & UI

The system SHALL be accessible through:

1. Mobile Application: Android 8.0+, iOS 14+. Built with Flutter. Mobile-first design.
2. Web Application: Responsive for Desktop, Tablet, Mobile browsers. Chrome, Firefox, Safari, Edge.
3. Administrative Portal: <http://React.js> web dashboard for all Admin roles and Enterprise users.

NFR-UI-1: System SHALL be responsive and support screen sizes from 320px to 4K.

3.7 Technology Stack

Layer	Technology	Justification
Frontend - Mobile/Web	Flutter	Single codebase for Android, iOS, Web
Frontend - Admin	http://React.js + Tailwind	Rich admin dashboards
Backend	NestJS on http://Node.js	Scalable, modular, TypeScript
API	REST + GraphQL	Flexibility for mobile + 3rd party
Primary Database	PostgreSQL on AWS RDS	ACID compliance, relational data
Cache	Redis	Session, rate limiting
Search	Elasticsearch	Fast athlete/scout search
Cloud Infrastructure	AWS: EC2, RDS, S3, CloudFront	Scalability, Compliance
Authentication	JWT, OAuth 2.0, 2FA	Security standard
Payments	Paystack, Flutterwave, Stripe	Africa + International coverage
Storage	AWS S3 + CloudFront CDN	Media delivery

3.8 Multi-Sport Architecture

Requirement: FR-ARCH-1

The platform SHALL be sport-agnostic. Sports configuration SHALL be data-driven, not code-driven.

V1.0 Supported Sports: Football, Basketball, Volleyball, Athletics, Swimming, Cycling, Handball, Tennis, Badminton, Table Tennis, Rugby, Cricket, Boxing, Wrestling, Taekwondo, Judo, Karate, African Wrestling in (Dambe)

Each sport SHALL be configurable for:

1. Competition formats: League, Knockout, Round Robin
2. Positions/Disciplines
3. Performance Metrics: Speed, Sprint, Jump, Goals, etc.
4. Rankings, Statistics, Rules, Awards

New sports SHALL be added via configuration with < 2 weeks effort.

3.9 Scalability & Performance Targets

Phase 1 Deployment Targets:

- 250,000 Registered Users
- 100,000 Concurrent Active Users
- 1,000 Clubs, 500 Competitions, 100 Schools, 100 Academies

Year 5 Architecture Target: Million+ of users across multiple countries.

NFR-SCALE-1: System SHALL support horizontal scaling via load balancers and stateless services.

NFR-PERF-1: 95% of API requests SHALL respond in < 300ms.

NFR-AVAIL-1: System SHALL have 99.9% uptime SLA.

3.10 Integration Strategy [SYS-10.0]

The system SHALL support secure, versioned integrations via API Gateway.

Phase 1 Integrations:

1. StatVest: Performance verification for Stage 3
2. Paystack, Flutterwave, Stripe: Payments
3. Firebase Cloud Messaging: Push notifications
4. Twilio/Termii: SMS
5. AWS S3: File storage

Phase 2-4 Integrations:

1. GPS Wearables, HRM Devices
2. Government Sports Systems
3. Federation Management Systems
4. Public 3rd Party APIs - Paid

All integrations SHALL use OAuth 2.0 and be monitored via audit logs.

CHAPTER 4: PRODUCT SCOPE, USER ROLES, ACCESS CONTROL & BUSINESS RULES

4.1 Introduction

KAFRIADA NET SHALL implement a role-based, multi-tenant access model.

Core Rule [BR-1.0]: One Person = One KAFRIADA Account. One Athlete = One Permanent KAFRIADA ID.

A single account MAY hold multiple approved roles. Access to all features SHALL be determined by Role-Based Access Control (RBAC) with principle of least privilege.

4.2 In Scope Phase 1:

database, and a single KAFRIADA ID system.

In Scope (Version 1.0 – KAFRIADA CORE)

- Athlete Identity
- Club Management
- Athlete Care
- Marketplace

Out of Scope (Version 1.0)

The following modules are scheduled for future phases:

- Full Tournament Automation
- Artificial Intelligence Performance Analytics
- Learning Management System (LMS)
- Enterprise Dashboards

Supported Sports (Version 1.0)

- Football
- Basketball
- Athletics
- Volleyball
- Swimming
- Cycling
- Handball
- Combat Sports

The platform is designed to remain sport-agnostic, allowing additional sports to be introduced without major architectural changes.

4.2 Key Features Phase 1:

1. KAFRIADA ID: 3 Stage Verification, QR, Digital Passport
2. Club Management: Registration, Teams, Contracts, Medical
3. Athlete Care: Medical Records, Injury, Emergency
4. Marketplace: Scholarships, Trials, Jobs, Transfers, Equipment

4.2 Core Access Principles

ID	Principle	Requirement
AC-2.1	Single Account	The system SHALL prevent duplicate accounts using email + phone + NIN
AC-2.2	Single Athlete ID	The system SHALL issue only one permanent KAFRIADA ID per athlete
AC-2.3	Least Privilege	Users SHALL only access data and functions required for their role(s)
AC-2.4	Audit Trail	The system SHALL log all critical actions with UserID, Timestamp, IP
AC-2.5	Authentication	All sensitive actions SHALL require re-authentication and/or 2FA
AC-2.6	Multi-Role	A user MAY request additional roles subject to approval workflow

4.3 User Categories & Roles

Total: 28 Roles grouped into 4 categories. (Note: For [3.5 User Roles & Access Control Overview](#))

Category	Role ID	Role Name	Account Type
A. Individual MODULE 1: KAFRIADA ID (ATHLETE IDENTITY)	R-01	Athlete	Individual
	R-02	Coach	Individual
	R-03	Club Administrator/Manager	Individual
	R-04	Team Assistant	Individual
	R-05	Scout	Individual, Paid
	R-06	Scout Manager	Individual, Paid
	R-07	Medical Personnel	Individual
	R-08	Volunteer	Individual
	R-09	Parent/Guardian	Individual
	R-10	Guest/Public User	Individual
	R-11	Referee	Individual
	R-12	Sports Agent	Individual, Paid

B. Organization MODULE 2: CLUB MANAGEMENT	R-11	Club Administrator	Organization
	R-12	Academy Administrator	Organization
	R-13	School Administrator	Organization
	R-14	University Admin	Organization
	R-15	Tournament Organizer	Organization
	R-16	League Organizer	Organization
	R-17	Sports Association Admin	Organization
	R-18	NGO Admin	Organization
	R-19	Sponsor Admin	Organization
C. Government	R-20	LGA Sports Officer	Government

	R-21	State Sports Council Admin	Government
	R-22	Federation Admin	Government
	R-23	Ministry of Sports Admin	Government
D. Platform Admin	R-24	Super Administrator	Platform
	R-25	Country/State/LGA Admin	Platform
	R-26	Finance, Support, QA, Auditor	Platform

(Note: Each category role will have digital wallet, see [5.2.6 ATHLETE DIGITAL PROFILE](#) and for the definition of roles, see [17.3 ROLE CATEGORIES & DEFINITIONS](#))

4.4 Assumptions & Dependencies:

Internet access, Paystack/Flutterwave integration, StatVest hardware for Stage 3

KEY DECISIONS BAKED INTO THIS VERSION

1. Revenue Target: ₦500M - ₦1B ARR
2. Sports: All sports in V1. Sport-agnostic architecture
3. App: 1 Mobile App, Role-based dashboards
4. KAFRIADA ID: 'KA-[COUNTRY]-[STATE]-[LGA]-[YEAR]-[6 DIGIT]'
5. Stage 2: No NIN. Stage 3: Official ID required
6. Minors: Guardian consent mandatory. Profile visibility based on stage
7. Tournaments: Super Admin approval only. Federation badge optional
8. Transfers: Digital contracts + expiry in V1
9. Wearables: Phase 2
10. APIs: Phase 4, Paid

4.4 Registration & Verification Workflow

4.4.1 Individual Registration

The system SHALL allow user registration via:

1. Register using Email or Mobile Number.
2. Create a secure password.
3. Verify account using One-Time Password (OTP).
4. Select one or more user roles.
5. Complete profile information.
6. Accept Terms & Conditions.
7. Account status becomes Active. Wallet Account Created

4.4.2 Organization Registration

1. Organization Name
2. Organization Type
3. Registration Number (CAC/RC – Optional)
4. Country

5. State
6. Local Government Area
7. Office Address
8. Official Representative
9. Logo Upload
10. Accept Terms & Conditions.
11. Account status becomes 'Pending Review'. Wallet Account Created

Business Rule: Organization status SHALL be 'Pending Review' until approved by Super Admin.
Verified orgs receive badge.

4.4.3 Athlete 3-Stage Verification

Stage	Fee	Requirements	Status	Badge	Features
1	Free	Basic profile, Email/Phone	'Unverified'	Grey	Basic profile, Join clubs
2	₦5,000	Photo, Height, Weight, History	'Under Verification'	Orange	Searchable by scouts
3	₦25,000	Official ID, StatVest Assessment	'Verified'	Blue	Full visibility, Priority in search

Business Rule: Under-18 athletes SHALL require Parent/Guardian consent before Stage 2.

4.5 Role Permission Matrix

'R = Read, C = Create, U = Update, D = Delete, A = Approve'

Function	Athlete	Coach	Club Admin	Scout	Tourn Org	Parent	Fed Admin	Super Admin
IDENTITY								
Manage Own Profile	CRUD	R	R	R	R	R	R	R
Verify Athlete Stage 2/3	-	-	-	-	-	-	-	A
View Verified Profiles	R	R	R	CRUD	R	R	R	R
CLUB								
Register Club	-	-	C	-	-	-	-	A
Manage Teams/Players	R	CRUD	CRUD	R	R	R	R	R
Upload Contracts	R	-	CRUD	-	-	R	-	A
COMPETITION								
Create Tournament	-	-	-	-	C	-	-	A
Approve Tournament	-	-	-	-	-	-	-	A
Generate Fixtures	-	-	-	-	CRUD	-	-	CRUD
SCOUTING								
Advanced Athlete Search	-	-	-	CRUD	-	-	R	R

Send Trial Invite	-	-	-	C	-	-	-	A
MARKETPLACE								
Post Scholarship/Job/Donation	-	-	C	CRUD	-	-	-	A
Apply to Opportunity	C	-	-	-	-	C	-	A
Post Event			C	CRUD				A
Post Advert			C	C				A
Admin								
Manage Users	-	-	-	-	-	-	R	CRUD
Manage Payments	-	-	R	R	R	R	R	CRUD
View Audit Logs	-	-	-	-	-	-	R	R

4.6 Account & Organization Status

- User Account Statuses: `Draft, Active, Pending Verification, Under Review, Verified, Suspended, Deactivated, Archived`
- Organization Statuses: `Draft, Submitted, Pending Review, Approved, Verified, Suspended, Archived`
- Business Rule [BR-4.0]: Only `Verified` organizations SHALL host official competitions.

4.7 Business Rules

ID	Rule
BR-7.1	One individual SHALL own only one KAFRIADA Account
BR-7.2	KAFRIADA IDs SHALL be permanent and non-reassignable
BR-7.3	Duplicate athlete registrations SHALL be prohibited and flagged
BR-7.4	Every tournament SHALL require Super Admin approval before publication
BR-7.5	Verification fees SHALL be non-refundable once process begins
BR-7.6	Under-18 athletes SHALL require linked Guardian consent for Stage 2+
BR-7.7	All payments SHALL generate digital receipts and audit records
BR-7.8	Suspended users SHALL lose access to all restricted services
BR-7.9	Transfer requests SHALL require digital contract + expiry date

4.8 Audit & Compliance

NFR-SEC-1: The system SHALL maintain immutable audit logs for: Registration, Profile Updates, Verification, Approvals, Payments, Contracts, Role Changes, Admin Actions.

Audit Record SHALL include: UserID, Timestamp, Action, ObjectID, IP, Device, Result.

Logs SHALL be retained for 7 years to comply with NDPR.

CHAPTER 5 - FUNCTIONAL REQUIREMENTS FOR KAFRIADA CORE

5.1 INTRODUCTION

KAFRIADA CORE is the Minimum Viable Product (MVP) and foundational platform of the KAFRIADA GROUP ecosystem.

The objective of Phase 1 is to establish the core digital infrastructure for African sports by:

1. Identity: Creating a permanent, verified athlete identity system
2. Operations: Digitizing club and academy management
3. Welfare: Supporting athlete health, education, and career development
4. Marketplace: Connecting athletes to scouts, trials, scholarships, and opportunities

Core Principle:

All modules in Phase 1 shall use the KAFRIADA Unique ID (KUID) as the primary and immutable identifier for all athletes, coaches, and organizations.

KAFRIADA CORE MODULES

5.2 PHASE 1 MODULES

KAFRIADA CORE consists of 4 integrated modules:

FR-1.0: KAFRIADA ID MODULE

- FR-1.1: User Registration
- FR-1.2: 3-Stage Verification
- FR-1.3: QR Code & Digital Passport

(See: [MODULE 1: KAFRIADA ID \(ATHLETE IDENTITY\)](#))

FR-2.0: KAFRIADA CLUB MODULE

- FR-2.1: Club Registration
- FR-2.2: Team & Player Management
- FR-2.3: Contract Management

(See: [MODULE 2: CLUB MANAGEMENT](#))

FR-3.0: KAFRIADA CARE MODULE

- FR-3.1: Medical Records
- FR-3.2: Injury & Rehab Tracking

(See: [MODULE 3: ATHLETE CARE](#))

FR-4.0: MARKETPLACE MODULE

- FR-4.1: Scholarships, Trials, Jobs, Transfers, Advertisements. (See: [MODULE 4: CONNECT MARKETPLACE](#))

This is what the developers will code from directly.

MODULES AND PURPOSE

Module ID	Module Name	Core Purpose
M1	KAFRIADA ID	Verified Athlete Identity & Career Tracking
M2	Club Management	Team, Player, Staff and Administrative Operations
M3	Athlete Care	Health, Education, Welfare and Development
M4	Connect Marketplace	Scout Access, Trials, Scholarships, Transfers

Shared Platform Services: Authentication, Payments, Notifications, File Storage, QR Verification, Reporting, Role-Based Access Control.

MODULE 1: KAFRIADA ID (ATHLETE IDENTITY)

5.2.1 OVERVIEW

The KAFRIADA ID module provides every athlete with a permanent, tamper-proof digital identity. This identity follows the athlete throughout their entire sporting journey from grassroots to professional and retirement.

Each athlete is issued a KAFRIADA Unique ID (KUID), a Digital Profile, and a Secure QR Code for instant verification.

5.2.2 OBJECTIVES

1. Eliminate duplicate registrations and age fraud
2. Establish a trusted, single source of truth for athlete data
3. Provide verified, searchable profiles for scouts and clubs
4. Enable automated career tracking and compliance reporting
5. Facilitate secure participation in tournaments and trials
6. Increase talent visibility for Africa + MENA

5.2.3 ATHLETE REGISTRATION WORKFLOW

The registration process is a 2-step flow:

Step 1: Account Creation

- Inputs: Email OR Mobile Number, Password, Country
- Validation: OTP verification
- Output: Active KAFRIADA Account

Step 2: Basic Profile - Stage 1 Unverified

- **Personal Data:** First Name, Middle Name, Last Name, Gender, DOB, Nationality, State/Province, LGA/District, Phone, Email.
- **Sport Data:** Primary Sport, Secondary Sport, Position/Event, Jersey #, Dominant Foot/Hand, Current Club/Academy, Years of Experience.
- **System Generated:** KUID, QR Code, Wallet Account Created, Stage 1 Status [Optional]

5.2.4 KAFRIADA UNIQUE ID (KUID) FORMAT

Structure: `KA-COUNTRY-STATE-YEAR-SERIAL`

Example: `KA-NG-JG-BKD-2026-000123`

Rules:

1. Auto-generated upon Stage 1 completion
2. Immutable and permanent
3. Globally unique across the platform
4. Used as the primary key in all modules and reports

5.2.5 VERIFICATION FRAMEWORK

STAGE	FEE	REQUIREMENTS	OUTCOME
Stage 1	Free	Basic Profile + OTP	Unverified Profile + KUID + QR
Stage 2	₦5,000 / \$3.5	Photo, Height, Weight, Sport History, Awards, Media Uploads	"In Verification" Status. Enhanced Profile
Stage 3	₦25,000 / \$18	ID Verification + StatVest Assessment + Officer Approval	"Verified" Blue Badge + Certificate + Priority Scout Access

Stage 3 Performance Metrics: Speed, Agility, Endurance, Jump, Strength, Sport-Specific Accuracy, Reaction Time.

5.2.6 ATHLETE DIGITAL PROFILE

Each profile consists of 4 sections:

1. **Personal:** Photo, Name, KUID, QR, Nationality, Age, Gender, Position
2. **Sporting:** Current Club/Academy, Career Timeline, History, Achievements, Awards, Preferred Foot/Hand, Performance Metrics, Injury History [Privacy Controlled]
3. **Portfolio:** Photos, Videos, Match Highlights, Certificates, Scouting Reports
4. **Career Timeline:** Auto-logged events including Registrations, Transfers, Tournaments, Awards, Injuries, Certifications, Trials, Scholarships, Retirement
5. **Digital Wallet:** Every verified user and organization shall automatically receive a KAFRIADA Digital Wallet.

Wallet holders include: (**Note: Users based on roles in the [4.3 User Categories & Roles](#)**)

Individual Wallets: Athlete, Club Administrator, Coach, Team Assistant, Referee, Scout, Agent, Parent/Guardian, Volunteer, Medical Personnel, Public User,

SUPPORTED PROFILES

The module supports 12 primary user profiles:

- | | |
|------------------------|----------------------|
| 1. Athlete | 8. Medical Personnel |
| 2. Club Administrator | 9. Volunteer |
| 3. Coach | 10. Referee |
| 4. Club/Team Assistant | 11. Guardian |
| 5. Referee | 12. Public User |
| 6. Sports Agent | |
| 7. Scout | |

5.2.7 QR CODE & VERIFICATION

Each athlete receives a secure QR code linked to their public profile.

Public Scan Shows: Name, KUID, Verification Status, Current Club, Primary Sport, Public Achievements.

Sensitive Data: Only visible to authorized roles based on privacy settings.

5.2.8 SEARCH & FILTERING

Authorized users can search and filter athletes by: Name, KUID, Country, State, LGA, Sport, Position, Age, Verification Status, Club, Academy.

Future: AI-based performance and recommendation search.

5.2.9 PRIVACY & VISIBILITY CONTROLS

Athletes control visibility with 5 levels: Public, Verified Users Only, Scouts Only, Clubs Only, Private.

For U18 Athletes: Guardian consent is mandatory. Default visibility = Private. Platform safeguarding rules apply.

5.2.10 NOTIFICATIONS

System will notify athletes via In-App, Push, Email, SMS for: Registration, Verification Status, Payments, Trials, Transfers, Scholarships, Tournaments, Medical Updates, Announcements.

5.2.11 FUNCTIONAL REQUIREMENTS SUMMARY - MODULE 1

REQ ID	DESCRIPTION	PRIORITY
FR-ID-001	System shall allow athlete account registration with OTP	High
FR-ID-002	System shall auto-generate unique KUID upon Stage 1 completion	High
FR-ID-003	System shall generate secure QR code linked to athlete profile	High
FR-ID-004	System shall support 3-stage verification with payments	High
FR-ID-005	System shall maintain an automated career timeline per KUID	High
FR-ID-006	System shall allow media portfolio uploads: photo, video, docs	High
FR-ID-007	System shall allow athletes to manage profile visibility settings	High
FR-ID-008	System shall provide advanced search and filters for profiles	High
FR-ID-009	System shall issue digital badges and certificates post-verification	High
FR-ID-010	System shall send multi-channel notifications to athletes	Medium

Next: Module 2 - Club Management: Club Registration, Team Structures, Player Rosters, Staff, Contracts, Attendance, Subscriptions & Admin Workflows.

5.5.12 SYSTEM ACTIONS POST-REGISTRATION:

1. Generate permanent KAFRIADA ID (KUID)
2. Generate unique QR Code
3. Create digital athlete profile
4. Assign status: 'Stage 1 - Unverified'

5.6 VERIFICATION FRAMEWORK

Stage	Status	Fee	Requirements	Outcome
Stage 1	Unverified	Free	Basic Profile + OTP	KUID + QR + Basic Profile
Stage 2	Under Verification	₦5,000	Profile Photo, Height, Weight, BMI, Career History, Awards, Certificates, Media Uploads	Enhanced Profile. Receipt Issued
Stage 3	Verified	₦25,000	Official ID Verification + StatVest Assessment +	

KAFRIADA Officer Approval Blue Badge + Certificate + Premium Scout Visibility
[Auto]

5.6.1 Stage 3 Performance Metrics:

Sprint Speed, Acceleration, Endurance, Jump Height, Agility, Balance, Strength, Reaction Time, Sport-Specific Accuracy, Match Fitness Score.

5.7 ATHLETE CAREER TIMELINE

The platform shall automatically maintain a chronological, immutable career timeline per KUID. Events include: Clubs Joined/Exited, Competitions, Awards, Goals/Stats, Injuries, Medical Clearances, Transfers, Education, Certifications.

MODULE 2: CLUB MANAGEMENT

5.8 OVERVIEW

The Club Management Module provides a comprehensive digital platform that enables registered sports clubs, academies, schools, and sports organizations to manage athletes, coaches, technical staff, contracts, transfers, competitions, finances, and administrative operations within the KAFRIADA ecosystem.

The module ensures transparency, accountability, regulatory compliance, and efficient management of sports organizations through centralized digital services.

Objectives:

The Club Management Module shall enable organizations to:

- Register and verify clubs and sports organizations.
- Manage teams, players, coaches, and technical staff.
- Maintain digital wallet and transactions.
- Digitally manage player contracts and renewals.
- Conduct secure player transfers.
- Monitor player eligibility and registrations.
- Participate in competitions.
- Maintain digital records and reports.
- Comply with KAFRIADA and recognized sports governing body regulations.

The module shall integrate with all KAFRIADA NET modules including Athlete Management, Club Management, Competition Management, Marketplace, Finance, Education, Sponsorship, Communication, Athlete Care, Scouting, Technical Personnel, and future modules.

5.9 SUPPORTED ORGANIZATION TYPES

Sports Clubs, Academies, Schools, Universities, Community Teams, Corporate Teams, Government Sports Centres. Others: see [4.3 User Categories & Roles](#)

5.10 CLUB REGISTRATION

Club registration should be, based on the following stages accordingly:

Stage 1 (unverified) Free	Stage 2 (Pending verification) N5000	Stage 3 (Verified) N100,000
Required Fields: • Organization Name, • Sports Type, • Country, • State, • LGA, • Office Address, • Logo,	• Team Formation Style • Trophies • Stadium Name • Stadium Address	• Upload links • Upload Media • Subdomain Name (N50,000)

• Colours, • Year Established, • Contact Info, • Authorized Representative,		
--	--	--

Workflow: Submission → Pending Review (can be used by the team with Pending Review Mark),
Wallet Account Created → Approval by KAFRIADA Super Admin.

CLUB STAGES BENEFITS

Stage 1 (unverified) Free	Stage 2 (Pending verification) N15000	Stage 3 (Verified) N100,000
• Register players • Transfers • Squad • League • Team categorizing (Senior Football) • Club Limit 50 Players • Year Established (Non-visible) • Club Colors (Non-visible)	• Club History • Trophy Record • Club Colors (Visible) • Jersey Colors • Club Limit 50 Players • 50% Sponsorship Benefit • Year Established (Visible) • Stadium Name & Address	• Club Media Page • Club Links (Social Media, Website & Others) • Subdomain Website (N50,000) • Club Limit 70 Players • 90% Sponsorship Benefit • Club News Page

CLUB PROFILE MANAGEMENT

Each club shall maintain according to subscription stages a digital profile containing:

General Information	Sporting Information	Club Statistics
• Club Name • Logo • History • Contact Information • Website • Social Media Links	• Sports Categories • Home Venue • Training Centres • League Participation • Age Categories	• Registered Athletes • Coaches (1,2,3) • Technical Staff • Medical Personnel • Competitions • Championships • Club Ranking

5.11 TEAM MANAGEMENT

Clubs can create multiple teams with maximum of 50 players for only single sport by: Sport, Age Category, Gender, Competition Level.

Example: `U-10 Football`, `Senior Football - Men`

Team Attributes: Team Code, Team Name, Assigned Coach, Team Captain, Squad List, Training Schedule.

Verified clubs may create multiple teams under different sports and age categories.

Examples include:

- Senior Team
- Women's Team
- U-23
- U-20

Functional Requirements

- Create Teams (by Club Administrator)
- Assign Athletes
- Assign Coaches
- Assign Technical Staff

- U-17
- Academy Team

- Manage Team Rosters
- Track Team Performance
- Archive Previous Seasons

Technical Staff Management

The system shall manage:

- Head Coach
- Assistant Coach
- Goalkeeping Coach
- Fitness Coach
- Physiotherapist
- Nutritionist
- Team Doctor
- Team Manager
- Media Officer
- Club Secretary

Functions include:

- Registration
- Assignment
- Certification Management
- Contract Tracking
- Performance Monitoring

Note: Other roles excluding Club Administrator, Coach & Medical Team should be TEAM ASSISTANT

Club Dashboard

The Club Dashboard shall display:

- | | |
|--|---|
| <ul style="list-style-type: none"> • Total Registered Players • Active Teams • Coaches • Technical Staff • Medical Staff • Active Competitions | <ul style="list-style-type: none"> • Pending Transfers • Active Transfers • Completed Transfers • Contract Expiry Alerts • Financial Summary • Club Verification Status |
|--|---|

5.12 PLAYER MANAGEMENT

Player Registration availability by Club from Stage 1.

Verified clubs may recruit players through:

- | | |
|---|--|
| <ul style="list-style-type: none"> • Direct Invitation • Athlete Application • Approved Transfer • Trial Selection • Academy Promotion | <p>Functional Requirements</p> <ul style="list-style-type: none"> • Invite Players • Register New Players • Approve Player Applications • Maintain Complete Player History • Validate Player Eligibility • Prevent Duplicate Registrations |
|---|--|

Every registered player must possess a valid KAFRIADA ID (KUID).

Authorized club admins/coaches can also:

1. Invite athletes via KUID
2. Register new athletes [Stage 1]
3. Assign jersey numbers

4. Transfer/Release athletes between teams
5. Track availability and attendance
6. View contract status

5.13 COACH MANAGEMENT

Features: Coach Assignment, License Tracking, History, Team Allocation, Performance Reviews, Verification Status.

5.14 CONTRACT MANAGEMENT

The system shall allow clubs to: Upload contracts, Record Start/Expiry dates, Upload loan agreements, Log transfer requests, Receive renewal reminders.

Rule: All contracts are securely stored and linked to the athlete's KUID.

5.14.1 DIGITAL PLAYER CONTRACTS

The platform shall support legally recognized electronic contracts.

Contract Information

- Contract Number
- Player
- Club
- Start Date
- End Date
- Salary (Level 1, Level 2, Level 3, this is the salary structure of KAFRIADA where by each team can specify what to pay at each level).
- Bonuses (where applicable)
- Benefits (where applicable)
- Release Clause (auto-generate, manual and optional)
- Performance Clause (auto-generate, manual and optional)
- Digital Signatures
- Supporting Documents (Player KAFRIADA Profile PDF (Recent))

Functional Requirements

- Create Contracts
- Edit Contracts
- Upload Supporting Documents
- Renew Contracts
- Track Contract Expiry
- Generate Renewal Notifications
- Archive Expired Contracts

5.14. 2 PLAYER TRANSFER MANAGEMENT

The KAFRIADA Player Transfer System shall provide a transparent, secure, and regulated digital transfer marketplace.

Every transfer shall comply with KAFRIADA regulations and the policies of recognized Sports Federations or governing organizations.

Transfer Workflow

The transfer process shall follow the sequence below:

1. Selling Club lists the player for transfer.
2. Player reviews and accepts transfer availability.
3. Interested club submits an official transfer request.
4. Required documents are uploaded.
5. The transfer is submitted for validation.
6. KAFRIADA Administrator or an officially recognized Sports Organization/Federation validates the transfer.
7. Upon approval, the Administrator sets the official Transfer Release Date and Time.
8. The transfer remains hidden until the approved release time.
9. Once released, the transfer becomes visible on the KAFRIADA Marketplace.
10. Payment is completed.
11. Revenue is automatically distributed.
12. Player registration is updated.
13. Transfer records are permanently archived.

Transfer Approval Policy

No player transfer shall become active without official validation.

Approval may only be granted by:

- KAFRIADA Administrator
- National Sports Federation
- State Sports Association
- League Management Authority
- Any organization officially approved by KAFRIADA

Pending transfers shall never appear publicly until approval has been granted.

Transfer Availability

Every approved transfer shall include:

- Transfer Start Date
- Transfer Start Time
- Transfer Closing Date
- Transfer Closing Time

The marketplace shall automatically publish and remove transfers according to the approved transfer schedule.

Every transfer listing shall contain:

- Transfer Fee
- Currency
- Market Value
- Negotiable Status
- Transfer Type
- Contract Expiry
- Selling Club
- Player Details
- Agent Information
- Governing Organization
- Approval Status

Revenue Distribution

Every completed transfer shall automatically distribute proceeds according to configurable percentages.

Distribution Components

- Player Percentage
- Selling Club Percentage
- Licensed Sports Agent Percentage
- Recommended Sports Organization/Federation Percentage
- KAFRIADA Platform Service Fee

The total allocation shall always equal 100%.

The Super Administrator shall configure all percentage allocations.

Every distribution shall generate financial records and payment receipts.

Payment Methods

The platform shall support:

Online Payments

- Paystack
- Flutterwave
- Stripe
- Bank Transfer Gateway

Offline Payments

- Cash Payment
- Certified Bank Transfer
- Cheque

Offline transactions shall remain Pending Verification until validated by a KAFRIADA Administrator.

Transfer Marketplace Listings (See [5.22 Athlete Transfer Marketplace](#))

The KAFRIADA Marketplace shall display transfer activities in real time.

Dashboard Categories

- Pending Transfers
- Approved Transfers
- Active Transfers
- Completed Transfers
- Cancelled Transfers
- Rejected Transfers
- Expired Transfers

Each listing shall display:

- Player Name
- KAFRIADA ID
- Photograph
- Club
- Position
- Sport
- Transfer Fee
- Approval Status
- Payment Status
- Remaining Contract
- Transfer Window
- Governing Organization

Functional Requirements

Requirement ID	Description
FR-CLUB-7.1	Verified clubs shall initiate player transfers.
FR-CLUB-7.2	Players shall approve transfer participation.
FR-CLUB-7.3	Every transfer shall require approval by KAFRIADA or an authorized Sports Organization.
FR-CLUB-7.4	Transfer listings shall remain hidden until the approved release date and time.
FR-CLUB-7.5	The system shall automatically distribute transfer proceeds based on configured percentages.
FR-CLUB-7.6	Online and offline payment methods shall be supported.
FR-CLUB-7.7	All transfer activities shall be permanently logged.
FR-CLUB-7.8	Club rosters shall automatically update after successful transfer completion.

MODULE 3: ATHLETE CARE

5.15 ATHLETE CARE OVERVIEW

To provide a secure, NDPR-compliant digital record of athlete health, medical assessments, injuries, and rehabilitation to support safer participation.

The Athlete Care Module provides a centralized digital healthcare and welfare management system for athletes throughout their sporting careers. It enables athletes, clubs, medical professionals, sports organizations, and KAFRIADA administrators to securely manage medical records, injury history, rehabilitation, insurance, wellness monitoring, emergency response, and healthcare compliance.

The module promotes athlete safety, improves medical decision-making, and ensures continuity of care while protecting confidential health information.

Objectives

The Athlete Care Module shall:

- Maintain secure electronic medical records.
- Record injuries and rehabilitation history.
- Monitor athlete health and wellness.
- Support emergency medical response.
- Manage medical clearances and return-to-play approvals.
- Integrate with sports insurance providers.
- Generate health analytics and reports.
- Ensure compliance with applicable health and data protection regulations.

5.16 INJURY MANAGEMENT

Record: Injury Type, Date, Context, [Training/Match], Body Part, Severity, Treatment Plan, Recovery Timeline, Return-to-Play Status.

The platform shall maintain a complete injury database. Each injury shall include:

Injury Record	Injury Categories
• Injury Identification Number • Athlete • Sport • Team • Date of Injury • Injury Type • Body Part • Severity • Cause • Match or Training Session • Initial Diagnosis • Treating Medical Officer • Estimated Recovery Period	• Minor • Moderate • Serious • Career Threatening
	Injury Status
	• Active • Recovering • Rehabilitation • Cleared • Closed

Emergency Response

Every athlete profile shall include emergency response information.

Emergency Information

- Primary Contact
- Secondary Contact
- Blood Group
- Allergies
- Medical Conditions
- Current Medication
- Preferred Hospital

An Emergency Alert feature shall allow authorized users to immediately notify emergency contacts and medical personnel.

Medical officers may issue:

- Fit to Play
- Temporarily Unfit
- Rehabilitation Required
- Permanently Restricted

Return-to-Play Clearance

No injured athlete shall participate in official competitions without medical clearance.

Approval Workflow

1. Medical Examination
2. Rehabilitation Review
3. Functional Assessment
4. Medical Recommendation
5. Final Clearance

Return-to-play status shall be digitally recorded and linked to the athlete's profile.

Rehabilitation Management

The rehabilitation module shall record:

- Rehabilitation Plan
- Assigned Physiotherapist
- Exercise Schedule
- Recovery Progress
- Follow-up Appointments
- Medical Reviews
- Rehabilitation Completion Date

The system shall automatically notify athletes and medical personnel of upcoming rehabilitation sessions.

Privacy and Medical Data Protection

Medical information shall be classified as highly confidential.

Access shall be restricted to:

- Athlete
- Authorized Medical Personnel
- Parent/Guardian (for minors)
- KAFRIADA Medical Administrator
- Approved Medical Organizations

Club officials, scouts, agents, and sponsors shall not access confidential medical records without the athlete's explicit consent.

Athlete Care Dashboard

The Athlete Care Dashboard shall display:

- Active Medical Cases
- Injury Statistics
- Recovery Progress
- Rehabilitation Schedule
- Medical Appointments
- Medical Clearance Status
- Insurance Status
- Emergency Alerts
- Wellness Indicators

Medical administrators shall also view:

- Injury Trends
- Medical Reports
- Healthcare Compliance
- Athlete Risk Analysis

Notifications

The system shall automatically notify relevant users about:

- Medical Appointments
- Injury Reports
- Rehabilitation Sessions
- Medical Clearance Decisions
- Insurance Expiry
- Medical Document Uploads
- Emergency Alerts

Notification Channels

- In-App Notification
- Email
- SMS
- Push Notification

5.17 MEDICAL RECORDS

Accessible only to authorized Medical Personnel. CORE Data includes: Blood Group, Allergies, Medical Conditions, Medications, Vaccination History, Surgeries, Notes, Attachments.

Electronic Medical Records (EMR) (Future Upgrade)

Each athlete shall have a secure digital medical profile linked to their KAFRIADA ID.

Medical Profile Information

Personal Medical Information

- Blood Group
- Genotype
- Height
- Weight
- BMI
- Allergies
- Existing Medical Conditions
- Disabilities
- Emergency Contacts

Medical History

- Previous Surgeries
- Chronic Illnesses
- Previous Injuries
- Vaccination Records
- Medication History
- Medical Certificates

Medical Documents

- Laboratory Results
- X-Ray Reports
- MRI Reports
- CT Scan Reports
- ECG Reports
- Physiotherapy Reports

NOTE: Only authorized medical personnel may update medical records.

5.18 MEDICAL CLEARANCE

Authorized personnel can issue 4 statuses: 'Fit to Train', 'Fit to Compete', 'Temporarily Unfit', 'Permanently Unfit'.

Rule: System auto-notifies Club Admin and Coach on any status change.

Medical Assessment

Authorized medical practitioners shall perform athlete assessments.

Assessments include:

- Personal Medical Information
- General Medical Examination
- Physical Fitness Assessment
- Cardiovascular Assessment
- Musculoskeletal Assessment
- Vision Assessment
- Nutrition Assessment
- Mental Health Screening
- Performance Evaluation

Sports Insurance (Future Version Upgrade)

The platform shall support athlete insurance management.

Insurance Information

- Insurance Provider
- Policy Number
- Coverage Type
- Coverage Amount
- Effective Date
- Expiry Date
- Claim Status

Medical officers may attach insurance claims directly to injury records.

Athlete Wellness Monitoring

The system shall monitor athlete wellness indicators including:

- Sleep Quality
- Fatigue Level
- Stress Level
- Nutrition
- Hydration
- Body Weight
- Recovery Score
- Training Load

Future releases may integrate wearable devices for automatic data synchronization.

Functional Requirements

Requirement ID	Description
FR-CARE-1.1	The system shall maintain electronic medical records for every athlete.
FR-CARE-1.2	Authorized medical personnel shall manage injury records.
FR-CARE-1.3	The system shall track rehabilitation progress.
FR-CARE-1.4	Medical clearance shall be required before return to competition.
FR-CARE-1.5	Athlete wellness indicators shall be recorded and monitored.
FR-CARE-1.6	Emergency medical information shall be securely available to authorized personnel.
FR-CARE-1.7	Medical records shall be protected using role-based access control.
FR-CARE-1.8	Every medical action shall generate a permanent audit log.

Business Rules

Rule ID	Business Rule
BR-CARE-1	Every athlete shall have a secure digital medical record linked to their KAFRIADA ID.
BR-CARE-2	Only authorized healthcare professionals may modify medical records.
BR-CARE-3	No athlete with an active injury status may participate in official competitions without medical clearance.
BR-CARE-4	Medical information shall remain confidential and protected by privacy regulations.
BR-CARE-5	Every medical assessment, diagnosis, treatment, and clearance shall be permanently recorded.
BR-CARE-6	Emergency medical information shall remain available during emergency situations.
BR-CARE-7	Rehabilitation progress shall be tracked until official discharge.
BR-CARE-8	Medical records shall be retained in accordance with applicable healthcare and data protection regulations.

Summary

The Athlete Care Module provides a comprehensive digital healthcare ecosystem that supports athlete safety, injury management, rehabilitation, medical compliance, insurance, emergency response, and wellness monitoring. By integrating secure electronic medical records with KAFRIADA ID, the module enables trusted collaboration between athletes, clubs, healthcare professionals, and governing organizations while ensuring confidentiality, regulatory compliance, and long-term athlete welfare.

MODULE 4: CONNECT MARKETPLACE

5.19 OVERVIEW

To connect athletes, clubs, and professionals to trackable athletes career movement, verified opportunities for career, education, and commercial growth.

Overview

The KAFRIADA Marketplace is the central digital marketplace connecting athletes, clubs, academies, schools, sports organizations, scouts, sponsors, agents, equipment vendors, employers, and governing bodies.

The Marketplace serves as the official platform for publishing verified sports opportunities, player transfers, scholarships, employment opportunities, sponsorships, sporting equipment, competitions, and athlete services.

Every marketplace listing shall be validated according to KAFRIADA policies to ensure transparency, credibility, and fairness.

Objectives

The Marketplace shall:

- Connect athletes with verified opportunities.
- Facilitate secure player transfers.
- Advertise scholarships and grants.
- Promote sports jobs and internships.
- Enable sponsorship opportunities.
- Support equipment sales and rentals.
- Provide digital payment services.
- Increase athlete visibility across Africa and globally.
- Create sustainable revenue for stakeholders.

5.20 Marketplace Categories

The platform shall support the following marketplace categories:

S/N	CATEGORY	CONTENTS
	Athlete Transfer Marketplace	• Professional Transfers • Amateur Transfers • Loan Transfers • Trial Opportunities
	Scholarships	• Local Scholarships • International Scholarships • Educational Grants • Sports Development Grants
	Employment	• Coaching Jobs • Administrative Jobs • Medical Positions • Refereeing Opportunities

		• Volunteer Positions
	Sports Equipment	• Jerseys • Football Boots • Balls • Training Equipment • Gym Equipment • Medical Equipment
	Sports Services	• Coaching Services • Physiotherapy • Nutrition • Sports Psychology • Video Analysis • Legal Services • Player Representation
	Sponsorship/Donation	• Athlete Sponsorship • Club Sponsorship • Competition Sponsorship • Event Sponsorship

5.21 Marketplace Listings

Every listing shall include:

- Listing ID
- Category
- Title
- Description
- Images or Videos
- Owner
- Organization
- Country
- State
- Price
- Currency
- Negotiable Status
- Availability Status
- Expiry Date
- Approval Status

Only verified users and organizations may publish marketplace listings.

5.21.1 Listing Approval

Marketplace listings shall follow the approval workflow below:

1. User submits listing.
2. Listing enters Pending Review.

3. KAFRIADA Administrator reviews the listing.
4. Approved listings become Active.
5. Rejected listings are returned with comments.
6. Expired listings are automatically archived.

Certain trusted organizations may receive automatic publishing privileges as determined by the Super Administrator.

5.22 Athlete Transfer Marketplace

(See [5.14. 2 PLAYER TRANSFER MANAGEMENT](#))

The Athlete Transfer Marketplace shall integrate directly with the Club Management Module. Every transfer listing shall display:

• Athlete Profile • KAFRIADA ID • Photograph • Position • Sport • Current Club • Contract Expiry • Transfer Fee • Transfer Window • Verification Status • Governing Organization • Transfer Status	Transfer statuses include: • Pending Approval • Approved • Active • Under Negotiation • Payment Pending • Completed • Cancelled • Expired Transfers shall only become visible after approval by KAFRIADA or an authorized governing organization.
---	---

Marketplace Payments

Supported payment methods include:

Online

- Paystack
- Flutterwave
- Stripe
- Bank Transfer

Offline

- Cash
- Certified Cheque
- Verified Bank Deposit

Offline payments shall require administrative verification before completion.

Every successful transaction shall generate:

- Electronic Receipt
- Invoice
- Audit Record
- Financial Ledger Entry

Reviews and Ratings

Verified users may rate and review:

- Clubs
- Vendors
- Coaches
- Sports Service

Providers

- Equipment Sellers

Review moderation shall be managed by KAFRIADA Administrators.

Fraudulent reviews may be removed after investigation.

Marketplace Dashboard

The dashboard shall provide:

Listings

- Active Listings
- Pending Listings
- Approved Listings
- Rejected Listings
- Expired Listings

Transactions

- Pending Payments
- Successful Payments
- Refund Requests
- Revenue Summary

Analytics

- Total Views
- Total Applications
- Total Sales
- Popular Categories
- Marketplace Revenue
- Conversion Rate

Search and Filters

Users shall search marketplace listings using:

- Sport
- Country
- State
- Club
- Position
- Category
- Price Range
- Age Group
- Gender
- Verification Status
- Organization
- Keyword

Advanced filtering shall support multiple criteria simultaneously.

Notifications

The Marketplace shall notify users regarding:

- Listing Approval
- Listing Rejection
- Buyer Interest
- Scholarship Deadline
- Job Applications
- Sponsorship Offers
- Payment Confirmation
- Listing Expiry
- Marketplace Promotions

Notifications shall be delivered through:

- In-App Notifications
- Email
- SMS (Important only, like OTP)
- Push Notifications

Functional Requirements

Requirement ID	Description
FR-MKT-1.1	The system shall support multiple marketplace categories.
FR-MKT-1.2	Only verified users shall publish listings.
FR-MKT-1.3	Marketplace listings shall require approval before publication unless exempted by policy.
FR-MKT-1.4	The system shall support secure online and offline payments.

FR-MKT-1.5	Athlete transfer listings shall integrate with the Player Transfer Module.
FR-MKT-1.6	Every transaction shall generate financial and audit records.
FR-MKT-1.7	Users shall perform advanced searches and filtering.
FR-MKT-1.8	Marketplace analytics shall be available to administrators.

Business Rules

Rule ID	Business Rule
BR-MKT-1	Only verified users and organizations may publish marketplace listings.
BR-MKT-2	All listings shall comply with KAFRIADA Marketplace Policies.
BR-MKT-3	Transfer listings shall only become public after regulatory approval.
BR-MKT-4	Every completed marketplace transaction shall generate a digital receipt and audit log.
BR-MKT-5	Listings automatically expire after their validity period unless renewed.
BR-MKT-6	Marketplace administrators may suspend or remove listings that violate platform policies.
BR-MKT-7	Financial transactions shall comply with applicable anti-fraud and financial regulations.
BR-MKT-8	Every marketplace activity shall be permanently logged for compliance and reporting purposes.

Summary

The Marketplace Module provides a secure, verified, and scalable digital commerce ecosystem for African sports. It enables athletes, clubs, organizations, sponsors, employers, vendors, and governing bodies to connect through trusted marketplace services while ensuring regulatory compliance, transparent transactions, integrated payment processing, advanced search capabilities, and comprehensive audit tracking. The module supports KAFRIADA's vision of becoming Africa's unified digital sports marketplace.

CHAPTER 6: DETAILED MODULE SPECIFICATIONS & USER WORKFLOWS

PHASE 1 – KAFRIADA CORE

Version: 1.2 | Effective Date: 1 July 2026

6.1 INTRODUCTION

This chapter defines the detailed functional behavior of all Phase 1 modules. It specifies user journeys, screen requirements, business logic, validation rules, and system responses for implementation by Developers, UI/UX, QA, and Product.

Design Principles:

1. Simplicity: Max 3 clicks to complete core tasks
2. Security: RBAC + OTP + Data Encryption for all PII
3. Scalability: All workflows support 10M+ users and multi-country localization

All workflows use the KAFRIADA Unique ID (KUID) as the primary key.

MODULE 1 (6.0): KAFRIADA IDENTITY

6.2 USER REGISTRATION WORKFLOW

UC-ID-001: Create KAFRIADA Account

Step	Actor Action	System Response
1	Launch App/Web and select 'Create Account'	Display Registration Type: Individual / Organization
2	Select 'Individual' and enter: Name, Phone, Email, Country, Password	Validate fields. Check for duplicate Email/Phone
3	Accept T&C and Privacy Policy	Enable 'Submit' button
4	Select Role(s): Athlete, Coach, Scout, etc.	Save role selection
5	Submit OTP received via Email/SMS	Verify OTP. OTP expires in 5 minutes
6	Complete Profile Setup	Generate KUID, QR Code, and redirect to Role Dashboard

Business Rules & Validations:

1. VR-ID-001: Email and Phone must be unique. System returns 'Error 409: Account already exists'
2. VR-ID-002: OTP expires after 5 minutes. 3 attempts max.
3. VR-ID-003: Password: Min 8 chars, 1 Upper, 1 Lower, 1 Number, 1 Special Char
4. VR-ID-004: Registration SLA: < 60 seconds under normal network

Acceptance Criteria: User lands on dashboard with KUID displayed within 60s of OTP verification.

6.3 ATHLETE PROFILE MANAGEMENT

UC-ID-002: Manage Athlete Profile

Athletes shall be able to C,R,U,D the following profile sections:

Section	Fields
Personal	Edit Name, DOB, Address, Contact
Media	Upload Profile Photo, Action Photos, Highlight Videos. Max 50MB per file
Contacts	Update Emergency Contact: Name, Relationship, Phone, Email
Social	Add/Manage Social Media Links
Status	View Verification Status: Stage 1/2/3
Export	Download Digital Profile as PDF

Feature: Profile Completion Indicator

System displays `% Completion Bar`. (Formula: `Completed Fields / Total Required Fields 100`.)

Target nudges at 50%, 80%, 100%.

6.4 KAFRIADA ID (KUID) GENERATION

Rule: Auto-generated upon Stage 1 completion.

Format: `KA-COUNTRY-STATE-YEAR-SERIAL`

Example: `KA-NG-JG-BKD-2026-000123`

Business Rules:

1. BR-ID-001: KUID is immutable and permanent for life
2. BR-ID-002: KUID is never reused, even after retirement or deletion
3. BR-ID-003: KUID is the FK in all 40+ Phase 1 tables

6.5 QR CODE VERIFICATION

UC-ID-003: Verify Athlete via QR

Each profile has a secure, signed QR Code.

Public Scan Data: Name, KUID, Verification Status, Current Club, Primary Sport, Profile Photo, Public Profile URL

Access Control: Sensitive data is hidden. Requires login + permission.

[Medical][Contact][Payment]

MODULE 2 (6.0): CLUB MANAGEMENT

6.6 CLUB REGISTRATION WORKFLOW

UC-CLUB-001: Register Organization

STEP	ACTOR	ACTION
1	Club Admin	Create Org Account + Submit Form: Name, Type, Address, Logo, Rep, Docs
2	System	Set Status = `Pending Review`
3	Super Admin	Review + Approve/Reject
4	System	Send Notification. On Approval, activate Club Dashboard

Club Dashboard KPIs: Teams, Athletes, Coaches, Upcoming Fixtures, Active Contracts, Membership Status, Subscription Status, Notifications

6.7 TEAM MANAGEMENT WORKFLOW

UC-CLUB-002: Manage Teams

- **Club Admin** can: Create Team, Assign Head/Assistant Coach, Invite Athletes via KUID, Set Training Schedule, Track Attendance.
- **Team Entity:** Team Code, Name, Sport, Age Category, Gender, Squad List, Staff, Stats, Competition History

6.8 MEMBERSHIP MANAGEMENT

Statuses: `Active`, `Pending`, `Suspended`, `Expired`, `Released`

Rule: System maintains full membership history log per KUID for audit and transfer purposes.

6.9 CONTRACT MANAGEMENT WORKFLOW

UC-CLUB-003: Manage Contracts

- Clubs can: Upload Contract PDF, Set Start/End Date, Create Loan Agreement, Submit Transfer Request.
- Automation: System sends renewal reminders at `T-90, T-60, T-30 days` before expiry to Club Admin + Athlete.
- Security: All contracts linked to KUID and stored encrypted.

MODULE 3 (6.0): ATHLETE CARE

6.10 MEDICAL PROFILE MANAGEMENT

Access Control: RBAC. Only Athlete + Assigned Medical Personnel + Club Doctor can view.

Data Fields: Blood Group, Height, Weight, Allergies, Chronic Conditions, Previous Injuries, Medications, Medical Notes, Attachments.

6.11 INJURY MANAGEMENT WORKFLOW

UC-CARE-001: Log Injury

Step	Actor	Action
1	Medical Personnel	Create Injury Record: Type, Severity, Body Part, Date, Context
2	Medical Personnel	Attach Docs + Set Recovery Timeline + Treatment Plan
3	Medical Personnel	Update Rehab Progress
4	Medical Personnel	Issue 'Return-to-Play' Clearance
5	System	Notify Athlete, Coach, Club Admin of Status Change

Clearance Statuses: 'Fit to Train', 'Fit to Compete', 'Temporarily Unfit', 'Permanently Unfit'

6.12 EMERGENCY INFORMATION

Fields: Emergency Contact Name, Relationship, Phone, Medical Alerts

Rule: This section is accessible to Authorized Medical Personnel offline during tournaments for emergency response.

MODULE 4 (6.0): CONNECT MARKETPLACE

6.13 TRIAL APPLICATION WORKFLOW

UC-MKT-001: Apply for Trial

1. Org publishes Trial → 2. Athlete Searches & Applies → 3. Org Reviews → 4. System sends Status Update

Application Statuses: 'Submitted', 'Under Review', 'Shortlisted', 'Invited', 'Selected', 'Rejected'

6.14 SCHOLARSHIP WORKFLOW

Provider Flow: Publish Opportunity → Define Eligibility → Review Applications → Publish Results

Athlete Flow: Search → Apply → Upload Docs → Track Progress

6.15 SPORTS JOBS

Organizations post: Coaching, Medical, Admin, Volunteer roles, Sport Tech Developers, Others.

Applicants: Submit Application + CV + Track Status.

6.16 EQUIPMENT MARKETPLACE - PHASE 1 SCOPE

- V1.0: Vendor Listing Only. Categories: Jerseys, Balls, Shoes, Training/Gym Equipment, Wearables & Others.
- V2.0 Future: Online Ordering, Payments, Logistics, Vendor Ratings.

6.17 NOTIFICATION ENGINE

The system shall support 4 channels: In-App, Push, Email, SMS.

Trigger Events: Registration, Verification Update, Contract Reminder, Trial Invite, Scholarship Deadline, Payment, Medical Clearance Change, Marketplace Message.

6.18 ROLE-BASED DASHBOARDS

Role	Key Widgets
Athlete	Profile %, Verification, Timeline, Trials, Medical Status, Notifications
Club Admin	Teams, Players, Contracts, Attendance, Membership, Reports, Subscription, Transfer
Coach	Assigned Teams, Schedule, Attendance, Performance Summary
Medical	Injury Queue, Pending Clearances, Athlete Medical History

Super Admin Total Users, Pending Verifications, Approvals, Revenue, System Health, Audit Logs

6.19 SEARCH & FILTERING

Global search with filters: Sport, Country, State, LGA, Club, Academy, Age, Gender, Verification, Position, Competition Level. Response time SLA: <2s for 100k records.

6.20 ERROR HANDLING & MESSAGING

ERROR SCENARIO	USER MESSAGE	SYSTEM ACTION
Invalid Login	"Incorrect email or password"	Log attempt. Lock after 5 tries
Duplicate Account	"This email/phone is already registered"	Suggest Password Reset
Expired OTP	"OTP expired. Please request a new one"	Invalidate OTP
Payment Failure	"Payment failed. Please try again"	Log transaction ID
Unauthorized	"You do not have permission"	Log to Audit Trail

All errors logged to `audit logs` table for troubleshooting.

CHAPTER 7: DATABASE DESIGN & DATA MODEL

PHASE 1 (7.0) – KAFRIADA CORE

Version: 1.2 | DBMS: PostgreSQL 15+ | Effective Date: 1 July 2026

7.1 INTRODUCTION

The KAFRIADA NET database is the single source of truth for the entire platform. It is designed as a centralized, secure, normalized, and horizontally scalable relational database to support 10M+ users across Africa + MENA.

Core Objectives:

1. Integrity: Enforce data consistency across Identity, Club, Care, and Marketplace domains
2. Performance: Support <2s search on 10M athlete records via indexing and caching
3. Compliance: NDPR, GDPR compliant audit trails and data retention
4. Scalability: Partition-ready by Country and Year

Technology Stack:

Layer	Technology	Purpose
Primary DB	PostgreSQL 15+	Transactional data, ACID compliance
Cache	Redis	Session, Dashboard, Search results
Search	Elasticsearch	Athlete/Club/Opportunity search & filters
Storage	AWS S3	Media, Documents, Contracts

7.2 DATABASE DESIGN PRINCIPLES

1. Normalization: All tables normalized to 3NF minimum
2. Keys: UUID v4 for internal PK. KAFRIADA ID (KUID) for business PK on Athletes/Orgs
3. Integrity: Foreign Key constraints enforced. No orphan records
4. Audit: All critical tables have `created\_at`, `updated\_at`, `created\_by`, `deleted\_at` for soft delete
5. Indexing: All FKs and search fields indexed. Composite indexes for multi-filter
6. Security: Field-level encryption for PII, TLS 1.3 in transit, AES-256 at rest
7. Scalability: Partitioning strategy by `country\_code` and `year` for large tables

7.3 HIGH-LEVEL DATABASE DOMAINS

Domain	Core Tables
1. Identity & Access	users, roles, permissions, user roles, athlete profiles, coach profiles, verifications records
2. Organization	organizations, teams, team memberships, coaches, coach assignments, attendance
3. Athlete Care	Medical records, injuries, rehabilitation logs, medical clearances
4. Marketplace	scholarships, trials, jobs, equipment listings, sponsorships, transfers, applications
5. Finance	payments, invoices, subscriptions, wallets
6. Communication	notifications, messages, announcements

7. Master Data	countries, states, lgas, sports, positions
8. System	Audit logs, activity logs, settings, files

7.4 CORE DATABASE TABLES & SCHEMA

DOMAIN 1: AUTHENTICATION & IDENTITY

Table: `users`

Stores all login accounts. One user can have multiple roles.

id [UUID, PK]

kafriada_id [VARCHAR(30), UNIQUE, NN] -- KA-NG-JG-BKD,2026-000123

first_name [VARCHAR(100), NN]

last_name [VARCHAR(100), NN]

middle_name [VARCHAR(100)]

email [VARCHAR(255), UNIQUE, NN]

phone [VARCHAR(20), UNIQUE, NN]

password_hash [TEXT, NN]

country_code [CHAR(2), FK->countries]

account_status [ENUM: active, suspended, deleted, NN]

email_verified [BOOLEAN, Default: false]

phone_verified [BOOLEAN, Default: false]

created_at [TIMESTAMP, NN]

updated_at [TIMESTAMP, NN]

deleted_at [TIMESTAMP]

Table: `roles`

`id, name, description`

Table: `user\_roles`

`user\_id[FK], role\_id[FK], assigned\_at` - M2M

Table: `permissions`

`id, code, description` e.g. `verify\_athlete`, `create\_club`

Table: `athlete\_profiles`

user_id [UUID, PK, FK->users]

kafriada_id [VARCHAR(30), UNIQUE, NN]

sport_id [FK->sports]

position_id [FK->positions]

height_cm [DECIMAL]

weight_kg [DECIMAL]

jersey_number [INT]

dominant_hand [ENUM]

preferred_foot [ENUM]
 current_club_id [FK->organizations]
 current_team_id [FK->teams]
 verification_stage [ENUM: stage1, stage2, stage3, NN]
 profile_completion_pct [INT]
 public_profile_url [VARCHAR]
 #DOMAIN 2: ORGANIZATION MANAGEMENT

Table: `organizations`

id [UUID, PK]
 name [VARCHAR(255), NN]
 org_type [ENUM: club, academy, school, federation, ngo, NN]
 country_code [CHAR(2), FK]
 state_id [FK->states]
 lga_id [FK->lga]
 address [TEXT]
 logo_url [VARCHAR]
 year_established [INT]
 subscription_status [ENUM]
 approval_status [ENUM: pending, approved, rejected]

Table: `teams`

`id, organization\_id[FK], name, sport\_id[FK], age\_category, gender, head\_coach\_id[FK]`

Table: `team\_memberships`

`id, team\_id[FK], user\_id[FK], jersey\_number, join\_date, exit\_date, status[ENUM]`

Table: `contracts`

`id, athlete\_id[FK->users], organization\_id[FK], type[ENUM], start\_date, end\_date, file\_url, status, renewal\_reminder\_sent`

DOMAIN 3 (7.0): ATHLETE CARE

Table: `medical\_records`

`id, athlete\_id[FK], blood\_group, allergies[TEXT], conditions[TEXT], medications[TEXT], recorded\_by[FK->users]`

Table: `injuries`

`id, athlete\_id[FK], injury\_type, body\_part, severity[ENUM], date\_occurred, treatment\_plan[TEXT], recovery\_status[ENUM]`

Table: `medical\_clearances`

`id, athlete\_id[FK], status[ENUM: fit\_train, fit\_compete, unfit\_temp, unfit\_perm], expiry\_date, issued\_by[FK->users]`

DOMAIN 4: MARKETPLACE

Table: `trials`

`id, organization\_id[FK], title, sport\_id, location, start\_date, application\_deadline, status`

Table: `trial\_applications`

`id, trial\_id[FK], athlete\_id[FK], status[ENUM], applied\_at`

Table: `scholarships`

`id, provider\_id[FK], title, eligibility\_criteria[JSONB], deadline, status`

Table: `transfers`

`id, athlete\_id[FK], from\_org\_id[FK], to\_org\_id[FK], type[ENUM: transfer, loan], status[ENUM], request\_date`

DOMAIN 5: FINANCE

Table: `payments`

id [UUID, PK]

user_id [FK->users]

amount [DECIMAL(12,2), NN]

currency [CHAR(3), NN] -- NGN, USD, SAR

gateway [ENUM: paystack, flutterwave]

gateway_ref [VARCHAR, UNIQUE]

payment_type [ENUM: verification, subscription, tournament]

status [ENUM: pending, success, failed]

paid_at [TIMESTAMP]

Table: `subscriptions`

`id, organization\_id[FK], plan\_name, start\_date, end\_date, status`

DOMAIN 6: SYSTEM

Table: `audit\_logs`

`id`, `user\_id`[FK], `action`[VARCHAR], `table\_name`, `record\_id`, `old\_data`[JSONB], `new\_data`[JSONB],
`ip\_address`, `user\_agent`, `created\_at`

Table: `countries`, `states`, `lgas`, `sports`, `positions`

Master data tables for localization and filtering.

7.5 KEY DATABASE RELATIONSHIPS

1:N users -> athlete_profiles

1:N organizations -> teams

N:M users <-> roles via user_roles

1:N teams -> team_memberships -> users

1:N users -> medical_records

1:N users -> contracts

1:N users -> payments

1:N trials -> trial_applications -> users

7.6 INDEXING STRATEGY

Primary indexes on all PKs and FKs. Additional indexes for performance:

Table	Indexed Columns
`users`	`kafriada_id`, `email`, `phone`
`athlete_profiles`	`kafriada_id`, `verification_stage`, `sport_id`, `country_code`
`organizations`	`name`, `org_type`, `country_code`, `approval_status`
`team_memberships`	`team_id`, `user_id`, `status`
`payments`	`user_id`, `status`, `gateway_ref`, `created_at`
`audit_logs`	`user_id`, `created_at`, `action`

Composite index example: `athlete\_profiles` (country_code, sport_id, verification_stage)

7.7 DATA RETENTION & ARCHIVAL POLICY

Data Type	Retention Period	Action
Athlete Profile	Permanent	Never deleted. Soft delete only
Competition Records	Permanent	Archive after 10 years
Contracts	10 Years post-expiry	Archive to cold storage
Payment Records	7 Years	For tax/audit compliance
Audit Logs	10 Years	Immutable
Notifications/Messages	2 Years	Auto-purge
Inactive Accounts	3 Years inactivity	Archive, anonymize PII

7.8 DATABASE SECURITY

1. Encryption: AES-256 at rest. TLS 1.3 in transit

2. Backups: Daily automated backups + Point-in-Time Recovery to S3

3. Access: DB credentials via AWS Secrets Manager. RBAC at DB level
4. PII: Sensitive fields encrypted: `phone`, `dob`, `medical\_records`
5. DR: Multi-AZ deployment. Read replica in AWS Bahrain for MENA

7.9 ESTIMATED DATABASE SIZE & SCALABILITY

Metric	Year 1-2 Target	Year 3-4 Target	Year 5 Target
States	Jigawa (27), Kano (44), Kaduna (23), Abuja (6ACs), Bauchi (20), Gombe (11), Plateau (17), Lagos (20)	Others	Others
Athletes (50x100xLGAs)	1,000,000	2,000,000	5,000,000+
Organizations (100xLGAs)	20,000	40,000	50,000
Records/Year	25 Million	50 Million	250 Million
Storage Est.	5 TB	10 TB	25 TB

Design supports horizontal partitioning by `country\_code` and sharding when >5TB.

CHAPTER 8: API ARCHITECTURE & SYSTEM INTEGRATIONS

PHASE 1 (8.0) – KAFRIADA CORE

Version: 1.2 | Effective Date: 1 July 2026

8.1 INTRODUCTION

KAFRIADA NET is designed as an API-First Platform. All client applications including Mobile Apps, Web Portal, Admin Dashboard, and Third-Party systems communicate exclusively through secure APIs.

Objectives:

1. Scalability: Microservices architecture to support 10M+ users
2. Interoperability: Standardized integration with Federations, Government, Wearables, and International Platforms
3. Maintainability: Decoupled services with versioned APIs
4. Security: Centralized authentication, rate limiting, and audit at API Gateway

The API Gateway is the single entry point for all traffic to backend microservices.

8.2 SYSTEM ARCHITECTURE

[Mobile App] [Web Portal] [Admin Dashboard] [Partners]

|

API Gateway [HTTPS/TLS 1.3]

|

Identity Service	Club Mgmt Service	Athlete Care Svc	Marketplace Service
Finance	Search	Notify	File Svc

|

[PostgreSQL] [Redis Cache] [Elasticsearch] [AWS S3]

API Gateway Responsibilities:

Authentication, Routing, Rate Limiting, Request/Response Logging, Token Validation, WAF, CORS

8.3 API STANDARDS

Standard	Specification
Protocol	RESTful API + Selective GraphQL
Data Format	JSON, UTF-8
Transport	HTTPS Only
Documentation	OpenAPI 3.1 / Swagger

Versioning	URI Versioning: `/api/v1/`
Base URL	`https://api.kafriada.net/v1/`

All timestamps in ISO 8601: `YYYY-MM-DDTHH:mm:ssZ`. All currencies in ISO 4217.

8.4 AUTHENTICATION & AUTHORIZATION

Auth Mechanism: JWT Bearer Token + Refresh Token. OAuth 2.0 for 3rd party. 2FA via OTP.

Endpoints:

Method	Endpoint	Description
`POST`	`/v1/auth/register`	Create new KAFRIADA Account + OTP
`POST`	`/v1/auth/login`	Login. Returns access_token + refresh_token
`POST`	`/v1/auth/logout`	Invalidate tokens
`POST`	`/v1/auth/refresh`	Get new access_token
`POST`	`/v1/auth/forgot-password`	Trigger password reset OTP
`POST`	`/v1/auth/reset-password`	Reset with OTP
`POST`	`/v1/auth/verify-email`	Verify email OTP
`POST`	`/v1/auth/verify-phone`	Verify phone OTP

8.5 CORE MODULE APIs

MODULE 1: ATHLETE IDENTITY API

Method	Endpoint	Description
`GET`	`/v1/athletes`	List athletes with filters
`GET`	`/v1/athletes/{kafriada_id}`	Get athlete profile by KUID
`POST`	`/v1/athletes`	Create Stage 1 Profile
`PUT`	`/v1/athletes/{id}`	Update profile
`POST`	`/v1/athletes/{id}/media`	Upload photo/video/certificate
`POST`	`/v1/athletes/{id}/verification/request`	Request Stage 2/3 Verification
`GET`	`/v1/athletes/{id}/qr`	Generate QR Code
`GET`	`/v1/athletes/{id}/profile.pdf`	Download Digital Profile

MODULE 2: CLUB MANAGEMENT API

Method	Endpoint	Description
`GET`	`/v1/organizations`	List clubs/academies
`POST`	`/v1/organizations`	Register new organization
`GET`	`/v1/organizations/{id}`	Get org details
`POST`	`/v1/organizations/{id}/teams`	Create team
`POST`	`/v1/teams/{id}/members/invite`	Invite athlete by KUID
`POST`	`/v1/organizations/{id}/contracts`	Upload contract
`GET`	`/v1/organizations/{id}/attendance`	Get attendance records

MODULE 3: ATHLETE CARE API

`GET/POST /v1/athletes/{id}/medical-records`

`GET/POST /v1/athletes/{id}/injuries`

`POST /v1/athletes/{id}/medical-clearance`

MODULE 4: MARKETPLACE API

`GET /v1/marketplace/trials`
`POST /v1/marketplace/trials/{id}/apply`
`GET /v1/marketplace/scholarships`
`GET /v1/marketplace/jobs`
`POST /v1/marketplace/transfers/request`

8.6 PAYMENT INTEGRATION

- **Supported Gateways:**

REGION	GATEWAY
Africa	Paystack, Flutterwave, Moniepoint
Global	Stripe, Flutterwave International

- **Supported Currencies:** NGN, USD, SAR, CNY
- **Payment Categories:** Verification Fees, Club/Academy Subscription (Stage 2), Tournament Fees, Scout Access, Course Payments, Marketplace Purchases
- **Endpoints:**
 - `POST /v1/payments/initialize`
 - `POST /v1/payments/verify`
 - `GET /v1/payments/{id}`
 - `GET /v1/subscriptions`

All payments are idempotent and logged to `payments` table.

8.7 STATVEST INTEGRATION

Purpose: Provide authenticated performance data for Stage 3 Verification.

Integration Method: Secure API Webhook + Signed Payload or Manual

Data Received: Sprint Speed, Acceleration, Vertical Jump, Endurance, Reaction Time, Sport-Specific Accuracy

Rules:

1. Data is read-only in KAFRIADA. Cannot be edited by users
2. Stored with timestamp and device_id for audit
3. Historical performance tracked per KUID

`POST /v1/integrations/statvest/webhook`

8.8 NOTIFICATION SERVICES

Providers: Firebase Cloud Messaging [FCM] for Push, SMTP/SES for Email, Twilio/Telco SMS for SMS

Channels: In-App, Push, Email, SMS, WhatsApp Business API[Future]

Event Categories: Registration, Verification, Trials, Contracts, Payments, Tournaments, Medical, Marketplace

`POST /v1/notifications/send`

8.9 FILE STORAGE & CDN

Storage: AWS S3 with bucket per domain: `/media/profiles/`, `/docs/contracts/`, `/medical/`

Delivery: CloudFront CDN for global low-latency access

Security: Pre-signed URLs. Max upload 50MB. Virus scan on upload.

`POST /v1/files/upload`

8.10 SEARCH ENGINE

Engine: Elasticsearch 8.x

Indexed Entities: Athletes, Clubs, Coaches, Trials, Scholarships, Jobs

Supported Filters: Country, State, LGA, Sport, Age, Gender, Position, Verification Status,

Performance Metrics

`GET /v1/search?q=&filters={}` Response SLA: < 300ms

8.11 API SECURITY

1. Transport: HTTPS + TLS 1.3 mandatory
2. Auth: JWT + RBAC. Scopes per role
3. Validation: Input validation on all endpoints. Prevent SQLi, XSS
4. Rate Limiting: 1000 req/min per user. 5000 req/min per API Key
5. Monitoring: All requests logged. WAF enabled
6. 2FA: Enforced for Admin, Medical, Finance roles

8.12 WEBHOOKS

KAFRIADA will push real-time events to partners.

Supported Events: `payment.success`, `athlete.verified`, `club.approved`, `contract.expiring`, `subscription.renewed`

Security: HMAC-SHA256 signed payload + `X-Kafriada-Signature` header + Retry with exponential backoff

`POST /v1/webhooks/subscribe`

8.13 API VERSIONING & DEPRECATION

Format: `/api/v1/`, `/api/v2/`

Breaking changes only in major versions. N-2 version support. 6-month deprecation notice.

8.14 THIRD-PARTY INTEGRATION ROADMAP

Phase	Integrations
Phase 1	Paystack, Flutterwave, Stripe, Firebase, AWS, StatVest (GPS Wearables, Heart Rate Monitors) , Email/SMS
Phase 2	AI Analytics (Performance), League & Livescore Management, (Scout Performance Analytics), Live Streamn Chat and Matches Live Streaan
Phase 3	LMS, Digital Certificate Providers
Phase 4	NFF, CAF, Government DBs, FIFA Connect ID

8.15 API PERFORMANCE & SLA TARGETS

Metric	Target
Avg Response Time	< 300ms for standard GET/POST
Auth Response Time	< 500ms
Concurrency	10,000 concurrent users Phase 1
Uptime	99.9%
Error Rate	< 0.1%

CHAPTER 9: USER INTERFACE (UI), USER EXPERIENCE (UX) & DESIGN SYSTEM

PHASE 1 – KAFRIADA CORE

Version: 1.2 | Effective Date: 1 July 2026

9.1 INTRODUCTION

The KAFRIADA NET UI/UX is designed to deliver a modern, inclusive, and trustworthy digital experience for users across Africa and the Middle East.

Design Goal:

Make elite-level sports technology accessible to a 14-year-old grassroots athlete and a federation administrator on the same platform.

Core Principles: Simplicity, Speed, Accessibility, and Brand Consistency across Mobile, Tablet, and Web.

All interfaces must reinforce the KAFRIADA brand and support low-bandwidth environments.

9.2 DESIGN PHILOSOPHY & PRINCIPLES

Principle	Application
1. User-Centered	Workflows completed in ≤ 3 clicks. Jargon-free language
2. Mobile-First	All screens designed for 360x800px first, then scaled up
3. Consistency	Single Component Library used across all 4 KAFRIADA products
4. Accessibility	WCAG 2.1 AA compliant. Usable by all levels of digital literacy
5. Performance	Core screens load $< 3s$ on 3G. Skeleton loaders used
6. Localization	Support for EN, AR, FR, ES. RTL support for Arabic
7. Trust & Security	Verification badges and secure indicators visible on all data

9.3 BRANDING STANDARDS

Element	Specification
Brand Name	KAFRIADA NET
Parent Company	KowaGuru Technology Limited
Official Slogan	African Sport Network Platform
Platform Vision	Building Africa's Digital Sports Ecosystem
Platform Motto	One Account • One Athlete Identity • One Ecosystem
Logo Usage: Logo must appear on Login, Splash, Dashboards, Certificates, Reports, QR Pages, Emails, PDFs, and Digital ID Cards. Clear space = 1x logo height. Do not distort.	

9.4 COLOR PALETTE

Role	Color Name	Usage
Primary	KAFRIADA Green `#0A7A3E`	Primary CTAs, Brand Header
Secondary	KAFRIADA Gold `#D4A017`	Achievements, Premium, Awards
Neutral	White `#FFFFFF`	Backgrounds

Text	Dark Gray `#1F2937`	Headings, Body Text
Status Colors: `Success: #16A34A` `Warning: #F59E0B` `Error: #DC2626` `Info: #2563EB` `Verified: #2563EB` `Under Verification: #F59E0B` `Unverified: #6B7280` Full HEX, RGB, CMYK defined in KAFRIADA Brand Guidelines v1.0		

9.5 TYPOGRAPHY

Font Family: `Inter` Primary. `Noto Sans` for AR/FR/ES multilingual support.

Hierarchy:

STYLE	SIZE	WEIGHT	USAGE
H1	32px	700	Page Titles
H2	24px	600	Section Headers
H3	20px	600	Card Titles
Body	16px	400	Paragraphs
Caption	12px	400	Helper Text
Button	16px	600	CTAs

Supports Latin + Arabic + French + Spanish character sets.

9.6 ICONOGRAPHY

Library: Custom + Material Design. Stroke 2px. 24x24px base grid.

Core Icons: Athlete, Coach, Club, Scout, Competition, Medical, Education, Marketplace, Payment, Verification, QR Code, Analytics, Reports.

9.7 NAVIGATION STRUCTURE

Global Navigation: Persistent bottom nav / Left sidebar

Menu Items: Home, Dashboard, Profile, KAFRIADA CORE, KAFRIADA COMPETE, KAFRIADA EDU, Marketplace, Notifications, Messages, Settings, Help [Mobile][Web]

Rule: Menu items are rendered based on RBAC. A Scout will not see "Club Admin" tools.

9.8 ROLE-BASED DASHBOARD SPECIFICATIONS

Role	Primary Widgets
Athlete	Profile Completion %, KUID Card, Verification Status, Career Timeline, Upcoming Trials/Scholarships, Medical Status, Notifications, My Wallet
Coach	Assigned Teams, Training Schedule, Attendance Summary, Player Performance, Upcoming Matches, My Wallet
Club Admin	Teams, Athletes, Active Contracts, Subscription Status, Revenue, Upcoming Fixtures, Reports, My Wallet
Scout	Verified Athlete Search, Watchlist, Recruitment Reports, Saved Searches, Subscription Status, Host Trial/Scout, My Wallet, Sponsorship/Donation.
Medical	Injury Queue, Pending Clearances, Athlete Medical History, My Wallet
Super Admin	Total Users, Total Active Countries, Total Active States & LGAs, Pending Verifications/Approvals, Revenue Analytics, System Health, Audit Logs

9.9 USER PROFILE COMPONENT

All profiles must display:

1. Header: Profile Photo, Name, Verification Badge, KUID, QR Code
2. Tabs: Personal Info, Sports Info, Career Timeline, Achievements, Media Gallery, Statistics
3. Controls: Privacy Settings, Contact Preferences, Download Profile PDF

Public visibility respects user consent and U18 safeguarding rules.

9.10 FORMS & DATA ENTRY STANDARDS

All forms must comply with:

1. Real-time Validation: Inline errors. Red border on invalid field
2. Clarity: Clear labels, placeholders, and helper text
3. Progress: Multi-step forms show step indicator. Auto-save drafts
4. File Upload: Drag & Drop. Max 50MB. Shows upload %

Key Forms: Athlete Registration, Club Registration, Verification Request, Payment, Tournament Registration

9.11 NOTIFICATIONS & MESSAGING

Channels: In-App, Push[FCM], Email[SES], SMS

Notification Center Categories: System, Competitions, Medical, Marketplace, Education, Payments, Security

Users can manage channel preferences per category.[Twilio]

9.12 ACCESSIBILITY REQUIREMENTS - WCAG 2.1 AA

Mandatory compliance:

1. Keyboard Navigation: All interactive elements accessible via Tab
2. Screen Readers: Semantic HTML + ARIA labels
3. Contrast: Min 4.5:1 for text
4. Text Resize: Supports 200% zoom without breaking layout
5. Alt Text: All images and icons

9.13 LOCALIZATION & INTERNATIONALIZATION (i18n)

V1.0 Languages: English, Arabic [RTL], French, Spanish

Future: Portuguese, Swahili[Default]

Localized Elements: Date Format, Timezone, Currency, Number Format, Address Structure, LGA/State

9.14 RESPONSIVE DESIGN

Single codebase. Responsive breakpoints:

`Mobile: 360-768px` | `Tablet: 768-1024px` | `Desktop: 1024px+`

All features must be functional on all breakpoints. No separate "mobile site".

9.15 OFFLINE EXPERIENCE - MOBILE APP

To support low-connectivity areas:

1. View Cached Data: Last synced dashboard, profile, schedule
2. Offline Drafts: Forms saved locally and auto-sync on reconnect
3. Sync Indicator: Clear "Online/Offline/Syncing" status bar

9.16 UX PERFORMANCE & SUCCESS METRICS

Metric	Target
Registration Completion Rate	$\geq 90\%$
Average Login Time	< 10 seconds
Dashboard Load Time	< 3 seconds
Task Completion Rate	$\geq 95\%$
User Satisfaction (CSAT)	$\geq 4.5 / 5$
Mobile Crash Rate	$< 1\%$ per release

Metrics tracked via Firebase + Mixpanel for continuous improvement.

CHAPTER 10: SECURITY, PRIVACY & COMPLIANCE

PHASE 1 (10.0) – KAFRIADA CORE

Version: 1.2 | Effective Date: 1 July 2026

10.1 INTRODUCTION

Security and privacy are foundational to KAFRIADA NET. The platform processes sensitive personal data of athletes, including minors, and financial data across multiple jurisdictions.

Approach: `Security-by-Design` and `Privacy-by-Default`. Controls are embedded throughout the SDLC, infrastructure, and operations.

Objective: Protect the Confidentiality, Integrity, and Availability [CIA Triad] of all platform data while ensuring lawful processing and regulatory compliance across Africa + MENA.

10.2 SECURITY OBJECTIVES

1. Prevent unauthorized access to user accounts and data
2. Protect PII, medical, and payment data from breach
3. Ensure secure and PCI-compliant payments
4. Prevent identity fraud, age fraud, and duplicate registrations
5. Safeguard children and vulnerable users
6. Maintain immutable audit trails for all critical actions
7. Detect, contain, and respond to cyber threats
8. Ensure 99.9% availability via Backup and DR
9. Comply with NDPA 2023, GDPR, and other applicable laws[U18]

10.3 SECURITY PRINCIPLES

1. Security by Design: Threat modeling in every sprint
2. Privacy by Default: Minimum data collection and visibility
3. Least Privilege: RBAC enforced at API and DB level
4. Zero Trust: No implicit trust. Authenticate every request
5. Defense in Depth: WAF + API Gateway + Service + DB security
6. SSDLC: SAST, DAST, Dependency Scan in CI/CD
7. Continuous Monitoring: 24/7 SOC alerts

10.4 AUTHENTICATION & IDENTITY SECURITY

Supported Methods:

1. Email + Password

2. Mobile Number + OTP
3. Two-Factor Authentication - Mandatory for Admin, Finance, Medical
4. OAuth 2.0
5. SSO [Phase 4 - Enterprise][2FA][Future]

Password Policy:

- Min 8 characters
- 1 Uppercase, 1 Lowercase, 1 Number, 1 Special Character
- No password reuse in last 5 cycles
- Stored as `Argon2id` hash. Plain text never stored

Account Security:

- OTP expiry: 5 minutes. 3 attempts max
- Account lockout after 5 failed logins
- Device fingerprinting and suspicious login alerts

10.5 AUTHORIZATION - RBAC

All access controlled via Role-Based Access Control enforced server-side.

Role	Scope Example
Athlete	CRUD own profile only
Coach	CRUD assigned team + attendance
Club Admin	CRUD own organization, members, contracts
Medical	Read/Write medical records for assigned athletes
Scout	Read verified athlete profiles only
Super Admin	Full platform access

10.6 DATA ENCRYPTION

State	Standard	Scope
In Transit	TLS 1.3	All API, Web, and App traffic
At Rest	AES-256	DB, S3, Backups
Field Level	AES-256	PII, ID Docs, Medical Records, Payment Refs

Key Management: AWS KMS. Key rotation every 90 days. No keys in code.

10.7 PRIVACY PROTECTION - PRIVACY BY DEFAULT

- Default Public Profile Data: Name, KUID, Sport, Position, Verification Status, Current Club, Public Media
- Always Private: Phone, Email, Address, DOB, Medical Data, Guardian Info, Payment History

User must explicitly consent to share private data with Scouts/Clubs. Consent logged.

10.8 CHILD PROTECTION & SAFEGUARDING [U18]

For athletes < 18 years:

1. Guardian Account Linkage: Mandatory during Stage 1

2. Guardian Consent: Required for registration, data sharing, and marketplace
3. Restricted Communication: Direct messages from adults blocked by default
4. Visibility: Default profile = 'Private'. Cannot be made public without guardian approval
5. Audit: All consent actions logged with timestamp and guardian KUID

Compliant with Child Rights Act 2003 and platform safeguarding policy.

10.9 FRAUD PREVENTION

Controls implemented:

1. Duplicate Detection: Email, Phone, Biometric + ID check on Stage 3
2. KUID Uniqueness: System enforced. Never reused
3. Manual Verification: Stage 2/3 requires officer review
4. Anomaly Detection: New device, new country, velocity checks
5. Payment Verification: Gateway fraud checks + webhook validation
6. Audit: All critical actions logged for investigation

10.10 PAYMENT SECURITY

1. No Card Storage: KAFRIADA does not store PAN. Tokenization via Paystack/Flutterwave/Stripe
2. PCI DSS: Dependent on payment gateway compliance
3. Security: HTTPS only, signed webhooks, idempotency keys
4. Records: Transaction ID, Amount, Status, Gateway Ref stored. Full audit log

10.11 AUDIT LOGGING

All critical events written to immutable 'audit_logs' table.

Logged Events: Registration, Login/Logout, Password Change, Verification Approval, Club Approval, Contract Upload, Medical Update, Payment, Admin Action

Log Fields: 'user_id, kafriada_id, action, timestamp, ip_address, device, old_data, new_data, result'

Logs retained for 10 years.

10.12 BACKUP & DISASTER RECOVERY

ITEM	FREQUENCY	RETENTION
Database	Daily Full + Hourly Incremental	30 Days
Media/S3	Daily	90 Days
Config	On Change	Versioned

RPO: 24 Hours | RTO: 8 Hours for critical systems

Backups encrypted and stored in geographically separate AWS Region. Quarterly DR drill.

10.13 BUSINESS CONTINUITY

Infrastructure: Multi-AZ deployment, Load Balancer, Auto-scaling, Health Checks, Automatic Failover

Monitoring: CloudWatch + Datadog. 24/7 alerting.

BCP tested annually.

10.14 COMPLIANCE FRAMEWORK

JURISDICTION	REGULATION	SCOPE
Nigeria	Nigeria Data Protection Act [NDPA] 2023	Data processing, consent, DPO
Nigeria	Child Rights Act 2003	U18 Safeguarding
Global	GDPR Principles	Data subject rights, DPIA
GCC	Country-specific PDPL	For KSA, UAE deployments

KAFRIADA will appoint a Data Protection Officer. DPIA conducted for high-risk processing.

10.15 SECURITY MONITORING

24/7 monitoring for:

- Failed Logins, API Abuse/Rate Limit Breach, Unauthorized Access, Malware, Server/DB Anomalies, Payment Fraud
- Alerts routed to SOC. SIEM integration.

10.16 INCIDENT RESPONSE PLAN

- 8-Stage Process: `Detection > Assessment > Containment > Eradication > Recovery > Communication > Post-Mortem > Remediation`
- Breach Notification: Per NDPA, affected users notified within 72 hours.

10.17 DATA RETENTION & DELETION

DATA TYPE	RETENTION
Athlete Profile	Permanent. Anonymized on deletion request
Payment Records	7 Years
Contracts	10 Years post-expiry
Audit Logs	10 Years
Notifications	2 Years

User `Right to Deletion`: Data deleted or anonymized unless legal hold applies.

10.18 SECURITY AWARENESS

All staff, contractors, and partners with data access must complete annual training on: Data Protection, Phishing, Password Hygiene, Incident Reporting.

10.19 SECURITY TESTING

Pre-production gates:

1. SAST: Code scan on every PR
2. DAST: Scan on staging
3. Dependency Scan: Weekly
4. Pen Test: Annually + After major releases
5. API Security Test: AuthN/AuthZ tests

`Critical/High` vulnerabilities must be remediated before release.

CHAPTER 11: NON-FUNCTIONAL REQUIREMENTS (NFR)

PHASE 1 (11.0) – KAFRIADA CORE

Version: 1.2 | Effective Date: 1 July 2026

11.1 INTRODUCTION

Non-Functional Requirements define the quality attributes and operational standards that KAFRIADA NET must meet.[NFRs]

While Functional Requirements define 'what' the system does, NFRs define 'how well' the system does it.

These requirements ensure KAFRIADA NET operates as a reliable, secure, scalable, and high-performance sports technology infrastructure for Africa and MENA.

All NFRs are measurable and will be used as part of UAT and Production Go-Live criteria.

11.2 PERFORMANCE REQUIREMENTS

The system must deliver fast interactions even on low-bandwidth networks.

11.2.1 Response Time Targets

Operation	Target SLA	Measured Under
User Login	≤ 3 seconds	4G / 3G Network
Dashboard Load	≤ 3 seconds	Cached + Fresh Data
Profile View	≤ 2 seconds	With media
Global Search	≤ 3 seconds	1M records, Elasticsearch
QR Code Verification	≤ 2 seconds	Scan to display
Payment Confirmation	≤ 10 seconds	Gateway callback
File Upload	Varies	Based on file size + network

11.2.2 Concurrency & Load

Phase	Target
Phase 1 Year 1	10,000+ Athletes, 300+ Clubs, 10,000 Concurrent Users
Phase 3 Year 3	3,000,000 Athletes, 10,000 Organizations
Phase 5 Year 5	10,000,000+ Athletes, 25,000 Organizations

Scaling Method: Horizontal scaling via stateless microservices + Load Balancer + Auto-scaling groups

11.3 AVAILABILITY & RELIABILITY REQUIREMENTS

11.3.1 Availability

`Target: 99.9% Uptime Annually` = Max 8.76 hours downtime/year

Excludes: Scheduled maintenance with 72hr notice, Emergency security patches

11.3.2 Reliability

System must provide:

1. Automatic service recovery and health checks
2. Database Multi-AZ failover < 60s
3. Error monitoring and alerting 24/7
4. Verified backup and restore capability

11.4 SCALABILITY REQUIREMENTS

The architecture must scale without redesign to support growth in Users, Countries, Data, and Transactions.

Scaling Methods:

1. Infrastructure: Cloud-native on AWS. Auto-scaling
2. Application: Microservices architecture
3. Data: DB Read Replicas + Partitioning by `country\_code`
4. Performance: Redis Cache + CDN for static assets
5. Delivery: CloudFront CDN for global media delivery

11.5 SECURITY REQUIREMENTS

See Chapter 10 for full details. Key NFRs:

1. Secure Authentication: JWT + 2FA + Argon2 hashing
2. Encryption: TLS 1.3 in transit, AES-256 at rest
3. Access Control: RBAC enforced at API Gateway
4. Audit: Immutable logs for all critical actions
5. Monitoring: WAF, SIEM, Fraud Detection active

11.6 USABILITY REQUIREMENTS

KAFRIADA must be usable by users with low digital literacy.

Requirements:

1. Simple Onboarding: Registration in ≤ 5 steps, < 3 minutes
2. Navigation: Max 3 clicks to reach any core feature
3. Mobile-First: Primary design for 360x800 screens
4. Localization: EN, AR, FR, ES in V1.0
5. Error Handling: Human-readable error messages + suggested action
6. Help: Contextual tooltips and Help Center

Acceptance Target: A grassroots athlete with basic smartphone skills can register without assistance.

11.7 ACCESSIBILITY REQUIREMENTS - WCAG 2.1 AA

1. Screen Readers: Full ARIA support
2. Keyboard Navigation: All actions accessible via keyboard
3. Contrast: Min 4.5:1 ratio
4. Text Scaling: Supports 200% zoom
5. Alt Text: All images and icons have alt text
6. Forms: Proper labels and error association

11.8 MAINTAINABILITY REQUIREMENTS

The codebase must be easy to support, debug, and extend.

Requirements:

1. Code: Clean, modular, documented. Follows PEP8/ESLint standards
2. Architecture: Microservices with clear bounded contexts
3. Testing: Automated Unit, Integration, and E2E tests
4. Source Control: Git with PR and Code Review mandatory
5. CI/CD: Automated build, test, and deployment pipelines
6. Documentation: API Docs via OpenAPI 3.1, Developer Wiki

11.9 DATA INTEGRITY REQUIREMENTS

Data must be accurate, consistent, and protected from corruption.

Rules:

1. Validation: Server-side validation on all inputs
2. Constraints: DB constraints and FKs enforced
3. Deduplication: Email, Phone, KUID uniqueness enforced
4. Workflows: Approval workflows for critical changes
5. Audit: All data modifications logged

Critical Rule: A KUID cannot be duplicated, deleted, or transferred.

11.10 COMPATIBILITY REQUIREMENTS

Platform	Supported Version
Android	8.0 Oreo +
iOS	14.0 +
Browsers	Chrome 100+, Safari 14+, Firefox 90+, Edge 100+
Devices	Smartphones, Tablets, Laptops, Desktops

11.11 LOCALIZATION & INTERNATIONALIZATION

V1.0 Languages: English, Arabic [RTL], French, Spanish

Localized Elements: Date Format, Time, Currency[NGN][USD][SAR], Number Format, Address Hierarchy[LGA][Country][State]

11.12 OFFLINE CAPABILITY REQUIREMENTS - MOBILE APP

To support areas with poor connectivity:

Feature	Offline Behavior
View Data	View last synced profile, schedule, roster
Data Entry	Draft registration, attendance, injury notes
Sync	Auto-sync on reconnect. Conflict resolution logic
Indicator	Clear "Offline / Syncing / Synced" status

11.13 MONITORING & OBSERVABILITY REQUIREMENTS

1. Application Monitoring: Track API response time, error rate, user journeys [Datadog / Sentry]
 2. Infrastructure Monitoring: CPU, RAM, Disk, Network, DB performance
 3. Security Monitoring: Failed logins, WAF events, suspicious activity [SIEM][CloudWatch]
- All dashboards accessible to DevOps and Super Admin.

11.14 LOGGING REQUIREMENTS

3 log streams maintained:

Log Type	Purpose	Retention
Application Logs	Errors, Warnings, System Events	90 Days
Security Logs	Auth events, Permission changes	1 Year
Audit Logs	User/Admin/Data changes	10 Years

Standard Fields: `timestamp, user\_id, kafriada\_id, action, component, status, ip`

11.15 DISASTER RECOVERY REQUIREMENTS

Metric	Target
RTO - Recovery Time Objective	≤ 8 Hours
RPO - Recovery Point Objective	≤ 24 Hours

Process: Daily DB backups, Hourly increments, Quarterly DR test, Geo-redundant storage

11.16 DEVELOPMENT QUALITY REQUIREMENTS

Area	Standard
Code Quality	Linting, Documentation, Peer Review required
Test Coverage	$\geq 70\%$ Overall. $\geq 90\%$ for Critical Modules
Critical Modules	Auth, Payments, KUID, Verification, Medical
Deployment	Automated CI/CD with Blue/Green deployment + Rollback

11.17 DEPLOYMENT ENVIRONMENTS

1. Dev: For active development and unit testing
2. Staging: Mirror of Prod. For UAT, Security, and Performance testing
3. Production: Live environment for end users

All deployments via CI/CD. Rollback to previous version < 15 minutes.

11.18 SUPPORT REQUIREMENTS

1. User Support: In-app Chat, Email support@kafriada.net, Help Center, Knowledge Base
 2. Technical Support: 24/7 monitoring, Bug triage, Security patching, Performance tuning
- SLA: P1 Critical Bug < 4hr response. P2 < 24hr response

11.19 PRODUCTION READINESS / QUALITY GATE CRITERIA

KAFRIADA NET is "Go-Live Ready" when:

1. 100% of P1/P2 Functional Requirements completed
2. Security Pen Test passed with 0 Critical/High findings
3. All Performance NFRs met under load test
4. DR and Backup restore successfully tested
5. UAT signed off by Product and Key Stakeholders
6. Monitoring and Alerting active in Production

CHAPTER 12: SYSTEM ARCHITECTURE & TECHNICAL DESIGN

PHASE 1 (12.0) – KAFRIADA CORE

Version: 1.2 | Effective Date: 1 July 2026

12.1 INTRODUCTION

This chapter defines the technical blueprint for KAFRIADA NET. The architecture is designed to support a continental-scale, multi-sport, multi-country platform with 10M+ users.

Architecture Drivers:

1. Scale: 1M+ Athletes Year 3, 10M+ Year 5
2. Security: Secure Identity + Medical + Payment data
3. Performance: <3s response, 99.9% uptime
4. Flexibility: API-First, Microservices, Multi-product ecosystem

Core Principles: Cloud-Native, API-First, Mobile-First, Security-by-Design, Data-Driven

12.2 HIGH-LEVEL SYSTEM ARCHITECTURE

[USERS]

|

[CLIENT LAYER]

| Mobile App [Flutter] | Web Portal [Next.js] | Admin Dashboard [Next.js]

|

[API GATEWAY LAYER]

| Auth, Routing, Rate Limit, WAF, Logging

|

[MICROSERVICES LAYER]

| Identity | Athlete | Club | Care | Marketplace | Payment | Notification | Search | Analytics |

|

[DATA LAYER]

| PostgreSQL [Primary] | Redis [Cache] | Elasticsearch [Search] | AWS S3 [Storage] |

|

[CLOUD INFRASTRUCTURE LAYER]

| AWS [Compute, DB, Storage, CDN, Security, Monitoring, Backup]

12.3 CLIENT APPLICATIONS ARCHITECTURE

12.3.1 Mobile Application

Technology: `Flutter 3.x`

Rationale: Single codebase for Android/iOS, excellent offline support, high performance, low maintenance

Core Modules:

Role	Key Features
Athlete	Registration, KUID, QR, Profile, Timeline, Trials, Notifications
Coach	Team Mgmt, Attendance, Schedule, Player Performance
Club Admin	Org Mgmt, Contracts, Reports, Membership
Guardian	Consent Mgmt, Minor Profile Monitoring
Scout	Athlete Search, Watchlist, Recruitment Reports

12.3.2 Web Portal

Technology: `React.js / Next.js`

Purpose: Dashboards for Professional users: Clubs, Scouts, Federations, Enterprise

12.3.3 Super Admin Dashboard

Purpose: Platform governance

Features: User Management, Verification Queue, Org Approvals, Payment Monitoring, Analytics, Security, Country/State/LGA Master Data

12.4 BACKEND ARCHITECTURE

12.4.1 Technology Stack

Primary: `Node.js 20 LTS + NestJS + TypeScript`

Rationale: Enterprise structure, high performance, strong typing, large ecosystem

Alternative: `Django REST Framework` for AI/Analytics services

12.4.2 Microservices Breakdown

SERVICE	RESPONSIBILITY	DB SCHEMA
1. Identity Service	Auth, Users, Roles, KUID Gen, Verification Workflow	`users`, `roles`
2. Athlete Service	Profiles, Career, Media, Stats, Public Profile	`athlete_profiles`
3. Club Service	Org, Teams, Membership, Coaches, Contracts	`organizations`, `teams`
4. Athlete Care Service	Medical Records, Injuries, Rehab, Clearance	`medical_records`
5. Marketplace Service	Trials, Scholarships, Jobs, Transfers, Sponsorships	`trials`, `scholarships`
6. Payment Service	Billing, Subscriptions, Invoices, Gateway Integration	`payments`
7. Notification Service	Push, Email, SMS orchestration	`notifications`
8. Search Service	Indexing and querying for ES	Connects to ES
9. Analytics Service	Reporting, KPIs, BI, Dashboards	Read Replica

All services communicate via REST/gRPC and async events via AWS SQS/SNS.

12.5 DATA ARCHITECTURE

Component	Technology	Purpose
Primary DB	PostgreSQL 15 Managed RDS	ACID transactions. Source of Truth
Cache	Redis ElastiCache	Sessions, Dashboard data, Rate limiting
Search	Elasticsearch 8	Athlete/Club/Opportunity search
Object Storage	AWS S3 + CloudFront CDN	Media, Docs, Contracts
Data Warehouse	AWS Redshift [Phase 2]	Analytics and BI

12.6 CLOUD INFRASTRUCTURE DESIGN

Primary Provider: `Amazon Web Services [AWS]`

Deployment Regions:

1. Africa: `AWS Lagos Region [af-south-1]` - For data sovereignty and low latency
2. MENA: `AWS Bahrain Region [me-south-1]` - For GCC compliance

Core Infrastructure Components:

Layer	AWS Service
Compute	ECS Fargate / EKS. Containerized microservices + Auto Scaling
Database	RDS PostgreSQL Multi-AZ with Read Replicas
Storage	S3 + CloudFront CDN
Security	IAM, WAF, KMS, Secrets Manager, Security Groups
Networking	VPC, ALB, Route53
Monitoring	CloudWatch, X-Ray, GuardDuty

12.7 AUTHENTICATION ARCHITECTURE

Flow: OAuth 2.0 + JWT Access Token + Refresh Token

User -> Mobile/Web -> API Gateway -> Identity Service

-> Validate Credentials -> Generate JWT -> RBAC Check -> Resource

Features: 2FA via OTP, Device Tracking, Session Invalidation, Token Blacklisting

12.8 DATA FLOW: ATHLETE REGISTRATION EXAMPLE

Athlete -> Mobile App -> API Gateway -> Identity Service

-> Validate -> Save to PostgreSQL -> Generate KUID

-> Generate QR -> Trigger Notification Service -> Return Dashboard

12.9 MEDIA PROCESSING ARCHITECTURE

Pipeline for Video/Image Upload:

1. Client Upload -> Pre-signed S3 URL
2. Validation: File type, size <50MB
3. Security: Antivirus scan via AWS Lambda
4. Processing: Compression and Thumbnail generation via AWS Lambda
5. Storage: Save to S3 bucket
6. Delivery: Serve via CloudFront CDN

7. Access Control: Permission check on every request

12.10 AI & DATA INTELLIGENCE ROADMAP

Phase 2:

1. AI Scouting Assistant: Talent recommendations, Player similarity, Performance prediction
2. AI Performance Analysis: Process video + StatVest data for insights

Phase 4:

3. AI Administrative Intelligence: Federation planning, Investment reports, Talent mapping

12.11 INTEGRATION ARCHITECTURE

Category	Phase 1 Integrations
Payment	Paystack, Flutterwave, Stripe
Communication	Firebase FCM, AWS SES, Twilio SMS
Performance	StatVest API
Cloud	AWS S3, CloudFront, SNS/SQS
Future	GPS Wearables, Government APIs, FIFA Connect

All external integrations via API Gateway with Webhook + Retry logic.

12.12 DEVELOPMENT & DEPLOYMENT ARCHITECTURE

Environments: `Local > Dev > Staging > Production`

Version Control: `GitHub`

CI/CD Pipeline: `GitHub Actions`

Developer Commit -> Automated Tests -> SAST/DAST Scan

-> Build Docker Image -> Deploy to Staging

-> UAT Approval -> Blue/Green Deploy to Production -> Rollback on Failure

12.13 TECHNICAL ARCHITECTURE GOALS & SLAs

Attribute	Target
Scalability	1M+ Athletes Phase 1, 10M+ Phase 5
Availability	99.9% Uptime
Performance	<300ms API Avg, <3s Page Load
Security	Enterprise-grade, NDPA/GDPR Compliant
Deployment	Multi-Region: Africa + MENA
Languages	4+ Languages, RTL Support
Extensibility	Unlimited Sports and Organizations

CHAPTER 13: DEVELOPMENT ROADMAP, SPRINT PLANNING & ENGINEERING IMPLEMENTATION PLAN

PHASE 1 (13.0) – KAFRIADA CORE

Version: 1.2 | Timeline: 24 Weeks | Effective Date: 1 July 2026

13.1 INTRODUCTION

This chapter defines the execution plan to deliver KAFRIADA CORE MVP from SRS to Production.

Approach: Agile Scrum with 2-week sprints, Continuous Integration, and User-Centric Delivery.

Core Principle: `One Account • One Athlete Identity • One Ecosystem`

Objective: Deliver a production-ready, secure, and scalable MVP in 6 months that validates the KAFRIADA identity system across Africa.

13.2 PHASE 1 MVP SCOPE

Item	Details
Product Name	KAFRIADA CORE
Duration	24 Weeks / 6 Months
Primary Goal	Launch Africa's first verified athlete identity and grassroots sports management platform
Target Users	Athletes, Coaches, Clubs, Guardians, Scouts, Admins

13.3 MVP CORE DELIVERABLES

The MVP will launch with 5 functional modules:

1. KAFRIADA Identity System: Registration, KUID Generation, QR, Profile, Verification Workflow Stage 1-3
2. Club Management System: Club/Academy Reg, Teams, Athlete Invite, Coach Assign, Membership, Attendance, Contracts
3. Athlete Care System: Medical Profile, Emergency Info, Injury, Rehab, Medical Clearance
4. Connect Marketplace: Scholarships, Trials, Jobs, Sponsorships, Transfer Listings
5. Admin & Finance System: User Mgmt, Approvals, Payment, Subscriptions, Reports, Audit Logs

13.4 DEVELOPMENT TEAM STRUCTURE

Recommended Team Size: 10 - 12 Engineers

Role	Count	Key Responsibilities
Product Owner	1	Vision, Backlog Prioritization, Stakeholder Mgmt
Project Manager / Scrum Master	1	Sprint Planning, Risk, Timeline, Delivery
Technical Lead / Architect	1	Architecture, Code Standards, Tech Decisions

Backend Engineers	3	APIs, Business Logic, DB, Security, NestJS
Mobile Engineers	2	Android/iOS App, Offline, Performance
Frontend Engineers [Next.js]	2	Web Portal, Admin Dashboard
UI/UX Designer	1	Research, Wireframes, Design System
QA Engineer	1	Test Plans, Functional, Security, UAT
DevOps / DB Engineer	1	AWS, CI/CD, Monitoring, DB, Backups [Flutter]

13.5 AGILE SPRINT STRUCTURE

- Methodology: Scrum
- Sprint Duration: 2 Weeks
- Sprint Ceremonies: Planning, Daily Standup, Code Review, QA, Sprint Review, Retrospective
- Artifacts: Product Backlog, Sprint Backlog, Burndown Chart

13.6 SPRINT DEVELOPMENT TIMELINE - 24 WEEKS

Sprint	Timeline	Theme / Deliverable	Key Features
Sprint 0	Week 1-2	Project Setup	Finalize Architecture, Setup AWS, GitHub, CI/CD, DB, Design System
Sprint 1	Week 3-4	Authentication	Registration, Login, OTP, Password, Roles, JWT
Sprint 2	Week 5-6	Athlete Identity V1	Athlete Profile, KUID Generation, QR Code, Profile Settings
Sprint 3	Week 7-8	Verification System	Stage 2 Workflow, File Upload, Admin Approval Queue
Sprint 4	Week 9-10	Club Management	Org Reg, Teams, Invite Athlete, Coach Assign, Membership
Sprint 5	Week 11-12	Athlete Care	Medical Profile, Injury, Rehab, Medical Clearance
Sprint 6	Week 13-14	Marketplace V1	Post/List Trials, Scholarships, Jobs. Athlete Apply
Sprint 7	Week 15-16	Payments & Subs	Integrate Paystack/Flutterwave. Verification Fees, Club Subs
Sprint 8	Week 17-18	Search & Notifications	Elasticsearch Integration, FCM Push, Email, SMS
Sprint 9	Week 19-20	Hardening & QA	Functional, API, Security, Performance, Bug Fixes
Sprint 10	Week 21-22	Closed Beta	Onboard 1,000 Athletes + 50 Clubs. Feedback + Iteration
Sprint 11	Week 23-24	Public MVP Launch	Prod Deployment, Monitoring, Marketing, Partner Onboarding

13.7 QUALITY ASSURANCE STRATEGY

Testing is integrated into every sprint.

TEST LEVEL	OWNER	SCOPE	TARGET
Unit Testing	Developers	Functions, Services	≥ 70% Coverage
Integration Testing	QA + Backend	API to DB, Service to Service	All APIs
System Testing	QA	End-to-End User Flows	All User Roles
Security Testing	QA + Tech Lead	Auth, RBAC, Pen Test, SAST/DAST	0 Critical Bugs
Performance Testing	DevOps	Load Test 10k Concurrent Users	<3s Response
UAT	Product + Pilot Users	Real Athletes, Clubs, Coaches	Sign-off Required

Bug Tracking: Jira. Severity: P1-Critical, P2-High, P3-Medium, P4-Low

13.8 MVP LAUNCH / GO-LIVE CRITERIA

KAFRIADA CORE MVP will be approved for public launch when:

1. Security: Pen Test passed. 0 Critical/High vulnerabilities
2. Functionality: All 5 MVP modules working per Acceptance Criteria
3. Data: KUID Generation is unique and stable. Verification workflow complete
4. Performance: Meets all NFRs from Chapter 11
5. Payments: End-to-end payment and subscription flow verified
6. Operations: Backup, Monitoring, Logging, and DR tested
7. UAT: Signed off by Product Owner and 3 Pilot Clubs

13.9 POST-MVP PRODUCT ROADMAP

After successful MVP launch, development continues in parallel tracks:

Phase	Product	Timeline	Key Features
Phase 2	KAFRIADA COMPETE	Months 7-12	Tournament Mgmt, Fixtures, League Tables, Match Stats, AI Scouting
Phase 3	KAFRIADA EDU	Months 10-15	LMS, Courses, Certificates, Academy Mgmt, School Sports
Phase 4	KAFRIADA ENTERPRISE	Months 13-18	Government Dashboards, Federation APIs, National DB, Pan-Africa Rollout

Each phase will follow the same Agile delivery model.

CHAPTER 14: TESTING STRATEGY, QUALITY ASSURANCE (QA) & ACCEPTANCE PLAN

KAFRIADA CORE MVP

Version 1.2 | Effective Date: 1 July 2026

14.1 INTRODUCTION

QA is a non-negotiable requirement for KAFRIADA NET. The platform will process sensitive PII, medical data, financial transactions, and national athlete records.

Objective: Ensure KAFRIADA CORE MVP is Secure, Reliable, Functional, and Scalable before public release.

Quality Mandate: `Zero Tolerance for Data Loss, Payment Failure, or Security Breach`

This chapter defines the end-to-end testing framework, environment standards, and acceptance criteria for all releases.

14.2 QA OBJECTIVES

The QA process shall validate 8 core pillars:

Pillar	Definition
1. Functional Accuracy	All features work per SRS v1.0
2. Security Protection	NDPA 2023, GDPR, PCI-DSS compliance. No data leaks
3. Performance Reliability	Meets all NFRs under load
4. Data Integrity	100% accuracy of KUID, payments, medical records
5. User Satisfaction	UAT NPS $\geq$ 8/10 from pilot users
6. Cross-Device Compatibility	Android 10+, iOS 14+, Chrome, Firefox, Safari, Edge
7. Compliance	Business rules, role permissions, audit trails enforced
8. Production Stability	99.9% uptime, DR < 4 hours

14.3 TESTING APPROACH & TEST PYRAMID

KAFRIADA shall use a Shift-Left, Multi-Level Testing Model:

Developer Testing



Unit Testing [70% Coverage Target]



Integration Testing [APIs + 3rd Party]



System / Functional Testing [E2E Flows]



Security Testing [SAST/DAST/PenTest]



Performance Testing [Load + Stress]



User Acceptance Testing [UAT]



Production Monitoring [Observability]

14.4 TESTING ENVIRONMENTS

Environment	Purpose	Access	Data
1. Development	Dev coding, unit tests	Developers Only	Synthetic Data
2. Staging	QA, Security, UAT, Demos	QA, Product, Pilot Users	Anonymized Prod-like Data
3. Production	Live Users	DevOps, Read-only for Support	Real User Data. Encrypted

Rule: No code moves to Prod without passing Staging. Staging must mirror Prod infra on AWS.

14.5 UNIT TESTING

Owner: Developers | Tool: Jest, Pytest, Flutter Test | Target: $\geq 70\%$ Coverage

Example Test Cases:

Module	Test Case ID	Scenario	Expected Result
Auth	UT-AUTH-01	Valid login with OTP	200 OK + JWT returned
Auth	UT-AUTH-02	5 Invalid password attempts	Account locked 15 mins
Identity	UT-ID-01	KUID Generation: NG, Jigawa, BKD, 2026	'KA-NG-JG-BKD-2026-000001'
Identity	UT-ID-02	Duplicate athlete with same NIN	409 Conflict. Blocked
Payments	UT-PAY-01	Stage 2 Fee: ₦5,000	Invoice amount = ₦5,000 + VAT

14.6 INTEGRATION TESTING

Owner: QA + Backend | Tool: Postman, Cypress

Key Integration Flows:

1. Mobile App → API → DB: Athlete registration data persists correctly
2. Payment Gateway → Webhook → System: Paystack/Flutterwave payment confirms, invoice generated, profile status updated
3. Notification Service: Club approves player → FCM Push + Email sent to Athlete
4. StatVest API → Verification: Performance data received updates Stage 3 status

14.7 FUNCTIONAL TESTING

E2E validation of all SRS requirements. Executed by QA.

Module	Test Case ID	Scenario	Expected Result
KUID	FT-ID-01	Scan valid QR	Correct athlete profile loads <2s
KUID	FT-ID-02	Scan invalid/tampered QR	Error: "Invalid Identity"
Verification	FT-VER-01	Complete Stage 2: Pay + Upload docs	Status = 'UNDER VERIFICATION'
Verification	FT-VER-02	Admin approves Stage 3 with StatVest data	Status = 'VERIFIED' + Blue Badge

Club	FT-CLUB-01	Club creates team + invites athlete	Athlete receives invite notification
Care	FT-CARE-01	Medical personnel updates injury	Only Athlete + Club Doctor can view
Marketplace	FT-MKT-01	Scout posts trial. Athlete applies	Application appears in Scout dashboard

14.8 SECURITY TESTING

Owner: QA + Tech Lead | Standard: OWASP Top 10, NDPA 2023 | Tool: BurpSuite, ZAP

Area	Test Type	Requirement
Authentication	Brute Force, Session Mgmt	Lockout after 5 fails. JWT expiry 1hr
Authorization	RBAC, IDOR	Athlete cannot access other athlete medical data
API Security	Injection, Rate Limiting	100 req/min per user. All inputs sanitized
Data Protection	Encryption at Rest + Transit	AES-256. TLS 1.3. Medical data field-level encryption
Penetration Test	External Audit	0 Critical/High findings before launch

14.9 PERFORMANCE TESTING

Owner: DevOps + QA | Tool: K6, JMeter | Environment: Staging

Metric	Target	Load Condition
Login	< 3 seconds	10,000 concurrent users
Profile Load	< 3 seconds	95th percentile
Search	< 3 seconds	5,000 req/min
QR Verification	< 2 seconds	1,000 scans/min
API Response	< 300ms avg	Normal Load
Stress Limit	System degrades gracefully	> 15,000 concurrent users

14.10 PLATFORM & COMPATIBILITY TESTING

- Mobile: Android 10+ on 2GB RAM devices. iOS 14+. Test: Install, Offline, Camera, QR, Battery, Upload
- Web: Chrome, Firefox, Safari, Edge - Last 2 versions. Test: Dashboards, Forms, Reports, Downloads

14.11 USER ACCEPTANCE TESTING (UAT)

Owner: Product Owner | Participants: 1,000 Pilot Athletes + 50 Clubs

Stakeholder	UAT Scenarios	Sign-off Criteria
Athletes	Register, Create Profile, Pay, Get Verified	>90% can complete without help
Clubs	Register Club, Create Team, Invite Players	Team roster management works
Coaches	Mark Attendance, View Player Profile	Data is accurate and timely
Medical	Add Injury, Issue Clearance	HIPAA-like access control enforced

Scouts Search, Filter, Contact Athlete Search returns relevant results

UAT Exit Criteria: $\geq 90\%$ Test Cases Pass + NPS $\geq 8/10$

14.12 BUG MANAGEMENT & SEVERITY MATRIX

Tool: Jira | SLA:

Severity	Definition	Examples	Resolution SLA
P1 - Critical	System down, Data loss, Payment fail, Security breach	Cannot login, Money deducted no credit	< 4 Hours. Hotfix
P2 - High	Major feature broken	Cannot register, QR not working	< 24 Hours. Before Release
P3 - Medium	Partial function issue	Slow search, Wrong report total	Next Sprint
P4 - Low	Cosmetic, UI	Spelling, Alignment	Backlog

14.13 RELEASE APPROVAL & GO-LIVE CHECKLIST

Production deployment is approved only when ALL are checked:

- ☐ ✓ All P1 and P2 bugs resolved and retested
- ☐ ✓ Security Pen Test passed. Report signed by Tech Lead
- ☐ ✓ Performance targets achieved under load test
- ☐ ✓ Backup and DR tested. RTO < 4hrs, RPO < 1hr
- ☐ ✓ Monitoring, Logging, Alerting active on Datadog
- ☐ ✓ UAT Sign-off received from Product Owner
- ☐ ✓ Deployment Plan and Rollback Plan approved
- ☐ ✓ Documentation updated: API Docs, User Manual

14.14 CONTINUOUS QA & MONITORING POST-LAUNCH

After Go-Live, QA continues via:

1. Observability: Datadog APM, Error Tracking, Uptime Monitoring
2. User Feedback: In-app feedback, Support tickets
3. Crash Analytics: Firebase Crashlytics for Mobile
4. Monthly Regression: Full test pack run before any minor release

CHAPTER 15: DEPLOYMENT, DEVOPS, CLOUD INFRASTRUCTURE & OPERATIONS MANAGEMENT

KAFRIADA NET – PHASE 1 (15.0)

KAFRIADA CORE | Version 1.2 | Effective Date: 1 July 2026

15.1 INTRODUCTION

This chapter defines the technical operations framework to deploy, run, and scale KAFRIADA NET as a 24/7, secure, and compliant digital sports ecosystem.

Objective: Build a production-grade AWS infrastructure that supports 10,000 users at MVP and scales to 10M+ users across Africa by Phase 4.

Operating Principle: `Automate Everything. Monitor Everything. Secure Everything.`

15.2 DEVOPS OBJECTIVES

Objective	Target
1. Delivery Speed	Deploy to Production 2x per week
2. Reliability	99.9% Platform Uptime
3. Quality	100% Automated Tests in CI Pipeline
4. Security	Security Scan on every commit. 0 secrets in code
5. Stability	<30 min MTTR for P1 incidents
6. Scalability	Auto-scale from 1k to 100k concurrent users

15.3 SOFTWARE DEVELOPMENT LIFECYCLE [SDLC]

KAFRIADA follows Agile DevOps SDLC:

`Plan → Design → Develop → Test → Build → Deploy → Operate → Monitor → Feedback`

15.4 SOURCE CODE MANAGEMENT [SCM]

Platform: GitHub Enterprise

Repository Structure: Monorepo with service folders

KAFRIADA-NET/

- ├── /mobile-app/ Flutter Application
- ├── /web-platform/ Next.js Admin + Public Portal
- ├── /backend-api/ NestJS Microservices
- ├── /database/ Prisma Migrations, Seeds
- ├── /infrastructure/ Terraform, Docker, K8s Manifests
- └── /documentation/ API Docs, Runbooks

15.5 BRANCHING STRATEGY: GITFLOW

Branch	Purpose	Deploys To
main	Production-ready code only	Production
develop	Integration branch for next release	Staging
feature/	`feature/athlete-kuid`, `feature/payments`	Dev
release/	`release/v1.0.0` - Final QA + Bug fixes	Staging
hotfix/	`hotfix/security-patch`	Production

Rule: All code merged via Pull Request + 2 Approvals + CI Pass

15.6 CI/CD PIPELINE

Tool: GitHub Actions + AWS CodePipeline

CI Pipeline - On Every Push to `develop`

1. Lint & Code Quality: ESLint, SonarQube
2. Automated Testing: Unit + Integration Tests. $\geq 70\%$ coverage gate
3. Security Scan: SAST - Snyk, Dependency Check
4. Build: Docker Image $\rightarrow$ Push to Amazon ECR
5. Artifact: Versioned Release Package

CD Pipeline - On Merge to `main`

1. Deploy to Staging: Auto-deploy to AWS Staging ECS
2. Acceptance Tests: Run Cypress E2E + Smoke Tests
3. Manual Approval: Required from Tech Lead + PM
4. Deploy to Production: Blue/Green deployment to AWS Prod ECS
5. Post-Deploy: Health checks + Rollback if failure

15.7 CLOUD INFRASTRUCTURE ARCHITECTURE

- Primary Cloud: Amazon Web Services [AWS]
- Region: `af-south-1` Cape Town + `eu-west-1` Ireland for DR
- Reason: GDPR compliance, Africa latency, Enterprise SLAs

15.8 AWS INFRASTRUCTURE COMPONENTS

Layer	AWS Service	Configuration	Purpose
1. Compute	ECS Fargate	Microservices Containers	Run Backend APIs. Serverless, auto-scale
2. Database	Amazon RDS PostgreSQL	Multi-AZ, http://db.r5.large	Primary DB. Automated backups, encryption
3. Storage	Amazon S3	Private Buckets + KMS	Athlete images, videos, docs. Signed URLs
4. CDN	Amazon CloudFront	Global Edge Locations	Deliver static assets <200ms globally
5. Load Balancer	ALB + Auto Scaling Group	Across 2 AZs	Distribute traffic. Health checks

6. DNS & SSL	Route53 + ACM	`kafriada.net`	Domains: `api.` , `admin.` , `www.` HTTPS only
7. Cache	Amazon ElastiCache Redis	Cluster Mode	Session, API response caching
8. Queue	Amazon SQS + SNS	Async Jobs	Payments, Notifications, Emails

15.9 CONTAINERIZATION & ORCHESTRATION

- Technology: Docker for all services
- Orchestration: AWS ECS Fargate for Phase 1-3
- Future: Migrate to Amazon EKS in Phase 4 for multi-country, multi-cluster deployment.[Kubernetes]

Core Containers:

`identity-service`, `club-service`, `payment-service`, `notification-service`, `web-app`, `admin-app`

15.10 ENVIRONMENT MANAGEMENT

Environment	Purpose	Isolation
Development	Dev coding	Separate AWS Account, Dev DB
Testing/QA	QA Automated Tests	Staging-like data
Staging	UAT, Demos, Final QA	1:1 Infra mirror of Prod
Production	Live Users	Locked down. 2FA required

Rule: Separate VPC, DB, Secrets, and IAM Roles for each environment.

15.11 MONITORING, LOGGING & OBSERVABILITY

Stack: AWS Native + 3rd Party

Area	Tool	Metrics Monitored
Application	Datadog + Sentry	API latency, Error rate, Crash reports, User flows
Infrastructure	AWS CloudWatch	CPU, Memory, Disk, Network, RDS metrics
Security	AWS GuardDuty + WAF	Failed logins, DDoS, SQL Injection attempts
Uptime	UptimeRobot	External check every 30s

Alerting: P1 alerts → PagerDuty → On-call Engineer < 5 min

15.12 BACKUP & DISASTER RECOVERY [DR]

Backup Strategy:

1. Database: RDS Automated Daily Snapshots + Point-in-time recovery. Retention: 35 days
2. Media: S3 Cross-Region Replication to Ireland. Versioning enabled
3. Infrastructure: All infra as Code via Terraform. Stored in GitHub

DR Targets:

- RTO: Recovery Time Objective ≤ 4 Hours
- RPO: Recovery Point Objective ≤ 1 Hour
- DR Drill: Full failover test quarterly

15.13 SYSTEM MAINTENANCE SCHEDULE

Frequency	Activity	Owner
Weekly	Security patching, Log review, Cost review	DevOps
Monthly	DB optimization, Dependency updates, Backup restore test	DB Eng + DevOps
Quarterly	DR Test, Pen Test, Infrastructure audit	Tech Lead + QA

15.14 TECHNICAL SUPPORT & ESCALATION MATRIX

Level	Team	Scope	SLA
L1: User Support	Customer Care	Password reset, How-to, Account issues	< 24 hours
L2: Technical Support	QA + DevOps	System errors, Integration issues, Data fix	< 4 hours
L3: Engineering	Tech Lead + Devs	Bugs, Architecture, Security Incidents	< 1 hour for P1

15.15 SERVICE LEVEL AGREEMENT [SLA]

KAFRIADA commits to the following for Production:

Priority	Definition	Response Time	Resolution Target
P1 - Critical	Platform down, Payment down, Data breach	15 Minutes	< 4 Hours
P2 - High	Major feature broken for >20% users	1 Hour	< 24 Hours
P3 - Medium	Minor feature issue	4 Hours	Next Release
P4 - Low	Cosmetic	24 Hours	Backlog

15.16 OPERATIONAL KPIs

KPI	Target	Measurement
Platform Availability	99.9%	CloudWatch
Deployment Frequency	2x per week	GitHub
Deployment Failure Rate	< 5%	CodePipeline
Mean Time To Recovery [MTTR]	< 30 Minutes	PagerDuty
Backup Success Rate	100%	AWS Backup
API P95 Response Time	< 300ms	Datadog

15.17 INFRASTRUCTURE ROADMAP

Phase	Infrastructure Addition	Trigger
Phase 2	Add: Live Match Servers, WebSocket, Analytics DB	100k MAU
Phase 3	Add: Video CDN, LMS Servers, Transcoding	500k MAU
Phase 4	Add: Multi-Region AWS, EKS, Data Lake, Public APIs	5M MAU, Govt Contracts

CHAPTER 16: BUSINESS MODEL, MONETIZATION ARCHITECTURE & REVENUE SYSTEM

KAFRIADA NET Version 1.2 | Effective Date: 1 July 2026

16.1 INTRODUCTION

KAFRIADA NET is positioned as 'Africa's Digital Sports Infrastructure Platform'. Beyond technology, it is an economic engine that connects and monetizes the entire sports value chain.

Core Thesis: By owning the 'One Account • One Athlete Identity • One Ecosystem', KAFRIADA creates a data and transaction moat that generates recurring revenue from athletes, clubs, scouts, and institutions.

Objective: Build a sustainable, multi-stream revenue business that funds free access for grassroots while scaling to enterprise.

16.2 BUSINESS MODEL VISION

Positioning: The OS for African Sports

Value Pillars:

1. Identity: Verified KAFRIADA ID as the passport for sport
2. Management: ERP for Clubs, Academies, Federations
3. Discovery: Data-driven Scout + Talent Network
4. Education: KAFRIADA EDU for upskilling
5. Intelligence: Analytics for Govts, Brands, and Investors

16.3 REVENUE MODEL OVERVIEW - 7 STREAMS

S/N	Revenue Stream	Primary Customer	Model
1	Verification Services	Athletes, Coaches	Transactional Fee
2	Club & Academy SaaS	Clubs, Schools, Academies	Subscription
3	Tournament Management	Organizers, LGAs, Associations	Fee + Commission
4	Scout & Agent Network	Scouts, Clubs, Agents	Subscription + Pay-per-contact
5	Education & Courses	Athletes, Coaches	Course Fee + Certification
6	Marketplace Commission	Sponsors, Vendors, Employers	Commission + Ads
7	Enterprise & Government	Federations, Ministries	Licensing + Data + API

16.4 REVENUE STREAM 1: ATHLETE & COACH VERIFICATION

Purpose: Monetize trust. A verified KUID is the core asset.

Tier	Cost	Includes	Status	Target
Stage 1: Basic	Free	KUID, QR, Basic Profile	'UNVERIFIED'	100% of users

Stage 2: Profile	₦5,000 / \$7	Photo, Bio, Measurements, History. Admin Review	'UNDER VERIFICATION'	Grassroots
Stage 3: Professional	₦25,000 / \$30	Stage 2 + StatVest Metrics, Video, Medical, Blue Badge	'VERIFIED'	Elite/Pro

Revenue Projection: 100,000 verifications @ avg ₦8,000 = ₦800,000,000 Year 1

16.5 REVENUE STREAM 2: CLUB, ACADEMY & SCHOOL SAAS

Tiered subscription for organization management tools.

Plan	Target	Annual Price	Key Features
Basic	Grassroots Clubs <50 players	₦50,000 / \$60	Roster, Attendance, Fixtures, Basic Reports
Professional	Academies, Pro Clubs	₦300,000 / \$350	Full ERP, Contracts, Medical, Performance, Finance
Enterprise	Federations, Universities	Custom: ₦5M+ / \$6,000+	Custom Dashboards, API, Data Export, SLA

Target: 8,000,000 Clubs Year 1 | 2,000,000,000 Clubs Year 3

16.6 REVENUE STREAM 3: TOURNAMENT MANAGEMENT

Digital infrastructure for competitions.

Revenue Sources:

1. Hosting Fee: ₦100,000 - ₦2,000,000 per tournament based on size
2. Registration Commission: 5% of all team/player registration fees
3. Premium Add-ons: Live Scoring ₦50,000, Stats ₦100,000, Livestream Integration ₦500,000

Target: 500 Tournaments Year 1

16.7 REVENUE STREAM 4: SCOUT & AGENT NETWORK

Monetize access to the largest verified athlete database in Africa.

Plan	Price	Access
Basic	Free	View UNVERIFIED public profiles only
Professional	\$100 / Year	Search VERIFIED DB, Filters, Save Lists, 10 Contact Requests/mo
Enterprise	\$1,200 / Year	Unlimited Contacts, Analytics, Watchlists, Direct API

Target: 1,000 Professional Scouts Year 2

16.8 REVENUE STREAM 5: KAFRIADA EDU

Online learning and certification for sports professionals.

Products:

1. Courses: Football, Nutrition, Coaching, Sports Psych. ₦10,000 - ₦50,000 per course
2. Certifications: CAF/NFF Aligned. ₦25,000 per certificate
3. Academy Partnerships: Revenue share 70/30 with content creators

Target: 20,000 Course Enrollments Year 2

16.9 REVENUE STREAM 6: MARKETPLACE & MEDIA

Commission-based economy inside KAFRIADA.

Categories:

1. Opportunities: Trials, Scholarships, Jobs. 10% success fee
2. Equipment: Jerseys, Boots. 8% commission
3. Services: Agents, Physios, Trainers. 15% commission
4. Advertising: Athlete Profile Sponsorship, Banner Ads, Tournament Sponsorship

16.10 REVENUE STREAM 7: ENTERPRISE & GOVERNMENT SOLUTIONS

B2G and B2B data and software licensing.

Products:

1. National Athlete Database: Licensed to Ministry/Federation. ₦50M+/year
2. Sports Intelligence Dashboard: Participation, Investment, KPI reports. ₦10M+/year
3. API Access: For Media, Betting, Research. Tiered API call pricing

16.11 FINANCIAL TRANSACTION & BILLING ARCHITECTURE

Component	Specification
Currencies	NGN, USD, EUR, SAR. Auto FX conversion
Gateways	Paystack, Flutterwave. Future: Stripe
Billing	Recurring subscriptions, Invoices, Receipts, Refunds
Admin Dashboard	Real-time revenue, MRR, Churn, ARPU, Commission reports
Compliance	PCI-DSS, NDPA 2023, VAT/GST handling

16.12 5-YEAR REVENUE TARGETS

Metric	Year 1	Year 3	Year 5
Verified Athletes	800,000	2,000,000	5,000,000
Paying Organizations	500	2,000	5,000
Annual Tournaments	1000	2,000	5,000
Annual Revenue Target	₦450M / \$550K	₦2.1B / \$2.5M	₦8.5B / \$10M

16.13 COMPETITIVE ADVANTAGE & MOAT

1. Network Effect: More athletes → More data → More scouts → More value
2. Data Moat: Only platform with verified, longitudinal athlete data in Africa
3. All-in-One: No competitor combines Identity + ERP + Scouting + Education
4. Grassroots to Pro: Free entry point ensures mass adoption, paid tiers monetize value

16.14 KEY PERFORMANCE INDICATORS [KPIs]

Business KPI	Target Year 1
Verification Conversion Rate	15% of registered athletes
Club MRR	₦15,000,000
Scout ARPU	\$100
Gross Merchandise Volume	₦200,000,000
Revenue Churn	<5% annually

CHAPTER 17: STAKEHOLDER MANAGEMENT, USER ROLES & PERMISSION ARCHITECTURE

KAFRIADA NET | RBAC v1.0 | Effective Date: 1 July 2026

17.1 INTRODUCTION

KAFRIADA NET operates as a multi-stakeholder ecosystem. To ensure data privacy, security, child protection, and accountability, access is governed by a centralized Role-Based Access Control [RBAC] system.

Core Principle: `One User, One KUID, Many Roles. Access = Role + Context + Consent`

This chapter defines all stakeholders, their permissions, data visibility rules, and approval workflows.

17.2 USER ACCOUNT & ROLE ARCHITECTURE

Rule 1: One Account: Every person has 1 KAFRIADA Account tied to 1 KUID

Rule 2: Multiple Roles: A single account can hold multiple approved roles.

Example: `Athlete + Coach + Scout`

Rule 3: Contextual Access: Permissions are granted based on Role + Organization + Relationship

17.3 ROLE CATEGORIES & DEFINITIONS

(Aligned with Section 4.3: 28 Roles across 4 Categories) See [4.3 User Categories & Roles](#)

Category	Role ID	Role Name	Account Type	Primary Purpose
A. Individual	R-01	Athlete	Individual	An individual sports participant who is the primary data subject and owner of their personal profile, performance metrics, and career trajectory.
	R-02	Coach	Individual	An individual sports instructor responsible for training athletes, planning sessions, conducting performance assessments, and managing assigned team rosters.
	R-03	Club Administrator /Manager	Individual	An individual operational manager responsible for day-to-day squad logistics, staff assignments, and team administrative tasks at the team level.
	R-04	Team Assistant	Individual	An individual support staff member who assists coaches and managers with operational tasks, attendance logs, equipment tracking, and session preparation.
	R-05	Scout	Individual , Paid	An individual talent evaluator authorized to search, identify, observe, and evaluate

				athletes across public and verified platform databases.
	R-06	Scout Manager	Individual , Paid	An individual head evaluator who leads scouting operations, coordinates scout networks, assigns evaluation tasks, and manages talent pipeline reports.
	R-07	Medical Personnel	Individual	A licensed healthcare professional (doctor, physio, medic) authorized to manage athlete health profiles, log injuries, upload treatment plans, and issue clearance reports.
	R-08	Volunteer	Individual	An individual event or program helper who assists with event operations, match-day coordination, check-ins, and local community activities.
	R-09	Parent/Guardian	Individual	A legally recognized parent or legal guardian responsible for granting consent, managing account permissions, and monitoring platform actions for minor athletes under 18.
	R-10	Guest/Public User	Individual	An unauthenticated or standard public visitor who browses public leaderboards, news, public event schedules, and open athlete profiles.
	R-11	Referee	Individual	An official match referee or umpire tasked with submitting verified match results, player statistics, disciplinary reports, and official match sheets.
	R-12	Sports Agent	Individual , Paid	A licensed representative who manages commercial opportunities, transfer inquiries, sponsorship deals, and contract negotiations on behalf of athletes.
B.	R-11	Club Administrator	Organization	An official representative of a registered sports club responsible for club-wide administration, squad rosters, organizational settings, and official affiliations.
	R-12	Academy Administrator	Organization	An official representative of a sports academy who manages youth development programs, training pipelines, player admissions, and academy rosters.
	R-13	School Administrator	Organization	An educational institution representative responsible for managing primary or secondary school athletic programs, student-athlete rosters, and inter-school competitions.
	R-14	University Admin	Organization	A collegiate sports administrator who manages university sports programs, varsity teams, student-athlete eligibility, and inter-collegiate leagues.

	R-15	Tournament Organizer	Organization	An entity or administrator responsible for creating, scheduling, managing, and executing sports tournaments, brackets, and fixtures.
	R-16	League Organizer	Organization	An administrative body or coordinator managing league structures, season schedules, division standings, and overall league compliance.
	R-17	Sports Association Admin	Organization	A governing sports body representative responsible for sport-specific regional charters, affiliate club oversight, and grassroots development.
	R-18	NGO Admin	Organization	A non-governmental organization representative managing sports-for-development initiatives, social impact programs, and grant-funded athletic projects.
	R-19	Sponsor Admin	Organization	A corporate or private sponsor representative who manages brand sponsorship campaigns, event funding, athlete endorsements, and grant allocations.
C. Government	R-20	LGA Sports Officer	Government	A Local Government Area official tasked with overseeing municipal grassroots sports development, community facilities, and local sports initiatives.
	R-21	State Sports Council Admin	Government	A state-level sports official responsible for state sports policy, regional sports council events, state team selections, and facility management.
	R-22	Federation Admin	Government	A national or a sports federation official who governs sport-specific national policies, official athlete rankings, national team selections, and sanctioning.
	R-23	Ministry of Sports Admin	Government	A national government official overseeing macro-level sports policies, sector budgeting, national sports analytics, and overall sports ministry initiatives.
D. Platform Admin	R-24	Super Administrator	Platform	An executive platform administrator with unrestricted access to global system configurations, user management, security protocols, and system-wide overrides.
	R-25	Country/State/LGA Admin	Platform	A regional platform system administrator responsible for localized verification approvals, regional user onboarding, and regional platform support.
	R-26	Finance, Support, QA, Auditor	Platform	A platform operations specialist responsible for financial reconciliation, system activity auditing, quality

				assurance testing, and technical user support.
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17.4 ROLE PERMISSION MATRIX

R = Read | C = Create | U = Update | D = Delete | A = Approve | - = No Access

Module / Data	R-01 Athlete	R-02 Coach	R-03 Club Mgr	R-04 Team Asst	R-05 Scout	R-06 Scout Mgr	R-07 Medical	R-08 Volunteer	R-09 Parent	R-10 Guest	R-11 Referee	R-12 Agent
Own Profile	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U
Public Profiles	R	R	R	R	R	R	R	R	R	R	R	R
Medical Records	R	-	-	-	-	-	C, R, U	-	R	-	-	-
Team Rosters	R	R, U	R, U	R	-	-	R	-	R	-	R	-
Match Reports & Stats	R	R	R	R	R	R	-	R (Entry)	R	R	C, R, U	R
Contracts & Transfers	R	-	-	-	-	-	-	-	R	-	-	C, R, U
Scout DB & Search	-	-	-	-	R*	R*	-	-	-	-	-	R*
Verification Approvals	-	-	-	-	-	-	-	-	-	-	-	-
Tournament / Fixtures	R	R	R	R	R	R	R	R	R	R	R	R
Wallet & Financials	R, U	R, U	-	-	R, U	R, U	-	-	R, U	-	-	R, U
User & Role Mgmt	-	-	-	-	-	-	-	-	-	-	-	-

*Note: Scope restricted to assigned entities/consent. Scout access is subject to active subscription tiers.

Module / Data	R-13 Club Admin	R-14 Academy Admin	R-15 School Admin	R-16 Varsity Admin	R-17 Tourn. Org	R-18 League Org	R-19 Assoc. Admin	R-20 NGO Admin	R-21 Sponsor Admin
Own Org Profile	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U
Athlete Profiles	R	R	R	R	R	R	R	R	R
Medical Records	-	-	-	-	-	-	-	-	-
Team Rosters	C, R, U, D	C, R, U, D	C, R, U, D	C, R, U, D	R	R	R	C, R, U	-
Contracts & Transfers	C, R, U	C, R, U	-	C, R, U	-	-	R	-	C, R, U
Scout DB & Search	R	R	-	R	-	-	R	-	R
Verification Request	C, R	C, R	C, R	C, R	-	-	-	-	-

Tournaments / League	R	R	C, R, U	C, R, U	C, R, U, D	C, R, U, D	R, A	C, R, U	R
Wallet & Financials	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U	R, U
User & Staff Mgmt	C, R, U	C, R, U	C, R, U	C, R, U	C, R, U	C, R, U	C, R, U	C, R, U	C, R, U

Module / Data	R-22 LGA Officer	R-23 State Council	R-24 Federation Admin	R-25 Ministry Admin	R-26 Super Admin	R-27 Regional Admin	R-28 Finance/Audit
All User Profiles	R (Agg)	R (Agg)	R	R (Agg)	C, R, U, D	R, U	R
Medical Records	-	-	-	-	R	-	-
Team Rosters	R	R	R	R (Agg)	C, R, U, D	R	R
Contracts & Transfers	-	R	R	-	C, R, U, D	R	R
Scout Database	-	-	-	-	R	R	-
Verification Approval	A	A	A	-	C, R, U, A	C, R, U, A	-
Tournament Approval	A	A	A	-	C, R, U, A	A	-
Wallet & Financials	-	-	-	-	C, R, U, D	R	C, R, U, A
User & Role Mgmt	R	R	R	R	C, R, U, D	C, R, U	-
Audit Logs & System	-	-	-	-	C, R, U, D	R	R

17.5 DETAILED ROLE PERMISSIONS

CATEGORY A: INDIVIDUAL (MODULE 1: KAFRIADA ID)

R-01: ATHLETE

- **Can:** Create and manage personal profile; update bio, social links, gallery, and highlights; view dynamic KUID QR code; apply for open opportunities/trials; control personal privacy settings; view verified performance assessments; manage individual digital wallet (make payments, receive grants/sponsorships).
- **Cannot:** Directly edit or alter verified biometric or performance test metrics post-approval; self-approve verification requests; modify medical status directly.
- **Data Ownership:** Full ownership of personal data under NDPA/GDPR compliance; holds right to export profile data or request account deletion/anonymization.

R-02: COACH

- **Can:** View tactical and physical metrics of assigned squad members; design and record session plans; submit player attendance and subjective performance assessments; request athlete transfers/additions to assigned teams.
- **Cannot:** Modify official athlete identification or demographic data; access confidential medical history without explicit athlete or parent consent; perform administrative club-wide balance operations.

R-03: CLUB ADMINISTRATOR / MANAGER (INDIVIDUAL)

- **Can:** Manage day-to-day squad logistics; create provisional match lineups; enter match-day rosters; log equipment and kit inventories; communicate directly with assigned squad members and parents.
- **Cannot:** Alter binding club contracts; modify top-level organization billing settings; sanction official transfer releases without organization-level approval.

R-04: TEAM ASSISTANT

- **Can:** Assist in recording attendance logs; log basic training metrics; upload match media/photos; update schedule notifications for assigned teams.
- **Cannot:** Submit formal evaluation scores; view restricted athlete personal contact details; edit contract or transfer details.

R-05: SCOUT

- **Can (Free Tier):** Search and view public, unverified athlete profiles; browse open competition results and public league standings.
- **Can (Paid/Premium Tier):** Search verified database using advanced physical filters (e.g., speed, height, jump); build custom talent watchlists; bookmark athletes; submit formal contact/inquiry requests.
- **Cannot:** Directly message minor athletes (U-18) without routing request to Parent/Guardian or Club Admin; access private medical files or financial records.

R-06: SCOUT MANAGER

- **Can:** Assign target events and geographic regions to subordinate scouts; aggregate scout reports into unified talent dossiers; manage team scout subscriptions and access tiers; submit agency/club scouting recommendations.
- **Cannot:** Override platform-verified athlete physical attributes; perform national federation-level athlete approvals.

R-07: MEDICAL PERSONNEL

- **Can:** Create and update injury logs; upload medical diagnostic documents and rehab schedules; issue "Fit-to-Play" medical clearance certificates; record field-level encrypted health records.

- **Cannot:** View financial data, commercial contracts, or scout assessments; share health records publicly without explicit patient/guardian consent.
- **Audit Standard:** All record access strictly audited under healthcare confidentiality protocols (HIPAA/NDPA).

R-08: VOLUNTEER

- **Can:** Scan athlete/attendee QR codes for event check-ins; record live score updates if assigned match operator role; view basic event schedules and venue maps.
- **Cannot:** View private athlete profiles, contact info, or medical histories; modify competition fixtures or standings.

R-09: PARENT / GUARDIAN

- **Can:** Provide mandatory digital consent for U-18 registration; link and manage child account privacy; view child performance trends, attendance, and injury logs; approve or reject scouting/agent inquiry requests; manage child's linked digital wallet.
- **Cannot:** Directly alter coach assessments or official match statistics recorded by referees.
- **Trigger:** Automatically mandatory and assigned whenever an athlete DOB indicates age under 18.

R-10: GUEST / PUBLIC USER

- **Can:** Browse public homepage news, open competition brackets, public rankings, leaderboards, and basic unverified public profiles.
- **Cannot:** View detailed physical/biometric breakdown metrics, private media, contact details, or historical verified trial data; apply for opportunities.

R-11: REFEREE

- **Can:** Access assigned match sheets; enter live match statistics (goals, assists, fouls, cards, substitutions); submit post-match official match reports and disciplinary flags.
- **Cannot:** Edit match results once finalized and signed off; edit athlete profile info or team structural details.

R-12: SPORTS AGENT

- **Can (Paid Tier):** Search full verified database; manage commercial representation portfolio for signed athletes; negotiate and draft sponsorship/transfer proposals on platform; access athlete performance analytics with consent.
- **Cannot:** Sign binding contracts on behalf of minors without mandatory R-09 Parent/Guardian co-signature; access medical data without explicit consent.

CATEGORY B: ORGANIZATION (MODULE 2: CLUB MANAGEMENT)

R-13: CLUB ADMINISTRATOR

- **Can:** Manage full organization profile; register and manage multiple squad rosters; invite/assign coaches and staff; create and manage player contracts; initiate athlete transfer requests; manage club digital wallet and pay subscription tiers.
- **Cannot:** Edit core athlete PII (e.g., legal name, date of birth) post-verification without submitting formal system override ticket.

R-14: ACADEMY ADMINISTRATOR

- **Can:** Manage youth intake cohorts; create player development track reports; manage academy trial events; manage player registrations and academy tuition subscriptions.
- **Cannot:** Bypass U-18 parental consent workflows for youth intake.

R-15: SCHOOL ADMINISTRATOR

- **Can:** Manage school athletic rosters by grade/age group; register school teams for inter-school cups and state school games; upload academic eligibility statuses for student-athletes.
- **Cannot:** Access non-school-related club contracts or commercial transfer data of students.

R-16: UNIVERSITY ADMIN

- **Can:** Manage university varsity sports teams; verify student-athlete collegiate eligibility; participate in inter-university leagues; issue sports scholarship awards via platform wallet.
- **Cannot:** Alter non-university career historical records of student-athletes.

R-17: TOURNAMENT ORGANIZER

- **Can:** Create and configure competitions (group stages, knockouts, round-robin); publish match schedules and venues; assign registered referees to matches; enter/validate official scores; manage tournament sponsorship banners.
- **Cannot:** Issue official national ranking points without R-24 Federation sanctioning/approval.

R-18: LEAGUE ORGANIZER

- **Can:** Create multi-division season structures; set promotion/relegation rules; manage season-long discipline and point deduction overrides; publish official league standings.
- **Cannot:** Modify club ownership structures or internal club financial data.

R-19: SPORTS ASSOCIATION ADMIN

- **Can:** Oversee affiliated regional clubs and academies; organize regional charter events; approve affiliate membership requests; view aggregated regional participant stats.
- **Cannot:** Delete individual athlete accounts or alter national policy settings.

R-20: NGO ADMIN

- **Can:** Create sports-for-development programs; track participant demographics and social impact metrics; distribute subsidized sports kits or grants via platform wallet.
- **Cannot:** Monopolize athlete representation rights or force commercial contracts on participants.

R-21: SPONSOR ADMIN

- **Can:** Create sponsorship opportunity listings; fund specific events, teams, or verified athletes; track promotional ROI and engagement stats; disburse financial grants directly to target wallets.
- **Cannot:** Access private identity documents, full phone numbers, or unconsented medical data of sponsored athletes.

CATEGORY C: GOVERNMENT

R-22: LGA SPORTS OFFICER

- **Can:** Access municipal sports participation dashboards; monitor facility usage and local grassroots events; verify grassroots community clubs within the LGA scope.
- **Cannot:** Modify state or national tournament structures; access individual private data without legal mandate.

R-23: STATE SPORTS COUNCIL ADMIN

- **Can:** Oversee state sports development initiatives; manage state festival team rosters and state selections; review aggregate state-wide talent analytics; sanction state-level competitions.
- **Cannot:** Alter platform-wide global system settings or direct software configurations.

R-24: FEDERATION ADMIN

- **Can:** Issue official sanctioning for leagues and tournaments; manage national team player eligibility pools; establish sport-specific physical verification standards; review national performance data.
- **Cannot:** Perform arbitrary system database overrides without auditing.

R-25: MINISTRY OF SPORTS ADMIN

- **Can:** Access high-level aggregated data analytics across all sports, regions, age brackets, and genders; generate national sports participation reports; oversee policy compliance and sector allocation.
 - **Cannot:** View individual user private messages, specific medical histories, or un-aggregated private records unless under explicit legal authorization.
-

CATEGORY D: PLATFORM ADMIN

R-26: SUPER ADMINISTRATOR

- **Can:** Unrestricted full platform access; configure core system settings, security rules, RBAC policies, payment gateways, and data models; perform emergency data overrides; manage all user roles.
- **Cannot:** Bypass immutable audit logging (all actions logged irrevocably).

R-27: COUNTRY / STATE / LGA ADMIN (REGIONAL PLATFORM ADMIN)

- **Can:** Manage regional verification pipelines; validate regional verification centers and officers; resolve localized platform user support requests; review regional compliance flags.
- **Cannot:** Alter global platform architecture, database schemas, or global monetization rules.

R-28: FINANCE, SUPPORT, QA, AUDITOR

- **Can:** Perform financial balance reconciliations and transaction audits; review global audit logs; manage user support tickets; conduct quality assurance and testing scenarios on staging data.
- **Cannot:** Arbitrarily edit user passwords, grant false verification badges, or modify underlying legal contract documents.

17.6 DATA VISIBILITY & ACCESS TIERS

Tier	Tier Level	Target Data Examples	Permitted Roles & Access Rules	Compliance & Safeguards
Tier 1	Public	Athlete Name, Primary Sport, Position, Unique KUID, Dynamic QR Code, Verification Badge, Open Competition Results, Public Leaderboards.	Accessible By: All Platform Roles (R-01 to R-28) & R-10 Guest/Public Users . Access Rule: Unrestricted public access. Does not expose personal contact info or sensitive identifier numbers.	Indexable by public search; privacy controls allow R-01/R-09 to toggle overall profile visibility.
Tier 2	Protected	Phone Number, Email Address, Historical Club Roster Data, Comprehensive Assessment Stats, Media Gallery, Scouting Dossiers.	Accessible By: R-01 (Self), R-09 (Parent/Guardian), R-02 (Assigned Coach), R-13 to R-16 (Org Admins), R-05/R-06/R-12 (Paid Scouts/Agents with active tiers). Access Rule: Assigned Scope / Active Membership / Paid Subscription + Target Consent.	Direct inquiries for U-18 athletes are automatically routed to R-09 Parent/Guardian or R-13/R-14 Club Admin.
Tier 3	Restricted	National Identification Number (NIN), Passport Number, Encrypted Medical Records, Injury History, Physical Clearance Certificates, Guardian Details.	Accessible By: R-01 (Self), R-09 (Parent/Guardian), R-07 (Assigned Medical Personnel), R-24 (Super Admin). Access Rule: Explicit Mandate + Active Patient Consent + Mandatory Immutable Audit Logging.	Field-level encryption (AES-256); access strictly logged to comply with HIPAA/NDPA healthcare confidentiality rules.
Tier 4	Confidential	Platform Digital Wallet Balances, Payment Tokens,	Accessible By: R-26 (Super Administrator), R-28 (Finance/Auditor),	High-security isolation; strictly

		Bank Details, System Audit Logs, Admin Flags, Transfer Dispute Notes.	Account Owner (R-01/R-09/R-13 for own financial records). Access Rule: System Administrative Authorization + Dual-Key Audit Enforcement.	prohibited from external APIs, commercial scouts, coaches, or public queries.
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DYNAMIC VISIBILITY OVERRIDES & CONSTRAINTS

- **U-18 Minor Protection Rule:** Any profile categorized as U-18 automatically locks **Tier 2 (Protected)** contact fields. All inquiry routes bypass the minor and divert exclusively to **R-09 (Parent/Guardian)** or the verified **R-13/R-14 (Club/Academy Admin)**.
- **Government Aggregation Rule:** Roles under Category C (**R-20 to R-23**) and Ministry Admins (**R-25**) access **Tier 1** and **Tier 2** metrics strictly as **anonymized, aggregated statistics** (e.g., total registered athletes in an LGA or state performance averages) without exposing PII.
- **Scout Tier Gating:** Unpaid/Free Tier **R-05 Scouts** are restricted solely to **Tier 1 (Public)** data. Upgrading to a paid subscription unlocks search filters for **Tier 2 (Protected)** verified physical metrics.

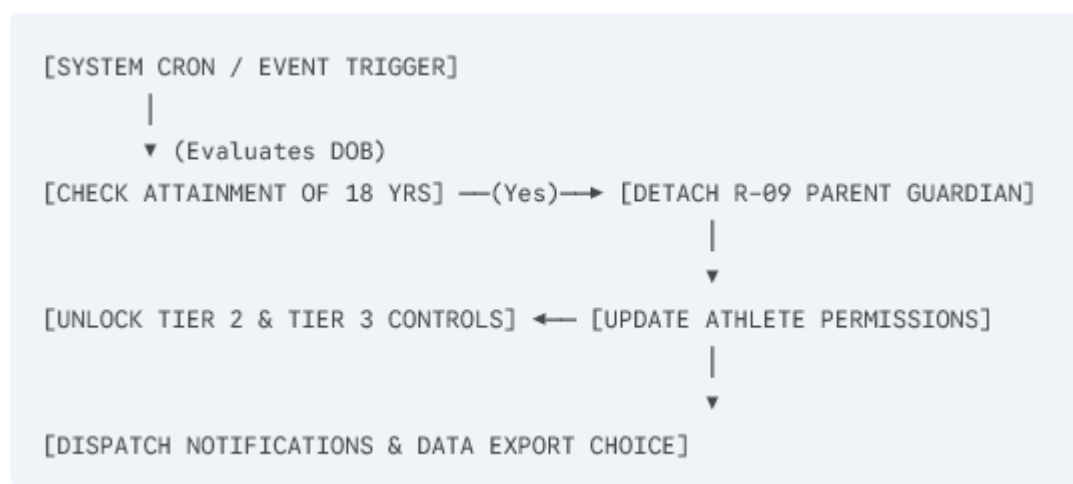
AUTOMATIC MINOR-TO-ADULT TRANSITION WORKFLOW (U-18 TO 18+)

When an athlete registered under an **R-09 (Parent/Guardian)** linked account reaches **18 years of age**, the system automatically executes a transitional event driven by a scheduled system **cron job (0 0 * * * UTC)**.

1. SYSTEM TRIGGER & DATE OF BIRTH CHECK

- **Evaluation Condition:** `CURRENT_DATE >= athlete.date_of_birth + INTERVAL '18 YEARS'`
- **Execution Frequency:** Daily at 00:00 UTC.
- **Transition State:** Account moves from `MINOR_PROTECTED` to `ADULT_PENDING_VERIFICATION` or `ADULT_DIRECT`.

2. AUTOMATED DATABASE & ROLE UPDATES



Database Schema Changes Executed On Transition:

1. **user_roles Update:**

- Updates status of linked R-09 Parent/Guardian association to ARCHIVED_TRANSITIONED.
- Unlinks parent override mandatory approval flags from users and athletes tables.

2. **athletes Table Update:**

- is_minor set to FALSE.
- consent_status set to TRANSITIONED_ADULT_DIRECT.

3. **data_visibility_tier Override:**

- Removes automatic redirect/filtering of **Tier 2 (Protected)** inquiry routes through parent handles.
- Direct profile messaging and commercial opportunity applications enable direct athlete interaction.

3. DATA PRIVACY & RE-CONSENT PROTOCOL

Upon unlocking the adult profile, the system triggers the following privacy workflows per **NDPA / GDPR** regulations:

- **Direct Data Ownership Transfer:** Full autonomy and ownership of personal profile history, media, contract approvals, and digital wallet assets are automatically assigned exclusively to the **R-01 Athlete**.
- **Historical Parental Access Revocation:**
 - The linked **R-09 Parent/Guardian** account loses edit, approve, and direct management access to the athlete's active profile and wallet.
 - Parent account converts to **Read-Only Historical Archive** mode for prior U-18 transactions (or gets decoupled entirely upon athlete choice).
- **Mandatory First-Logon Re-Consent (Adult Terms of Service):**
 - When the newly turned 18-year-old athlete logs into the platform, a required popup modal prompts them to review and sign:
 1. **Adult Terms of Service & Privacy Policy.**
 2. **Data Sharing & Scouting Database Opt-In Preferences** (enabling/disabling Tier 2 visibility to paid Scouts and Agents).
 3. **Parental Visibility Preference:** Option to allow or permanently revoke the parent's read-only viewer access to non-sensitive profile updates.

4. NOTIFICATION & AUDIT DISPATCH

On the date of transition, automated notifications are immediately sent across channels:

- **To Athlete (Email + In-App Push):**

"Happy 18th Birthday! Your KAFRIADA ID profile has been automatically upgraded to a full Adult Account. You now have full direct control over your opportunities, scouting inquiries, data privacy, and digital wallet."

- **To Parent/Guardian (Email + In-App Push):**

"Notice of Account Transition: Athlete [First Name & last Name] has reached 18 years of age. Primary administrative control and consent management for this account have transitioned directly to the athlete."

- **Audit Log Entry (audit_logs table):**
 - Records ACTION: AGE_OF_MAJORITY_TRANSITION, USER_ID, OLD_ROLE_MAPPING, NEW_ROLE_MAPPING, TIMESTAMP.

5. CORE API ENDPOINTS FOR TRANSITION WORKFLOW

- CRON /api/v1/system/jobs/age-transition — System-level batch execution script run daily.
- POST /api/v1/auth/transition-reconsent — Endpoint called by the athlete on first login to accept adult Terms of Service and update data privacy toggles.
- DELETE /api/v1/athletes/me/parent-link — Allows the 18+ athlete to manually decouple the parent account link at any time post-transition.

17.7 CHILD PROTECTION POLICY

Mandatory for NDPA & COPPA compliance.

1. Guardian Consent: Required for registration and data processing
2. Communication Gate: Scouts/Agents cannot DM minors. All requests route via Parent/Club
3. Data Minimization: Location, school name hidden by default
4. Audit: All access to U18 profiles flagged and reviewed monthly[U18]

17.8 ROLE APPROVAL WORKFLOWS

Example 1: Scout Onboarding

`Application → ID + Credential Verification → Payment → R16 Approval → Account Activated`

Example 2: Athlete Verification Stage 3

`Athlete Submits → Club Confirms → StatVest Data → R16 Admin Review → Status = VERIFIED`

Example 3: Medical Record Access

`Doctor Requests → Athlete/Parent Approves → Time-bound Access Granted → Audit Log Created`

17.9 AUDIT, LOGGING & ACCOUNTABILITY

All critical actions are recorded immutably:

- What: Profile edit, Verification, Data export, Payment, Role change
- Who: User ID, Role, IP Address
- When: Timestamp
- Where: Device, Location

Retention: 7 years. Used for compliance, dispute resolution, and security.

17.10 ECOSYSTEM GROWTH FLYWHEEL

`Athletes Join → Clubs Form Teams → Tournaments Run → Scouts Discover → Data Value Increases → More Stakeholders Join`

RBAC ensures this growth happens securely and compliantly.

CHAPTER 18: DATABASE DESIGN, ENTITY RELATIONSHIP MODEL [ERD] & DATA STRUCTURE

KAFRIADA NET | DB Schema v1.0 | PostgreSQL 15 | Effective Date: 1 July 2026

18.1 INTRODUCTION

The KAFRIADA database is the system of record for Africa's sports ecosystem. It must securely store 10M+ identities and support analytics, compliance, and real-time operations.

Design Pillars:

1. Identity-First: KUID is the immutable primary key
2. Normalized: 3NF to prevent duplication and ensure integrity
3. Secure-by-Design: Field-level encryption, RBAC, audit trails
4. Scalable: Partitioned, indexed, and ready for multi-region
5. Multi-Tenant: Supports countries, federations, clubs in one DB

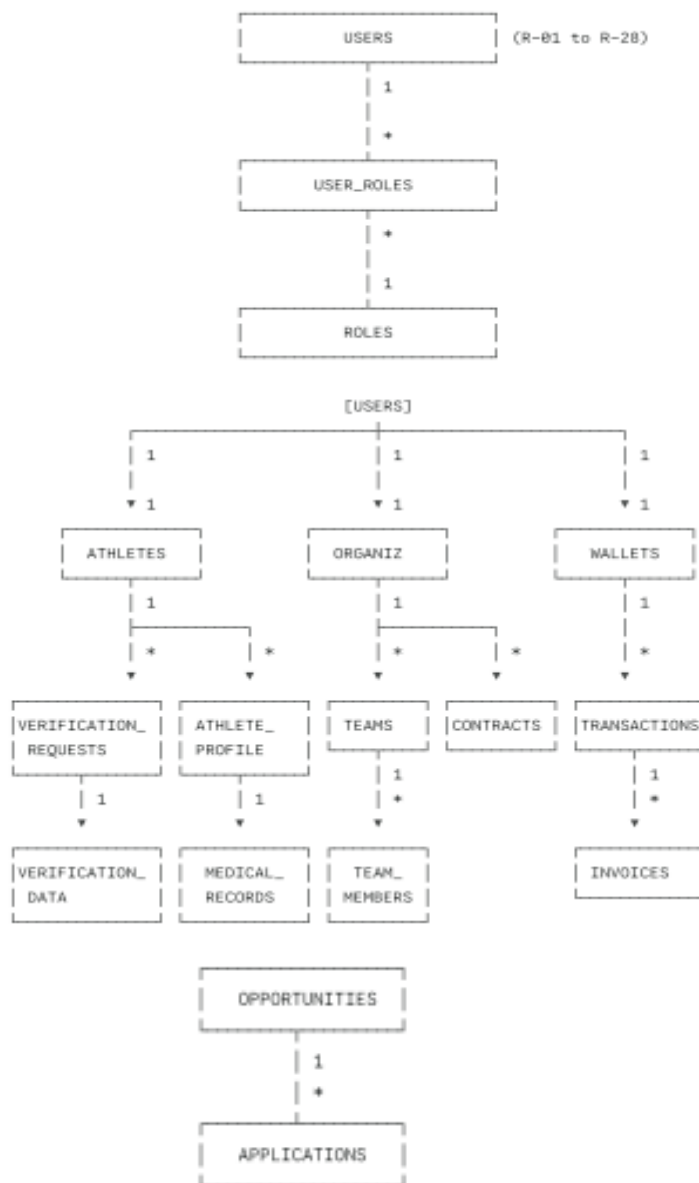
18.2 TECHNOLOGY STACK

Component	Technology	Purpose
Primary DB	PostgreSQL 15	Transactional data, ACID compliance
Cache	Redis	Sessions, API caching, rate limiting
Search	Elasticsearch	Athlete, Club, Opportunity discovery
Object Storage	AWS S3 + KMS	Documents, images, videos
Data Warehouse	AWS Redshift	Analytics, Phase 3+

18.3 DATABASE DESIGN PRINCIPLES

1. Unique Identity Principle: `kafriada\_id` is permanent and global. `KA-NG-JG-BKD-2026-000123`
2. Data Ownership Principle: Athletes own PII. Organizations own team/competition data
3. Privacy by Design: PII and Medical data stored in separate encrypted tables. Access logged
4. Auditability: All tables have `created\_at`, `updated\_at`, `created\_by`

18.4 HIGH-LEVEL ERD



18.5 CORE TABLE SCHEMAS

MODULE A: MODULE A: USER AUTHENTICATION & ACCESS CONTROL (RBAC)

Table 1: `users`

Column	Type	Constraints	Description
user_id	UUID	PK, DEFAULT gen_random_uuid()	Global unique user identifier
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email address
phone_number	VARCHAR(20)	UNIQUE	Phone number for login & OTP verification

password_hash	TEXT	NOT NULL	Hashed password (bcrypt/Argon2)
account_status	ENUM	ACTIVE, PENDING, SUSPENDED, DELETED	Account lifecycle state
is_minor	BOOLEAN	DEFAULT FALSE	Flag indicating U-18 status
created_at	TIMESTAMPTZ	DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last record update timestamp

Table 2: `roles`

Column	Type	Constraints	Description
role_id	VARCHAR(10)	PK	Unique code (R-01 to R-28)
role_name	VARCHAR(100)	UNIQUE, NOT NULL	Human-readable role title
category	ENUM	INDIVIDUAL, ORGANIZATION, GOVERNMENT, PLATFORM	Primary role classification
description	TEXT		Role definition and scope

Table 3: `user\_roles`

Column	Type	Constraints	Description
user_role_id	UUID	PK, DEFAULT gen_random_uuid()	Unique junction ID
user_id	UUID	FK → users.user_id, NOT NULL	Target user
role_id	VARCHAR(10)	FK → roles.role_id, NOT NULL	Assigned role
org_id	UUID	FK → organizations.organization_id, NULLABLE	Scope context (if assigned to an org)
status	ENUM	ACTIVE, PENDING_APPROVAL, ARCHIVED_TRANSITIONED	Role assignment status
assigned_at	TIMESTAMPTZ	DEFAULT NOW()	Role assignment timestamp

MODULE B: ATHLETE IDENTITY & PROFILE (KAFRIADA ID)

Table 4: `athletes`

Column	Type	Constraints	Description
athlete_id	UUID	PK, DEFAULT gen_random_uuid()	Unique athlete record identifier
user_id	UUID	FK → users.user_id, UNIQUE	1:1 user linkage

kafriada_id	VARCHAR(30)	UNIQUE, NOT NULL	Formatted KUID (KA-NG-JG-BKD-2026-000123)
first_name	VARCHAR(255)	NOT NULL	Encrypted field
last_name	VARCHAR(255)	NOT NULL	Encrypted field
gender	ENUM	MALE, FEMALE, OTHER	Gender identification
date_of_birth	DATE	NOT NULL	Date of birth for age & minor logic
nationality	VARCHAR(100)	NOT NULL	Country of origin
state_of_origin	VARCHAR(100)		State / Province
lga	VARCHAR(100)		Local Government Area
sport_type	VARCHAR(100)		Primary sport
primary_position	VARCHAR(100)		Primary field position
verification_status	ENUM	UNVERIFIED, UNDER_VERIFICATION, VERIFIED	Profile verification badge state
parent_user_id	UUID	FK → users.user_id, NULLABLE	Linked R-09 Parent for U-18 minors
consent_status	ENUM	PENDING_PARENT, PARENT_APPROVED, TRANSITIONED_ADULT_DIRECT	NDPA legal consent state

Table 5: `athlete\_profile`

Column	Type	Constraints	Description
profile_id	UUID	PK, DEFAULT gen_random_uuid()	Profile ID
athlete_id	UUID	FK → athletes.athlete_id, UNIQUE	Target athlete
height_cm	DECIMAL(5,2)		Self-reported or baseline height
weight_kg	DECIMAL(5,2)		Self-reported or baseline weight
preferred_foot	ENUM	LEFT, RIGHT, BOTH	Dominant foot/hand

bio	TEXT		Personal statement
achievements_json	JSONB		List of honors and career awards
privacy_tier	ENUM	PUBLIC, PROTECTED, RESTRICTED	Visibility tier set by user

Table 6: `athlete\_career\_history`

Column	Type	Constraints	Description
career_id	UUID	PK, DEFAULT gen_random_uuid()	Career entry ID
athlete_id	UUID	FK → athletes.athlete_id	Associated athlete
club_name	VARCHAR(255)	NOT NULL	Name of club/school
start_date	DATE	NOT NULL	Start of tenure
end_date	DATE		End of tenure (NULL if current)
appearances	INT	DEFAULT 0	Match appearances
goals_scored	INT	DEFAULT 0	Performance metric

Table 7: `athlete\_media` | `media\_id` PK, `athlete\_id` FK, `s3\_url`, `media\_type`, `is\_public`

Column	Type	Constraints	Description
media_id	UUID	PK, DEFAULT gen_random_uuid()	Media file ID
athlete_id	UUID	FK → athletes.athlete_id	Associated athlete
s3_url	TEXT	NOT NULL	Encrypted storage object URL
media_type	ENUM	PROFILE_PHOTO, HIGHLIGHT_VIDEO, DOCUMENT	Asset classification
is_public	BOOLEAN	DEFAULT TRUE	Tier 1 vs Tier 2 visibility

MODULE C: PHYSICAL VERIFICATION

Table 8: `verification\_requests`

Column	Type	Constraints	Description
request_id	UUID	PK, DEFAULT gen_random_uuid()	Request ID
athlete_id	UUID	FK → athletes.athlete_id	Target athlete
verification_stage	INT	NOT NULL	Stage (1: Identity, 2: Bio, 3: Physical)
payment_status	ENUM	PENDING, PAID, FAILED	Fee status
status	ENUM	SUBMITTED, IN_REVIEW, APPROVED, REJECTED	Pipeline status
approved_by	UUID	FK → users.user_id	R-24/R-27 Admin approver ID

Table 9: `verification\_data`

Column	Type	Constraints	Description
verification_id	UUID	PK, DEFAULT gen_random_uuid()	Metric set ID
request_id	UUID	FK → verification_requests.request_id	Parent request
verified_height_cm	DECIMAL(5,2)	NOT NULL	Measured height
verified_weight_kg	DECIMAL(5,2)	NOT NULL	Measured weight
sprint_10m_sec	DECIMAL(4,2)		Acceleration speed metric
sprint_30m_sec	DECIMAL(4,2)		Top speed metric
vertical_jump_cm	DECIMAL(5,2)		Explosive power metric
raw_metrics_json	JSONB		Complete sensor & test dump
verified_at	TIMESTAMPTZ	DEFAULT NOW()	Verification timestamp

Table 10: `qr\_codes`

Column	Type	Constraints	Description
qr_id	UUID	PK, DEFAULT gen_random_uuid()	Unique QR token record ID
athlete_id	UUID	FK → athletes.athlete_id, NOT NULL	Target athlete linkage
qr_data_token	TEXT	UNIQUE, NOT NULL	Encrypted dynamic payload / HMAC token
qr_type	ENUM	IDENTITY_VERIFICATION, MATCH_CHECKIN, PROFILE_VIEW	Purpose of generated QR code
status	ENUM	ACTIVE, EXPIRED, REVOKED	Validity status of the code
expires_at	TIMESTAMPTZ	NOT NULL	Expiration timestamp for dynamic refresh
created_at	TIMESTAMPTZ	DEFAULT NOW()	Generation timestamp

MODULE D: ORGANIZATION & CLUB

Table 11: `organizations`

Column	Type	Constraints	Description
--------	------	-------------	-------------

organization_id	UUID	PK, DEFAULT gen_random_uuid()	Unique org identifier
organization_type	ENUM	CLUB, ACADEMY, SCHOOL, UNIVERSITY, TOURNAMENT_ORG, LEAGUE_ORG, ASSOCIATION, NGO, SPONSOR	Org classification
name	VARCHAR(255)	NOT NULL	Legal name
registration_no	VARCHAR(100)	UNIQUE	Government/CAC registration number
country	VARCHAR(100)	NOT NULL	Country location
state	VARCHAR(100)		State / Province
approval_status	ENUM	PENDING, APPROVED, SUSPENDED	Platform sanction status

Table 12: `teams`

Column	Type	Constraints	Description
team_id	UUID	PK, DEFAULT gen_random_uuid()	Unique team ID
organization_id	UUID	FK → organizations.organization_id	Parent organization
team_name	VARCHAR(100)	NOT NULL	Squad title (e.g., U-17 Boys)
age_group	VARCHAR(50)		Category (U-13, U-15, Senior)
sport	VARCHAR(100)	NOT NULL	Associated sport

Table 13: `team\_members`

Column	Type	Constraints	Description
membership_id	UUID	PK, DEFAULT gen_random_uuid()	Roster entry ID
team_id	UUID	FK → teams.team_id	Assigned team
athlete_id	UUID	FK → athletes.athlete_id	Assigned athlete
jersey_number	INT		Assigned jersey #
join_date	DATE	NOT NULL	Joining date
status	ENUM	ACTIVE, INACTIVE, LOANED	Roster membership status

Table 14: `coaches`

Column	Type	Constraints	Description
coach_id	UUID	PK, DEFAULT gen_random_uuid()	Unique coach profile identifier

user_id	UUID	FK → users.user_id, UNIQUE	Linkage to R-02 Coach user account
coaching_license	VARCHAR(100)		Official certification (e.g., CAF A/B/C, UEFA)
specialization	VARCHAR(100)		Primary sport / tactical area
experience_years	INT	DEFAULT 0	Years of active coaching experience
bio	TEXT		Professional statement

Table 15: `coach\_assignments`

Column	Type	Constraints	Description
assignment_id	UUID	PK, DEFAULT gen_random_uuid()	Unique assignment entry ID
coach_id	UUID	FK → coaches.coach_id, NOT NULL	Target coach profile
team_id	UUID	FK → teams.team_id, NOT NULL	Assigned team
role_title	ENUM	HEAD_COACH, ASSISTANT_COACH, FITNESS_COACH	Specific role within squad staff
assigned_at	TIMESTAMPTZ	DEFAULT NOW()	Start of coaching tenure
status	ENUM	ACTIVE, TERMINATED	Assignment state

MODULE E: CONTRACTS , HEALTH AND INJURIES

Table 16: `contracts`

Column	Type	Constraints	Description
contract_id	UUID	PK, DEFAULT gen_random_uuid()	Unique contract ID
athlete_id	UUID	FK → athletes.athlete_id	Contracted athlete
organization_id	UUID	FK → organizations.organization_id	Contracting entity
agent_id	UUID	FK → users.user_id, NULLABLE	Representing R-12 Agent
start_date	DATE	NOT NULL	Contract start date
expiry_date	DATE	NOT NULL	Expiry date
contract_type	ENUM	PROFESSIONAL, YOUTH_SCHOLARSHIP, SPONSORSHIP	Contract class
document_url	TEXT	NOT NULL	Encrypted PDF link

Table 17: `medical\_records`

Column	Type	Constraints	Description
--------	------	-------------	-------------

medical_id	UUID	PK, DEFAULT gen_random_uuid()	Record identifier
athlete_id	UUID	FK → athletes.athlete_id	Target athlete
created_by	UUID	FK → users.user_id	R-07 Medical Personnel ID
data_encrypted	TEXT	NOT NULL	Field-level AES-256 encrypted payload
clearance_status	ENUM	FIT, INJURED, PENDING_CLEARANCE	Playing clearance state
created_at	TIMESTAMPTZ	DEFAULT NOW()	Record creation date

Table 19: `medical\_clearance`

Column	Type	Constraints	Description
clearance_id	UUID	PK, DEFAULT gen_random_uuid()	Unique clearance certificate ID
athlete_id	UUID	FK → athletes.athlete_id, NOT NULL	Target athlete
doctor_id	UUID	FK → users.user_id, NOT NULL	R-07 Medical Personnel issuing certificate
clearance_type	ENUM	PRE_SEASON, POST_INJURY_RETURN, TRANSFER_FITNESS	Context of clearance
status	ENUM	PASSED, FAILED, CONDITIONAL	Clearance decision
document_url	TEXT		Link to signed medical certificate
issued_at	TIMESTAMPTZ	DEFAULT NOW()	Issuance timestamp
valid_until	DATE	NOT NULL	Expiration date of clearance

Table 18: `injuries`

Column	Type	Constraints	Description
log_id	UUID	PK, DEFAULT gen_random_uuid()	Audit entry ID
user_id	UUID	FK → users.user_id, NULLABLE	User executing action
action	VARCHAR(100)	NOT NULL	Event code (e.g., MEDICAL_READ, AGE_TRANSITION)
module	VARCHAR(100)	NOT NULL	Affected module
record_id	UUID		Target entity ID
ip_address	INET		Client IP address
timestamp	TIMESTAMPTZ	DEFAULT NOW()	Immutable event timestamp

MODULE F: MARKETPLACE & PAYMENTS

Table 20: `transfer\_market\_listings` (*Club Transfers & Market Window*)`

Column	Type	Constraints	Description
listing_id	UUID	PK, DEFAULT gen_random_uuid()	Unique marketplace listing ID
athlete_id	UUID	FK → athletes.athlete_id, NOT NULL	Athlete available on transfer market
selling_org_id	UUID	FK → organizations.organization_id	Offering club (R-13/R-14)
listing_type	ENUM	PERMANENT_TRANSFER, LOAN, FREE_TRANSFER	Transfer nature
asking_price	DECIMAL(15,2)		Listed asking valuation
currency	VARCHAR(3)	FK → currencies.currency_code	Transaction currency (e.g., NGN, USD, EGP)
market_window_id	UUID	FK → transfer_windows.window_id	Associated seasonal window
status	ENUM	OPEN, NEGOTIATING, COMPLETED, CANCELLED	Status of listing

Table : transfer_windows

Column	Type	Constraints	Description
window_id	UUID	PK, DEFAULT gen_random_uuid()	Transfer window ID
region_id	UUID	FK → regions.region_id	Associated geographic region
window_name	VARCHAR(100)	NOT NULL	e.g., "Summer 2026 MENA Transfer Window"
start_date	TIMESTAMPTZ	NOT NULL	Opening timestamp
end_date	TIMESTAMPTZ	NOT NULL	Closing timestamp
is_active	BOOLEAN	DEFAULT FALSE	Window state enforced by system cron

Table : sponsorship_deals (*Clubs, Athletes, Events & Leagues*)

Column	Type	Constraints	Description
sponsorship_id	UUID	PK, DEFAULT gen_random_uuid()	Sponsorship record ID
sponsor_org_id	UUID	FK → organizations.organization_id	R-21 Sponsor Admin organization
target_type	ENUM	ATHLETE, CLUB, EVENT, LEAGUE	Sponsored entity category

target_id	UUID	NOT NULL	ID of target entity
funding_amount	DECIMAL(15,2)	NOT NULL	Sponsorship value
currency	VARCHAR(3)	FK → currencies.currency_code	Currency code
contract_url	TEXT		Signed sponsorship terms
status	ENUM	PROPOSED, ACTIVE, EXPIRED, TERMINATED	Sponsorship status

Table : advertisements (*Promotions & Media Banners*)

Column	Type	Constraints	Description
ad_id	UUID	PK, DEFAULT gen_random_uuid()	Campaign ID
advertiser_id	UUID	FK → users.user_id, NOT NULL	Sponsor/User creating ad
headline	VARCHAR(255)	NOT NULL	Campaign headline
banner_url	TEXT	NOT NULL	Media asset link
target_url	TEXT		Click-through destination
target_audience_json	JSONB		Filters (e.g., Age U-18, Region Africa, Sport Football)
impressions_bought	INT	NOT NULL	Purchased view count
impressions_delivered	INT	DEFAULT 0	Delivered impressions counter
status	ENUM	DRAFT, ACTIVE, PAUSED, COMPLETED	Campaign status

Table : subscriptions (*Monetization Plans*)

Column	Type	Constraints	Description
subscription_id	UUID	PK, DEFAULT gen_random_uuid()	Active subscription record ID
subscriber_id	UUID	NOT NULL	User or Organization ID
subscriber_type	ENUM	INDIVIDUAL, ORGANIZATION	Account type
plan_type	ENUM	SCOUT_PREMIUM, AGENT_PRO, CLUB_ENTERPRISE, ACADEMY_PRO	Plan tier
billing_cycle	ENUM	MONTHLY, ANNUALLY	Billing frequency
amount_paid	DECIMAL(15,2)	NOT NULL	Fee charged

starts_at	TIMESTAMPTZ	DEFAULT NOW()	Subscription activation time
expires_at	TIMESTAMPTZ	NOT NULL	Expiration timestamp
auto_renew	BOOLEAN	DEFAULT TRUE	Renewal toggle

Table : fee_structures (*Platform Monetization Schedule*)

Column	Type	Constraints	Description
fee_id	UUID	PK, DEFAULT gen_random_uuid()	Fee item ID
fee_code	VARCHAR(100)	UNIQUE, NOT NULL	Code (e.g., VERIFICATION_STAGE_1, TRANSFER_COMMISSION_PCT)
description	TEXT		Explanation of charge
amount_or_percentage	DECIMAL(10,4)	NOT NULL	Fixed value or percentage multiplier
is_percentage	BOOLEAN	DEFAULT FALSE	Indicator for percentage-based charge
currency	VARCHAR(3)	FK → currencies.currency_code	Fee currency

Table 21: `opportunities`

Column	Type	Constraints	Description
opportunity_id	UUID	PK, DEFAULT gen_random_uuid()	Post ID
creator_id	UUID	FK → users.user_id, NOT NULL	User/Org posting trial/scholarship
opportunity_type	ENUM	TRIALS, SCHOLARSHIP, SCOUTING_CAMP, JOB_VACANCY	Post category
title	VARCHAR(255)	NOT NULL	Opportunity title
description	TEXT	NOT NULL	Requirements and scope
application_fee	DECIMAL(15,2)	DEFAULT 0.00	Fee to apply (0 if free)
expiry_date	DATE	NOT NULL	Deadline

Table 22: `applications`

Column	Type	Constraints	Description
application_id	UUID	PK, DEFAULT gen_random_uuid()	Application entry ID
opportunity_id	UUID	FK → opportunities.opportunity_id	Linked post
applicant_id	UUID	FK → users.user_id, NOT NULL	Applying user (R-01 Athlete , Coach, etc.)
parent_approval_status	ENUM	APPROVED, REJECTED, NOT_REQUIRED	R-09 Parent clearance for U-18
payment_status	ENUM	FREE, PENDING, COMPLETED, REFUNDED	Application fee transaction state
status	ENUM	SUBMITTED, SHORTLISTED, ACCEPTED, REJECTED	Review status

Table 23: `subscriptions` (*Monetization Plans*)

Column	Type	Constraints	Description
subscription_id	UUID	PK, DEFAULT gen_random_uuid()	Active subscription record ID
subscriber_id	UUID	NOT NULL	User or Organization ID
subscriber_type	ENUM	INDIVIDUAL, ORGANIZATION	Account type
plan_type	ENUM	SCOUT_PREMIUM, AGENT_PRO, CLUB_ENTERPRISE, ACADEMY_PRO	Plan tier
billing_cycle	ENUM	MONTHLY, ANNUALLY	Billing frequency
amount_paid	DECIMAL(15,2)	NOT NULL	Fee charged
starts_at	TIMESTAMPTZ	DEFAULT NOW()	Subscription activation time
expires_at	TIMESTAMPTZ	NOT NULL	Expiration timestamp
auto_renew	BOOLEAN	DEFAULT TRUE	Renewal toggle

MODULE G: REGIONAL CONFIGURATION (AFRICA & MENA) AND SYSTEM

Table 24: `notifications`

Column	Type	Constraints	Description
notification_id	UUID	PK, DEFAULT gen_random_uuid()	Unique notification record ID
user_id	UUID	FK → users.user_id, NOT NULL	Target recipient
sender_id	UUID	FK → users.user_id, NULLABLE	Sender ID (System if NULL)

category	ENUM	SECURITY, ACCOUNT, VERIFICATION, MATCH, MEDICAL, FINANCIAL, MARKETPLACE	Functional domain
channel	ENUM	IN_APP, EMAIL, SMS, WHATSAPP	Delivery channel
title	VARCHAR(255)	NOT NULL	Headline/Subject
message	TEXT	NOT NULL	Main notification body
action_url	TEXT		Deep-link / CTA route
metadata_json	JSONB		Context data (e.g., athlete_id, amount, ref_id)
is_read	BOOLEAN	DEFAULT FALSE	Read status flag
read_at	TIMESTAMPTZ		Timestamp when read by user
delivery_status	ENUM	PENDING, DELIVERED, FAILED	Dispatch state
created_at	TIMESTAMPTZ	DEFAULT NOW()	Timestamp generated

Table 24.1: 'notification_preferences'

Column	Type	Constraints	Description
preference_id	UUID	PK, DEFAULT gen_random_uuid()	Unique preference ID
user_id	UUID	FK → users.user_id, UNIQUE	Target user settings
in_app_enabled	BOOLEAN	DEFAULT TRUE	Toggle for in-app push
email_enabled	BOOLEAN	DEFAULT TRUE	Toggle for transactional emails
sms_enabled	BOOLEAN	DEFAULT TRUE	Toggle for SMS alerts
whatsapp_enabled	BOOLEAN	DEFAULT FALSE	Toggle for WhatsApp messages
marketing_opt_in	BOOLEAN	DEFAULT FALSE	Opt-in for ads & promos
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last update timestamp

Table 24.2: 'notification_templates'

Column	Type	Constraints	Description
template_id	UUID	PK, DEFAULT gen_random_uuid()	Template ID
template_code	VARCHAR(100)	UNIQUE, NOT NULL	Code (e.g., PARENT_CONSENT_REQ, AGE_18_TRANSITION)
channel	ENUM	IN_APP, EMAIL, SMS, WHATSAPP	Target format channel

subject_template	TEXT		Dynamic subject string with handlebars ({{var}})
body_template	TEXT	NOT NULL	Dynamic message body with handlebars ({{var}})
is_active	BOOLEAN	DEFAULT TRUE	Active switch toggle

PERFORMANCE & INDEXING FOR NOTIFICATIONS

- **Composite Index:** CREATE INDEX idx_notifications_user_read ON notifications(user_id, is_read, created_at DESC);

(Optimizes unread inbox queries and real-time counter fetches)

- **Partitioning:** The notifications table is range-partitioned quarterly by created_at to prevent table bloat over time.

Table 25: `audit\_logs`

Column	Type	Constraints	Description
log_id	UUID	PK, DEFAULT gen_random_uuid()	Audit entry ID
user_id	UUID	FK → users.user_id, NULLABLE	User executing action
action	VARCHAR(100)	NOT NULL	Event code (e.g., MEDICAL_READ, AGE_TRANSITION)
module	VARCHAR(100)	NOT NULL	Affected module
record_id	UUID		Target entity ID
ip_address	INET		Client IP address
timestamp	TIMESTAMPTZ	DEFAULT NOW()	Immutable event timestamp

Table 26: `countries`

Column	Type	Constraints	Description
country_id	UUID	PK, DEFAULT gen_random_uuid()	Country ID
region_id	UUID	FK → regions.region_id, NOT NULL	Parent region
iso_code_2	VARCHAR(2)	UNIQUE, NOT NULL	ISO 3166-1 alpha-2 code (e.g., NG, EG, KE, SA)
iso_code_3	VARCHAR(3)	UNIQUE, NOT NULL	ISO 3166-1 alpha-3 code (e.g., NGA, EGY, KEN, SAU)
country_name	VARCHAR(100)	NOT NULL	Name of nation
phone_prefix	VARCHAR(10)	NOT NULL	Calling code (e.g., +234, +20)

Table 27: `regions` | For KUID generation and reporting

Column	Type	Constraints	Description
region_id	UUID	PK, DEFAULT gen_random_uuid()	Region identifier

region_code	VARCHAR(20)	UNIQUE, NOT NULL	Code (e.g., AFRICA_WEST, MENA_NORTH_AFRICA, GCC)
region_name	VARCHAR(100)	NOT NULL	Regional zone title

Table : currencies

Column	Type	Constraints	Description
currency_code	VARCHAR(3)	PK	ISO 4217 code (NGN, USD, EGP, KES, SAR, AED)
currency_name	VARCHAR(50)	NOT NULL	Full title (e.g., Nigerian Naira, Egyptian Pound)
symbol	VARCHAR(10)	NOT NULL	Symbol (e.g., ₦, \$, E£, KSh)
exchange_rate_to_usd	DECIMAL(12,6)	NOT NULL	Base exchange rate conversion factor for platform wallet settlements

MODULE H: FINANCIAL WALLETS & TRANSACTIONS

Table 21: digital_wallets (*Standalone Schema*)

Column	Type	Constraints	Description
wallet_id	UUID	PK, DEFAULT gen_random_uuid()	Unique digital wallet ID
owner_id	UUID	NOT NULL	FK to users or organizations
owner_type	ENUM	INDIVIDUAL, ORGANIZATION, GOVERNMENT, PLATFORM	Owner category classification
balance	DECIMAL(15,2)	DEFAULT 0.00, CHECK (balance >= 0)	Liquid account balance
currency	VARCHAR(3)	DEFAULT 'NGN'	ISO currency code
status	ENUM	ACTIVE, FROZEN, RESTRICTED	Account operational status
pin_hash	VARCHAR(255)		Encrypted transaction PIN
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last ledger update timestamp

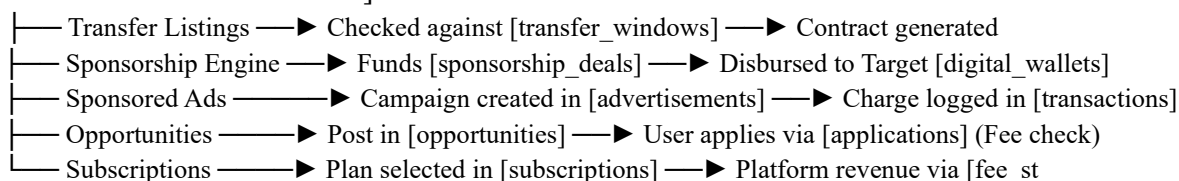
Table 22: 'transactions'

Column	Type	Constraints	Description
transaction_id	UUID	PK, DEFAULT gen_random_uuid()	Transaction ID
wallet_id	UUID	FK → digital_wallets.wallet_id	Source/Target wallet
transaction_type	ENUM	SUBSCRIPTION, VERIFICATION_FEE, SPONSORSHIP_GRANT, TRANSFER_FEE	Transaction category

amount	DECIMAL(15,2)	NOT NULL	Transaction value
gateway_ref	VARCHAR(255)	UNIQUE	Payment gateway reference
status	ENUM	PENDING, SUCCESSFUL, FAILED	Payment status
created_at	TIMESTAMPTZ	DEFAULT NOW()	Timestamp

MARKETPLACE & TRANSACTION WORKFLOW INTEGRATION

[MARKETPLACE ENGINE]



18.6 INDEXING & PERFORMANCE OPTIMIZATION STRATEGY

o maintain high throughput, low latency (<50ms for profile reads), and rapid search execution across millions of athlete profiles and metrics, the database employs the following indexes and partition rules:

1. Primary B-Tree Indexes (Core Foreign Keys & Lookups)

- **idx_users_email_phone:** B-Tree index on users(email, phone_number) for login authentication.
- **idx_athletes_kuid:** B-Tree unique index on athletes(kafriada_id) for quick QR scanning and lookups.
- **idx_user_roles_composite:** Composite index on user_roles(user_id, role_id, status) for fast authorization middleware evaluation.

2. GIN / JSONB Indexes (Scouting & Advanced Filters)

- **idx_verification_metrics_json:** GIN index on verification_data USING gin (raw_metrics_json) enabling scouts to run high-speed queries across dynamic physical data points (e.g., top speed ranges, jump heights).
- **idx_athlete_achievements_json:** GIN index on athlete_profile USING gin (achievements_json).

3. Spatial & Geographic Indexes

- **idx_athletes_location:** B-Tree composite index on athletes(nationality, state_of_origin, lga) to accelerate regional government dashboard analytics (**R-22 LGA Officer, R-23 State Council**).

4. Automated Partitioning Strategy

- **audit_logs Table Partitioning:** Range-partitioned by month on timestamp (e.g., audit_logs_2026_07) to maintain fast insert speeds and keep recent logs performant.
- **transactions Table Partitioning:** Range-partitioned by year on created_at.

And, can consider:

Critical indexes for scale:

```
CREATE INDEX idx_athletes_kuid ON athletes(kafriada_id);
CREATE INDEX idx_athletes_country ON athletes(country, state, lga);
CREATE INDEX idx_verification_status ON athletes(verification_status);
CREATE INDEX idx_team_members ON team_members(team_id, athlete_id);
Partitioning: `audit_logs`, `transactions`, `medical_records` by month
```

18.7 SECURITY CONTROLS

1. Encryption at Rest: RDS AES-256
2. Encryption in Transit: TLS 1.3
3. Field Encryption: Medical, NIN, Passport via AWS KMS
4. RBAC: DB roles map to App roles from Chapter 17
5. Backup: Daily snapshots, 35-day retention, PITR

18.8 DATA MIGRATION & VERSIONING

All schema changes managed via `Prisma Migrate`.

Migration files stored in `/database/migrations/` and tied to release tags.

18.9 DATABASE ACCEPTANCE CRITERIA

Database is production-ready when:

- All 27 core tables created with FK constraints
- KUID generation logic tested at 10k TPS
- Security scan: 0 critical vulnerabilities
- Load test: <300ms query p95
- Backup restore tested. RPO < 1hr[x]

18.10 NOTIFICATION

Here is a complete, production-ready **Notification Content & Messaging Catalog** for the Kafriada platform.

This catalog covers automated trigger events across all **28 Roles**, **4 Categories**, **Data Privacy/Consent Transitions**, and **Marketplace/Transaction Workflows**.

19. NOTIFICATION CONTENT & MESSAGING CATALOG

Notifications are categorized into four delivery channels:

- **IN-APP:** Real-time push notifications inside the application.
 - **EMAIL:** Formatted transactional and summary email templates.
 - **SMS:** Concise text messages for urgent verification, security OTPs, and alerts.
 - **WHATSAPP:** Rich media updates for match notifications and official status updates.
-

19.1 MODULE A: USER AUTHENTICATION & SECURITY

1. OTP Verification / Account Activation

- **Trigger:** User registers or attempts phone verification (users).
- **Target Role:** All Roles (**R-01 to R-28**)
- **SMS / WhatsApp:**

[Kafriada ID] Your verification code is {{otp_code}}. It expires in 10 minutes. Do not share this code with anyone.

- **Email Subject:** Verify your Kafriada ID Account
- **Email Body:**

Hi {{first_name}},

Welcome to Kafriada ID! Use the code below to complete your registration:

{{otp_code}}

If you did not initiate this request, please ignore this email.

2. Account Security Alert (New Device / IP)

- **Trigger:** System detects login from a new IP address or device (audit_logs).
- **Target Role:** All Roles (**R-01 to R-28**)
- **In-App:**

Security Alert: New login detected from {{device_name}} in {{location}}. If this wasn't you, review your account security immediately.

- **Email Subject:** Security Alert: New Login to Your Account

19.2 MODULE B: ATHLETE IDENTITY & MINOR TRANSITION WORKFLOW

3. Parent Consent Required (U-18 Athlete Registration)

- **Trigger:** Athlete under 18 signs up; account requires R-09 Parent link.
- **Target Role:** R-09 Parent/Guardian
- **In-App / Email Subject:** Action Required: Parental Consent for {{minor_first_name}}
- **In-App:**

Parental Consent Request: {{minor_first_name}} {{minor_last_name}} has registered a Kafriada ID profile and listed you as their guardian. Please review and grant consent.

- **Email Body:**

Dear Parent/Guardian,

{{minor_first_name}} {{minor_last_name}} has created a youth athlete profile on Kafriada ID. Under data protection regulations (NDPA), your explicit consent is required before their profile and performance data can be active.

[Click Here to Review and Approve Consent]

4. Automated 18th Birthday Transition (Minor to Adult)

- **Trigger:** Scheduled daily system cron job detects athlete reaching 18 years of age.

- **Target Role: R-01 Athlete & R-09 Parent/Guardian**
- **In-App (To Athlete):**

 **Happy 18th Birthday!** Your Kafriada ID profile has automatically upgraded to a full Adult Account. You now have direct control over your opportunities, scouting inquiries, and digital wallet.

- **Email Subject (To Athlete):** Your Account Has Been Upgraded to a Full Adult Profile!
- **In-App (To Parent):**

Notice of Account Transition: {{minor_first_name}} has reached 18 years of age. Primary administrative control and consent management for this account have transitioned directly to the athlete.

- **Email Subject (To Parent):** Account Management Update: {{minor_first_name}} Has Reached Age 18

19.3 MODULE C: VERIFICATION & PHYSICAL TESTING

5. Verification Assessment Approved

- **Trigger: R-24 Federation Admin or R-27 Regional Admin** approves physical testing metrics (verification_requests).
- **Target Role: R-01 Athlete, R-09 Parent, R-13 Club Admin**
- **In-App:**

 **Verification Approved!** Your physical metrics have been officially verified. Your **Verified Badge** is now active on your public profile and scouting databases.

- **SMS:**

Kafriada ID: Congratulations {{first_name}}! Your physical verification has been approved. View your badge at {{profile_url}}

6. Verification Assessment Rejected

- **Trigger:** Verification request fails validation or biometric consistency checks.
- **Target Role: R-01 Athlete, R-09 Parent**
- **In-App:**

 **Verification Update:** Your verification request (Stage {{stage_number}}) was not approved. Reason: {{rejection_reason}}. You may re-apply after {{retry_date}}.

19.4 MODULE D: SQUAD MANAGEMENT, COACHING & ROSTERS

7. Team Roster Invitation

- **Trigger: R-13 Club Admin or R-14 Academy Admin** adds an athlete to a squad roster (team_members).
- **Target Role: R-01 Athlete (or R-09 Parent if U-18)**
- **In-App:**

Team Invite: {{club_name}} has invited you to join their official squad roster for {{team_name}}.

- **Email Subject:** Invitation to Join {{club_name}} Roster

8. Coach Assessment Submitted

- **Trigger: R-02 Coach** logs a new performance evaluation or session assessment.
- **Target Role: R-01 Athlete, R-09 Parent**
- **In-App:**

 **New Performance Review:** Coach {{coach_last_name}} logged a new tactical & physical assessment for your training session on {{session_date}}.

19.5 MODULE E: HEALTH, INJURIES & MEDICAL CLEARANCE

9. Injury Reported

- **Trigger:** R-07 Medical Personnel logs a new injury record (injuries).
- **Target Role:** R-01 Athlete, R-09 Parent, R-02 Coach, R-13 Club Admin
- **In-App (Protected Tier):**

 **Medical Update:** A new medical log ({{injury_type}}) has been submitted by Dr. {{doctor_name}}. Expected recovery: {{expected_recovery_date}}.

- **Audit Constraint:** Accessible strictly by authorized roles under Tier 3 rules.

10. Medical Clearance Issued

- **Trigger:** R-07 Medical Personnel issues a "Fit-to-Play" certificate (medical_clearance).
- **Target Role:** R-01 Athlete, R-02 Coach, R-13 Club Admin
- **In-App:**

 **Medical Clearance Issued:** {{athlete_first_name}} has been certified FIT TO PLAY for upcoming matches.

19.6 MODULE F: MARKETPLACE, SPONSORSHIP & TRANSACTIONS

11. Transfer Listing Created / Window Open

- **Trigger:** R-13 Club Admin lists an athlete on the marketplace during an active transfer window (transfer_market_listings).
- **Target Role:** R-05/R-06 Scouts, R-12 Sports Agents, R-13/R-14 Org Admins
- **In-App:**

 **Transfer Market Alert:** {{athlete_first_name}} {{athlete_last_name}} ({{position}}, {{age}} yrs) has been listed for transfer by {{selling_club_name}}.

12. Sponsorship Proposal Received

- **Trigger:** R-21 Sponsor Admin creates a funding proposal (sponsorship_deals).
- **Target Role:** R-01 Athlete, R-09 Parent, R-13 Club Admin, R-17 Tournament Org
- **In-App:**

 **New Sponsorship Offer!** {{sponsor_company_name}} has submitted a sponsorship proposal valued at {{currency}} {{funding_amount}}.

- **Email Subject:** Sponsorship Offer Received from {{sponsor_company_name}}

13. Digital Wallet Deposit / Grant Credit

- **Trigger:** Wallet credit transaction succeeds (transactions).
- **Target Role:** All Wallet Holders (R-01, R-09, R-13 to R-28)
- **In-App:**

 **Wallet Credit:** Your digital wallet has been credited with {{currency}} {{amount}}. Ref: {{gateway_ref}}.

- **SMS:**

Kafriada Wallet: Credited with {{currency}} {{amount}} on {{timestamp}}. New Balance: {{currency}} {{new_balance}}. Ref: {{gateway_ref}}

14. Subscription Expiring Soon

- **Trigger:** System cron detects subscription expiration in 3 days (subscriptions).
- **Target Role:** R-05/R-06 Scouts, R-12 Agents, R-13 to R-21 Org Admins

- **In-App:**

 **Subscription Alert:** Your {{plan_type}} subscription expires on {{expiration_date}}. Renew now to keep access to premium scouting databases and transfer market listings.

19.7 MODULE G: GOVERNMENT & PLATFORM ADMINISTRATION

15. Tournament Sanctioning Approved / Rejected

- **Trigger:** R-24 Federation Admin sanctions a tournament (organizations).
- **Target Role:** R-17 Tournament Organizer, R-18 League Organizer
- **In-App:**

 **Tournament Sanctioned:** Your event {{tournament_name}} has been officially sanctioned by the Federation. Official ranking points and certified referees are now enabled.

16. Platform System Audit Flag

- **Trigger:** High-risk role override or unauthorized data query attempted.
- **Target Role:** R-26 Super Admin, R-28 Auditor
- **In-App (Critical Admin Alert):**

 **Security Flag:** High-risk action {{action_code}} performed by User ID {{user_id}} on module {{module_name}}. Review audit log ID {{log_id}}.

CHAPTER 19: API ARCHITECTURE, INTEGRATION DESIGN & TECHNICAL DOCUMENTATION

KAFRIADA NET | API v1.0 | REST + JSON | Effective Date: 1 July 2026

19.0 INTRODUCTION

The KAFRIADA API is the single source of truth for all platform interactions. It powers Mobile, Web, Admin, and 3rd Party integrations.

API-First Principle: Every feature is built as an API before any UI is developed.

Objectives:

1. Performance: P95 < 300ms
2. Security: Zero trust, RBAC, encrypted
3. Scalability: 10M RPS, versioned for multi-country
4. Interoperability: OpenAPI 3.0 for partners, govt, federations
5. Observability: 100% requests logged and monitored

9.1 ARCHITECTURAL PRINCIPLES & INTEGRATION STRATEGY

KAFRIADA NET operates as a centralized multi-stakeholder sports platform connecting 28 distinct roles across Individual (Category A), Organization (Category B), Government (Category C), and Platform Admin (Category D) tiers. To support real-time physical telemetry, digital wallet settlements, U-18 minor protection, dynamic QR identification, and regional analytics across Africa and the MENA region, the API architecture strictly adheres to five core design principles:

1. **Contract-First & Identity-Bound Architecture:** Every integration touchpoint is designed around OpenAPI Specification 3.1 contracts bound to the immutable `kafriada_id` format (e.g., KA-NG-JG-BKD-2026-000123).
2. **Contextual Role-Based Authorization:** Endpoints evaluate explicit combinations of **Role + Organization Context + Consent** at the gateway prior to executing business logic.
3. **Stateless Scale & Multi-Region Resilience:** API instances remain strictly stateless, delegating session states, dynamic rate limits, and lookup token caching to clustered Redis tiers.
4. **Privacy-by-Design & Field-Level Security:** Strict separation of data tiers (Tier 1 Public through Tier 4 Confidential) enforces AES-256 field-level encryption (for health records, NIN, passport data) directly at the API controller boundary.
5. **Event-Driven Decoupling:** Long-running computations, national statistics aggregations, dynamic U-18 to 18+ age transitions, and push notification dispatches are offloaded to asynchronous broker topologies (Apache Kafka / RabbitMQ).

19.2 API ARCHITECTURAL STYLES & SELECTION MATRIX

Different integration contexts within the KAFRIADA ecosystem demand specific interface styles based on payload complexity, transport overhead, latency budgets, and consumer execution environments.

Protocol Paradigm	Transport & Payload	Primary Platform Target Use Cases	Security & Governance Controls	Performance Target
REST (OpenAPI 3.1)	HTTP/2, JSON	Core mobile/web apps, public leaderboards, wallet operations, user management, transfer market listings.	OAuth 2.0 / OIDC Bearer JWTs, HMAC-SHA256 query signatures, rate limiting.	Latency $< 50 \text{ ms}$ (p95 Profile Reads)
gRPC	HTTP/2, Protocol Buffers	Low-latency microservice-to-microservice calls, biometric sensor data ingestion, real-time StatVest telemetry.	Mutual TLS (mTLS), internal service-mesh proxies (Envoy/Istio).	Latency $< 15 \text{ ms}$ (p95 Ingestion)
GraphQL	HTTP/2, JSON	Dynamic scout discovery pipelines, multi-metric physical filter builder (R-05/R-06 Scouts), administrative dashboards.	Query depth limiting, complexity scoring caps, N+1 query prevention resolvers.	Dynamic query budget enforcement
Event-Driven / Webhooks	HTTP/2, TCP, AMQP / JSON	Real-time notifications (SMS, Push, Email, WhatsApp), external payment gateway Webhooks, automated cron workflows.	Signature validation (X-Kafriada-Signature), exponential backoff retries with jitter, idempotency keys.	Async processing sub-second execution

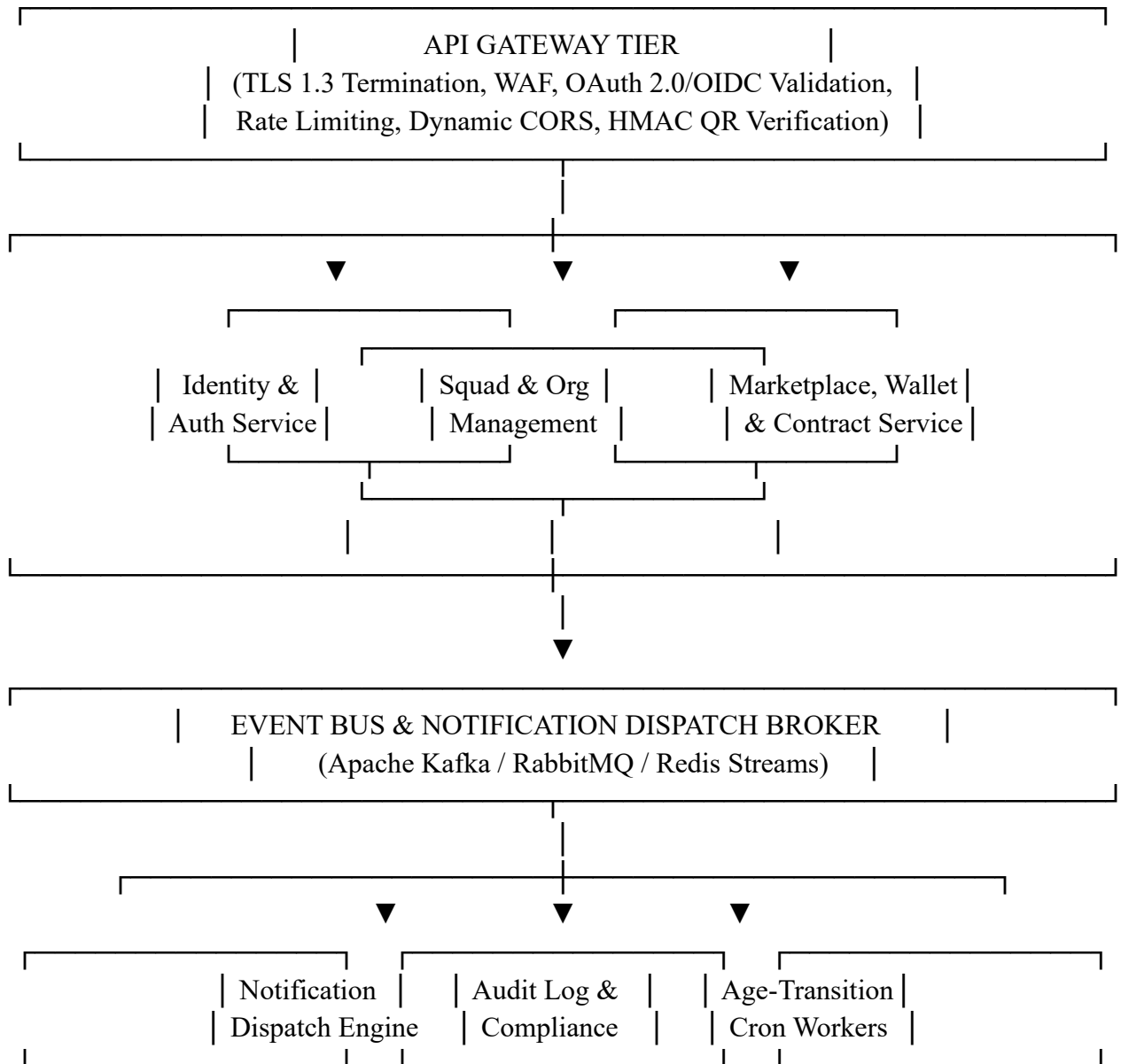
19.3 ENTERPRISE INTEGRATION TOPOLOGY & SERVICE

MESH

The platform routes all inbound traffic through a centralized, high-throughput Enterprise API Gateway layer, decoupling external client interfaces from internal domain microservices and shared persistence tiers.

[Mobile Apps / External Clients / Scouting Web Portals / 3rd-Party APIs]





ANOTHER ALTERNATIVE

All traffic routes through AWS API Gateway → WAF → Lambda Authorizer → Microservices
[Mobile App] [Web] [Partner]

↓

[API Gateway + WAF + Rate Limit]

↓

[Auth Service] [Athlete Service] [Club Service] [Payment Service] [Notification Service]

↓

[PostgreSQL] [Redis] [S3] [Elasticsearch]

19.4 SECURITY ARCHITECTURE & AUTHORIZATION FRAMEWORK

1. Unified Authentication (OAuth 2.0 & OIDC)

All API endpoints mandate standard HTTP Authorization headers bearing short-lived JSON Web Tokens (JWTs) issued by the central Identity Provider:

HTTP

Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

Token Payload Claims Context Structure:

JSON

```
{
  "sub": "u-9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
  "kuid": "KA-NG-JG-BKD-2026-000123",
  "roles": ["R-01", "R-02"],
  "active_context": {
    "role_id": "R-02",
    "org_id": "org-550e8400-e29b-41d4-a716-446655440000"
  },
  "is_minor": false,
  "iat": 1785593047,
  "exp": 1785596647
}
```

2. Field-Level Encryption & Data Tier Access Mapping

API controllers interface with data layer repositories by enforcing strict access policies based on the Data Visibility Tiers defined in **Section 17.6**:

- **Tier 1 (Public):** Accessible to all authenticated roles (R-01 to R-28) and unauthenticated public users (R-10). Includes public names, sport type, dynamic KUID QR codes, and open competition rankings.
- **Tier 2 (Protected):** Requires an active context match (e.g., Assigned Coach R-02, Club Admin R-13, or Paid Scout R-05/R-06 with an active subscription tier). For U-18 minor profiles, contact channels are programmatically redirected to the linked R-09 Parent/Guardian or R-13/R-14 Organization Admin.
- **Tier 3 (Restricted):** High-sensitivity payload elements (Encrypted Health Records, National Identification Numbers, Passport documents). Access requires explicit patient/guardian consent, a mandatory R-07 Medical Personnel role context, and generates an immutable record in `audit_logs`. Fields are stored using KMS-backed AES-256 field-level encryption.
- **Tier 4 (Confidential):** Platform balances, payment gateway tokens, system audit logs. Enforces dual-key approval workflows; strictly accessible only by platform operational

CHAPTER 20: USER INTERFACE [UI/UX] DESIGN SYSTEM & APPLICATION EXPERIENCE

KAFRIADA NET | Design System v1.0 | Effective Date: 1 July 2026

20.1 INTRODUCTION

The KAFRIADA NET UI/UX is the human layer of Africa's Digital Sports Infrastructure. It must be simple enough for a grassroots U15 athlete in rural Jigawa, and professional enough for a FIFA scout in London.

Design Mission: `Make every African athlete look and feel professional on day one.`

Target Users: Athletes, Coaches, Clubs, Scouts, Admins, Governments with varying digital literacy and device specs.

20.2 UI/UX DESIGN OBJECTIVES

	<u>Objective</u>	<u>Success Metric</u>
<u>1</u>	<u>Simplicity</u>	<u>Athlete registration < 3 minutes</u>
<u>2</u>	<u>Trust</u>	<u>Verification badge visible on 100% of athlete surfaces</u>
<u>3</u>	<u>Performance</u>	<u>App loads < 3s on 3G, <50MB install</u>
<u>4</u>	<u>Inclusivity</u>	<u>Support 4 languages + WCAG 2.1 AA</u>
<u>5</u>	<u>Professionalism</u>	<u>Profiles ready to send to clubs/agents</u>
<u>6</u>	<u>Consistency</u>	<u>One design system across Mobile + Web + Admin</u>

20.3 BRAND EXPERIENCE GUIDELINES

<u>Element</u>	<u>Definition</u>
<u>Brand Name</u>	<u>KAFRIADA NET</u>
<u>Slogan</u>	<u>`African Sport Network Platform`</u>
<u>Core Message</u>	<u>`One Account • One Athlete Identity • One Ecosystem`</u>
<u>Brand Colors</u>	<u>`Primary: KAFRIADA Green #0A7A0A` `Accent: Gold #FFC107` `Neutral: Charcoal #1F2937`</u>
<u>Typography</u>	<u>`Headings: Inter Bold` `Body: Inter Regular` `Supports Latin + Arabic + French`</u>
<u>Iconography</u>	<u>Sports-focused, outline style. Ball, whistle, trophy, shield</u>

20.4 DESIGN PRINCIPLES

1. Athlete-First: The `Athlete Profile` is the hero. Every flow leads back to it
2. Trust Through Verification: Verification status is persistent UI. Never hidden
3. Mobile-First: Designed for 6-inch Android first, then tablet, then web
4. Low-Bandwidth Ready: Image compression, lazy loading, offline support
5. Clarity over Cleverness: No jargon. Plain language + icons

20.5 VERIFICATION BADGE SYSTEM

Always visible on profiles, cards, and search results.

<u>Badge</u>	<u>Color</u>	<u>Status</u>	<u>Meaning</u>
<u>Grey</u>	<u>`#9CA3AF`</u>	<u>`UNVERIFIED`</u>	<u>Basic profile. Self-declared</u>
<u>Yellow</u>	<u>`#F59E0B`</u>	<u>`UNDER VERIFICATION`</u>	<u>Documents submitted. In review</u>
<u>Blue</u>	<u>`#2563EB`</u>	<u>`VERIFIED`</u>	<u>Professional identity. StatVest data</u>

20.6 APPLICATION STRUCTURE

Platform: 1 Mobile App + 1 Web Portal [Next.js]

Navigation: Role-based dashboards after login[Flutter]

Global Bottom Nav - Mobile

`Home | Discover | Opportunities | Messages | Profile`

20.7 ROLE-BASED DASHBOARDS

A. ATHLETE DASHBOARD

Purpose: Identity + Opportunity

Key Components:

1. Profile Card: Photo, Name, `KA-NG-JG-BKD-2026-000123`, Blue Badge, Sport, Position
2. Digital Sports Passport: QR Code, Share Link, Download PDF
3. Career Timeline: Clubs, Tournaments, Transfers. Visual timeline
4. Performance: Matches, Goals, Assists, StatVest metrics
5. Opportunities Feed: Trials, Scholarships, Jobs. 1-tap Apply

B. CLUB / ACADEMY DASHBOARD

Purpose: Team Operations

Key Components:

1. Club Overview: Teams, Player Count, Subscription Status
2. Player Management: Add, View Profiles, Contracts, Medical alerts
3. Team Dashboard: Squad, Fixtures, Attendance, Training Plan
4. Finance: Invoices, Payments, Subscription Renewal

C. COACH DASHBOARD

Purpose: Development

Key Components:

1. My Teams: Quick access to assigned squads
2. Training: Create schedule, Mark attendance, Rate players
3. Assessments: Submit performance reports. Triggers verification data

D. SCOUT DASHBOARD

Purpose: Discovery

Key Components:

1. Talent Search: Advanced filters: Country, Age, Position, Verification, Metrics
2. Watchlist: Save players, Add notes, Compare side-by-side
3. Reports: Generate PDF scouting report. Request contact

E. ADMIN DASHBOARD [WEB]

Purpose: Governance

Key Components:

1. Platform Overview: KPIs: Athletes, Verified %, Revenue, Tournaments
2. Verification Queue: Approve/Reject with document viewer
3. User & Org Management: Roles, Suspension, Audit logs
4. Content Moderation: Reports, Marketplace listings

20.8 KEY USER JOURNEYS

Journey 1: Athlete Registration

`Download App → Create Account → Select "Athlete" → Enter Bio → Generate KUID + QR → Stage 1 Complete → Upload Docs → Stage 2/3`

Journey 2: Club Onboarding

`Create Account → Register Organization → Upload Docs → Admin Approval → Create Team → Invite Players`

Journey 3: Scout Finds Player

`Login → Talent Search → Filter "Verified + Football + CM + 16-18 + Nigeria" → View Profile → Add to Watchlist → Request Contact`

20.9 DESIGN SYSTEM COMPONENTS

<u>Component</u>	<u>Usage</u>
<u>Buttons</u>	<u>Primary, Secondary, Ghost, Danger. 48px touch target</u>
<u>Cards</u>	<u>Athlete, Opportunity, Team, News. Consistent shadow</u>
<u>Forms</u>	<u>1 column, inline validation, clear error states</u>
<u>Badges</u>	<u>Verification, Status, Plan type</u>
<u>Charts</u>	<u>Line/Bar for performance. Powered by Recharts</u>
<u>Empty States</u>	<u>Friendly illustration + CTA. "No trials yet"</u>

20.10 LOCALIZATION & ACCESSIBILITY

1. Languages: English, French, Arabic, Hausa. RTL support for Arabic
2. Accessibility: WCAG 2.1 AA. Min contrast 4.5:1. Screen reader labels. Font scaling

3. Device Support: Android 8+, iOS 14+. Optimized for 2GB RAM devices

20.11 UI/UX TESTING PLAN

Testing conducted before each release with:

1. Athletes: Test registration friction and profile clarity
2. Coaches: Test team management and attendance flow
3. Scouts: Test search speed and data density
4. Admins: Test approval workflow efficiency

20.12 UI/UX ACCEPTANCE CRITERIA

Design is approved for release when:

- [] Athlete can register and get KUID in < 3 min without help
- [] Verification badge is visible on all athlete touchpoints
- [] All dashboards load < 3s on 3G
- [] 90% task success rate in user testing
- [] Brand guidelines applied 100% consistently
- [] Accessibility audit passes WCAG 2.1 AA

CHAPTER 21: SECURITY, PRIVACY, DATA PROTECTION & COMPLIANCE FRAMEWORK

KAFRIADA NET | ISMS v1.0 | Effective Date: 1 July 2026

21.1 INTRODUCTION

KAFRIADA NET processes highly sensitive data: athlete PII, medical records, financials, and U18 data. A breach would destroy trust.

Core Principle: 'Security-by-Design and Privacy-by-Default'

Security is not an afterthought. It is embedded in architecture, code, operations, and culture.

21.2 SECURITY OBJECTIVES - CIAA

Objective	Definition	KAFRIADA Target
Confidentiality	Only authorized access	Field-level encryption + RBAC
Integrity	Data is accurate and unmodified	DB constraints + Audit logs + Checksums
Availability	Platform is accessible	99.9% Uptime. RTO < 4 hours
Accountability	Every action is traceable	100% audit logging of critical actions

21.3 COMPLIANCE FRAMEWORK

KAFRIADA commits to the highest applicable standard in every market.

Regulation	Scope	Key Requirements Implemented
NDPA 2023 - Nigeria	All Nigerian users	Lawful basis, Consent, DPO, Data Subject Rights, Breach notification 72hrs
GDPR	EU users & partners	Right to Access, Erasure, Portability, Data Minimization, DPA
Child Online Protection	U18 Athletes	Guardian Consent, Limited data collection, No direct DMs
PCI-DSS SAQ-A	Payments	No card data stored. Tokenization via Paystack/Flutterwave

21.4 DATA CLASSIFICATION & HANDLING

Level	Classification	Examples	Controls
L1	Public	Name, Sport, KUID, Badge	No auth required
L2	Internal	Profile photo, Career history	Auth + Relationship
L3	Confidential	Contracts, Stats, Payments	RBAC + Encryption at rest
L4	Restricted	Medical, NIN, Passport, Guardian	L3 + Field encryption + Audit + Consent

Default: All new accounts start as 'L1 Public'. User must opt-in to share L2/L3.

21.5 PRIVACY & CONSENT ARCHITECTURE

1. Data Ownership: Athlete owns their data. KAFRIADA is Data Processor. Club/Federation is Data Controller
2. Consent Management: Granular consent stored in DB. User can revoke anytime
3. Purpose Limitation: Data collected only for: Identity, Verification, Competition, Opportunity
4. Data Minimization: We do not collect NIN for Stage 1. Only for Stage 3

21.6 CHILD PROTECTION FRAMEWORK [U18]

Mandatory for NDPA + COPPA alignment.

Workflow:

`Athlete Registration <18 → Guardian Phone/Email Required → OTP Consent → Account Activated with Restricted Mode`

Controls:

1. Communication Gate: Scouts/Agents cannot DM U18. Requests route to Guardian/Club
2. Data Masking: School name, exact location hidden by default
3. Reporting: In-app "Report Abuse" button. Escalates to DPO + Guardian
4. Deletion: Guardian can request full deletion of U18 data

21.7 TECHNICAL SECURITY CONTROLS

1. Authentication & Access

- Passwords: bcrypt/Argon2 hashing. Min 8 chars. Breach list check
- MFA: Required for R16 SuperAdmin, R13 Federation, R09 Scout, R04 Medical. TOTP + SMS OTP
- Session: JWT 1hr. Refresh 7 days. Device management page. Force logout

2. Encryption

- In Transit: TLS 1.3 enforced on all APIs and web
- At Rest: RDS AES-256, S3 SSE-KMS
- Field Level: L4 data encrypted via AWS KMS before DB write

3. Application Security

- API: Auth token, RBAC, Rate limit 100/min, Input validation, WAF
- File Upload: Type validation, ClamAV virus scan, stored in private S3 with signed URLs
- Payments: PCI scope reduced. No card data touches KAFRIADA servers

4. Infrastructure Security

- Cloud: AWS. VPC, Private subnets, Security Groups, GuardDuty
- Secrets: AWS Secrets Manager. No secrets in code/env
- Monitoring: CloudWatch + Datadog + PagerDuty for anomalies

21.8 AUDIT & ACCOUNTABILITY

Audit Log captures: `Who, What, When, Where, From Which IP`

Events Logged: Login, Profile Edit, Verification Approve, Medical Access, Data Export, Payment

Retention: 7 years. Immutable in S3.

21.9 INCIDENT RESPONSE & BREACH PROTOCOL

Target: <72hr notification to NDPC per NDPA

Phase	Action	Owner	SLA
1. Detect	Alert from SIEM/CloudWatch	DevOps	<15 min
2. Contain	Block access, Isolate system	Tech Lead	<1 hour
3. Assess	Scope, Data involved, Users affected	DPO + Security	<4 hours
4. Notify	Internal, NDPC, Users if high risk	CEO + DPO	<72 hours
5. Recover	Patch, Restore, Post-mortem	Engineering	<7 days

21.10 BACKUP & DISASTER RECOVERY

- Frequency: RDS automated daily + PITR. S3 versioning
- Encryption: Backups encrypted
- Testing: Restore drill quarterly
- RTO: 4 Hours | RPO: 1 Hour

21.11 SECURITY TESTING & GOVERNANCE

Test	Frequency	Owner
Vulnerability Scan	Weekly	DevOps
Penetration Test	Quarterly	3rd Party
Code Review	Every PR	2 Engineers
DPO Audit	Monthly	Data Protection Officer

Responsibility Matrix:

1. KowaGuru Tech Ltd: Data Controller. Overall governance
2. Data Protection Officer: Compliance, training, breach reporting
3. Engineering: Secure coding, patching, testing

21.12 SECURITY ACCEPTANCE CRITERIA

Framework is approved when:

- [] MFA enforced for all privileged roles
- [] All L4 data field-encrypted and access audited
- [] U18 guardian consent flow tested and working
- [] Penetration test: 0 Critical, 0 High findings
- [] Backup restore test successful
- [] NDPA compliance checklist 100% complete

CHAPTER 22: TESTING STRATEGY, QUALITY ASSURANCE [QA] & PRODUCT ACCEPTANCE

KAFRIADA NET | QA Strategy v1.0 | Effective Date: 1 July 2026

22.1 INTRODUCTION

Quality Assurance ensures KAFRIADA NET is reliable, secure, and trusted by athletes, clubs, and governments before it goes live.

QA Philosophy: `Shift Left. Test Early. Test Often. Automate Everything.`

Testing runs in parallel with development and continues post-launch.

22.2 QA OBJECTIVES

	Objective	Measurement
1	Requirement Coverage	100% of SRS items have test cases
2	Usability	90% task success rate in UAT
3	Data Accuracy	0 Duplicate KUIDs. 0 Data loss in migration
4	Performance	API P95 < 300ms. App load < 3s
5	Security	0 Critical/High vulnerabilities at launch
6	Stability	<1% crash rate on mobile

22.3 TESTING METHODOLOGY: AGILE QA

Testing is embedded in every 2-week sprint.

`Sprint Planning → Development → Unit Test → Integration → System Test → UAT → Release`

22.4 TESTING LEVELS & TOOLS

Level	Who	Scope	Tools
1. Unit	Developers	Functions, Components	Jest, Pytest
2. Integration	Developers + QA	Module communication	Postman, Jest
3. System	QA Team	Full E2E flows	Cypress, Appium
4. UAT	Real Users	Business validation	TestFlight, Feedback forms
5. Production	DevOps + QA	Smoke, Monitoring	Sentry, Datadog

22.5 CORE TEST AREAS

A. FUNCTIONAL TESTING

Test every module from SRS.

Module	Key Test Cases
Athlete Identity	Registration, KUID Generation, QR Scan, Profile Update, Verification Flow

Club Management	Org Registration, Team Creation, Invite Player, Contract Upload, Attendance
Athlete Care	Medical Record Create, Injury Log, Rehab Update, Clearance Approval
Marketplace	Create Opportunity, Apply, Sponsor Ad, Notifications
Payments	Create Invoice, Paystack Webhook, Receipt, Subscription Renewal

B. API TESTING

Tool: Postman + Newman + Swagger

Tests: Auth, RBAC, Rate Limit, Response time, Error handling, Contract validation

C. MOBILE APP TESTING

Devices: Low-end Android 8, Mid-range Android 12, iPhone 12

Areas: Installation, Offline mode, Performance, Battery, Memory, UX, Notifications

D. BROWSER & COMPATIBILITY TESTING

Browsers: Chrome, Firefox, Safari, Edge. Mobile: Chrome Android, Safari iOS

E. PERFORMANCE & LOAD TESTING

Tool: k6, Locust

Targets:

Metric	Target
API Response P95	< 300ms
Page Load	< 3 seconds
Concurrent Users	10,000 MAU for MVP
Uptime	99.9%

Scenarios: Peak registration, Tournament day traffic, Scout search spike

F. SECURITY TESTING

Types: SAST, DAST, Penetration Test, AuthZ testing

Critical Checks: Scout cannot access medical. U18 DM blocked. SQLi, XSS, IDOR

G. DATA QUALITY TESTING

Checks: No duplicate KUID, Valid date formats, Payment reconciliation, Referential integrity

22.6 USER ACCEPTANCE TESTING [UAT]

Goal: Business sign-off before launch

Participants:

1. Athletes: Test registration + profile + opportunities
2. Clubs: Test roster + contracts + attendance
3. Coaches: Test training + assessment
4. Scouts: Test search + watchlist
5. Admins: Test verification + approval workflow

UAT Process:

`Test Plan → Stakeholder Testing → Feedback Collection → Bug Fixes → Sign-off → Go-Live`

22.7 BUG MANAGEMENT & SEVERITY

All bugs tracked in Jira

Severity	Definition	SLA to Fix
P1 - Critical	Platform down, Data breach, Payment failure	< 4 Hours
P2 - High	Core feature broken	< 24 Hours
P3 - Medium	Workaround exists	Next Sprint
P4 - Low	Cosmetic	Backlog

22.8 MVP RELEASE ACCEPTANCE CHECKLIST

Technical

- ☐ All P1/P2 bugs resolved
- ☐ APIs 100% documented in Swagger
- ☐ DB backup + restore tested
- ☐ Monitoring + Alerts active

Business

- ☐ Payment flow tested end-to-end
- ☐ RBAC matrix validated
- ☐ Verification workflow approved by Product

Security & Compliance

- ☐ Pen test passed. 0 Criticals
- ☐ NDPA consent flow tested for U18
- ☐ Audit logs verified

UX

- ☐ Registration < 3 minutes
- ☐ Mobile app < 50MB, < 3s load
- ☐ Accessibility WCAG 2.1 AA pass

22.9 MVP LAUNCH ACCEPTANCE CRITERIA

KAFRIADA CORE v1.0 is approved for launch when:

1. 10,000 athletes can register and generate KUID
2. 300 clubs can onboard and manage teams
3. QR verification works offline
4. Payment for Stage 2 verification is successful
5. Security and Performance targets are met
6. Product, Tech, and DPO give final sign-off

22.10 CONTINUOUS QUALITY

Post-launch QA continues via:

1. Monitoring: Sentry for crashes, Datadog for performance
2. User Feedback: In-app feedback + Support tickets
3. Release Cadence: Monthly patches, Quarterly feature releases
4. Regression Suite: Automated tests run on every deploy

CHAPTER 23: DEVELOPMENT TEAM STRUCTURE, ROLES & PROJECT MANAGEMENT FRAMEWORK

KAFRIADA NET | Org v1.0 | Agile Scrum | Effective Date: 1 July 2026

23.1 INTRODUCTION

KAFRIADA NET is a sports infrastructure project delivered as software. Success requires both world-class engineering and deep sports domain expertise.

Delivery Model: Cross-functional product team. Mission-driven. Agile.

23.2 PROJECT MANAGEMENT APPROACH

Framework: Agile Scrum

Sprint Length: 2 Weeks

Release Cadence: Monthly Patch | Quarterly Minor Release

Why Scrum:

1. Fast MVP delivery with stakeholder feedback
2. Flexibility for sports industry requirements
3. Clear accountability across Tech + Sports teams

23.3 DEVELOPMENT WORKFLOW

Product Vision

↓

Roadmap & Epics

↓

Sprint Planning → Sprint 1-2 Weeks → Daily Standup

↓

Dev → QA → UAT

↓

Sprint Review + Retro

↓

Release → Monitor → Improve

23.4 PROJECT GOVERNANCE STRUCTURE

[KOWAGURU BOARD]

↓

[KAFRIADA EXECUTIVES]

↓

[PROJECT STEERING COMMITTEE] Product, Tech, Sports, Legal, Finance



[PROJECT MANAGEMENT OFFICE - PMO]



ENGINEERING TEAM	PRODUCT TEAM	SPORTS ADVISORY	BUSINESS & OPS TEAM
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23.5 KEY ROLES & RESPONSIBILITIES

A. EXECUTIVE LEADERSHIP

Role	Responsibility
Project Sponsor	CEO KowaGuru. Funding, partnerships, strategic direction
Project Director	Overall delivery, budget, risk, stakeholder alignment

B. PRODUCT TEAM

Role Responsibility

Product Manager Owns product vision, roadmap, prioritization, KPIs

Product Owner Manages backlog, writes user stories, accepts stories

Business Analyst Gathers requirements, process mapping, SRS maintenance

C. ENGINEERING TEAM

Role	Responsibility	Stack
Tech Lead / Architect	System design, code standards, tech decisions	http://Node.js, PostgreSQL
Backend Developers x2	APIs, business logic, payments, auth	http://Node.js, Express, Prisma
Mobile Developers x2	Athlete + Club App	Flutter
Frontend Web Dev x1	Admin + Federation Dashboards	http://React.js, http://Next.js
Database Engineer	DB design, optimization, migrations, backups	PostgreSQL
DevOps Engineer	AWS, CI/CD, Monitoring, Security	Terraform, GitHub Actions

D. DESIGN TEAM

Role	Responsibility
UI/UX Designer	User research, wireframes, prototypes, design system
Product Designer	Brand consistency, accessibility, UX QA
Graphic Designer	Marketing assets, illustrations

E. QA & SECURITY TEAM

Role	Responsibility
QA Lead	Test strategy, release sign-off, quality metrics

QA Engineer	Manual testing, test cases, UAT coordination
Automation QA	API tests, E2E tests, regression suite
Security Engineer	Pen testing, vulnerability management
Data Protection Officer [DPO]	NDPA/GDPR compliance, privacy, audit

F. SPORTS ADVISORY TEAM (Domain Experts)

Role	Responsibility
Head of Sports	Liaison between Tech and Sports. Validates workflows
Former Athlete Advisor	Athlete journey, needs, pain points
Coach Advisor	Training, assessment, team management logic
Scout Manager	Talent search, evaluation criteria
Tournament Lead	Fixtures, competition rules
Medical/Sports Science Advisor	Medical module, welfare, clearance

G. BUSINESS & OPERATIONS TEAM

Role	Responsibility
Partnerships Manager	NFF, CAF, Clubs, Governments
Sales Manager	Club subscriptions, Scout licenses
Marketing Manager	Growth, brand, community
Regional Coordinators x3	Country onboarding, training, support
Sports Registration Officers	Field support for verification

23.6 MVP TEAM STRUCTURE - PHASE 1

To deliver KAFRIADA CORE in 6 months:

Team	Role	Count
Leadership	Project Manager, Product Owner	2
Engineering	Tech Lead, Backend x2, Mobile x2, Frontend x1, DB, DevOps	8
Design	UI/UX Designer	1
QA	QA Engineer	1
Sports	Head of Sports + 4 Advisors	5
Business	Partnerships, Sales, Marketing	3
Total		20

23.7 SPRINT CEREMONIES

Ceremony	Frequency	Duration	Purpose
Sprint Planning	Every 2 weeks	2 hours	Commit to sprint goals
Daily Standup	Daily	15 min	Blockers, progress
Sprint Review	End of Sprint	1 hour	Demo to stakeholders
Retrospective	End of Sprint	1 hour	Process improvement

23.8 TOOLS & COMMUNICATION

Category	Tool
Code	GitHub
Project Tracking	Jira

Documentation	Notion, Confluence
Communication	Slack, Google Meet
Design	Figma
CI/CD	GitHub Actions
Monitoring	Datadog, Sentry

23.9 DOCUMENTATION STANDARDS

Every release must have:

1. SRS: Software Requirements Specification
2. Product Roadmap: 6-month view
3. API Docs: OpenAPI 3.0
4. DB Docs: ERD + Migration scripts
5. Design System: Figma + Storybook
6. Test Reports: QA sign-off
7. Security Report: Pen test summary
8. Deployment Runbook

23.10 TEAM PERFORMANCE KPIs

KPI	Target
Sprint Velocity	90% commitment met
Bug Escape Rate	< 2 P1/P2 bugs to production
Deployment Frequency	2x per month
Code Review Coverage	100%
Test Coverage	> 80% backend, > 60% frontend

23.11 PROJECT SUCCESS CRITERIA

KAFRIADA Phase 1 is successful when:

- ☐ MVP delivered within 6 months
- ☐ 10,000 athletes onboarded with verified KUID
- ☐ 300 clubs actively using platform
- ☐ Sports stakeholders validate product value
- ☐ Platform is secure, compliant, and stable
- ☐ Team is ready to scale to Phase 2

CHAPTER 24: KAFRIADA CORE PHASE 1 MVP FEATURE SPECIFICATION & DEVELOPMENT BACKLOG

KAFRIADA NET | MVP v1.0 | Effective Date: 1 July 2026 | Duration: 6 Months

24.1 INTRODUCTION

Phase 1 Goal: 'Launch Africa's First Verified Grassroots Athlete Identity System'

KAFRIADA CORE MVP establishes the foundation: Identity + Club + Care + Marketplace. It must work on low-end devices and be trusted by federations.

MVP Success Metrics:

1. 100,000 Athletes Registered with KUID
2. 2000 Clubs Onboarded
3. 3000 Coaches Active
4. 1,000 Athletes Stage 2 Verified
5. QR Verification Live in 5 States

KAFRIADA CORE is designed as a high-reliability, modular digital platform built to streamline user workflows, core data transactions, and operational management. **Phase 1 (MVP)** focuses on establishing the foundational infrastructure, identity and access management, core transactional features, real-time activity tracking, and baseline admin capabilities needed to validate product-market fit and ensure enterprise readiness.

1.1 Key Objectives

- **Deliver Core Value Loop:** Provide seamless onboarding, central workflow execution, and transactional processing.
- **Security & Compliance:** Establish Role-Based Access Control (RBAC), multi-factor authentication, and encrypted audit logging.
- **Performance Baseline:** Guarantee page load times under 1.5 seconds and system uptime of 99.9%.

24.2 SYSTEM SCOPE & BOUNDARIES

In-Scope (Phase 1 MVP)	Out-of-Scope (Deferred to Phase 2+)
User Registration, Authentication & Profile Management	Advanced AI/ML Predictive Analytics & Recommendations
Primary Core Module Workflow & Data Capture	Third-party Custom Plugin Marketplace
Role-Based Access Control (Admin, Standard User, Auditor)	Multi-Currency Automated Tax Compliance Engine
Automated Email/In-App Notifications	Native Mobile Apps (iOS / Android)

System Audit Logging & Administrative Dashboard	Automated Offline-First P2P Syncing
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24.3 USER PERSONAS & TARGET ROLES

1. **System Administrator (Admin):** Manages user access, configuration settings, system logs, and platform health.
2. **Core Operational User (Standard):** Executes primary workflows, inputs data, manages tasks, and generates standard reports.
3. **External Guest / Auditor (Read-Only):** Reviews transaction histories, exportable reports, and compliance logs.

24.4 PHASE 1 MVP FEATURE SPECIFICATIONS

Module 1: Identity & Security Engine (ISE)

- **ISE-01: Identity Provider & SSO Support**
 - *Description:* Secure sign-up/sign-in via Email/Password, OAuth 2.0 / OIDC (Google Workspace, Microsoft Entra ID).
 - *Requirements:* Enforce password complexity (min 8 chars, upper/lower/number/symbol), mandatory email verification, and MFA support via TOTP.
- **ISE-02: Role-Based Access Control (RBAC)**
 - *Description:* Granular access management mapped to user privileges.
 - *Requirements:* Roles: SuperAdmin, OrgAdmin, StandardUser, Auditor. Permission gates on API and UI routes.

Module 2: Core Data & Workflow Engine (CWE)

- **CWE-01: Workspace & Entity Management**
 - *Description:* Central hub for users to create, view, update, and manage core entity records.
 - *Requirements:* CRUD operations with field-level validation, status state machine (Draft > Submitted > Approved > Archived), auto-saving drafts.
- **CWE-02: Search & Filtering Framework**
 - *Description:* Sub-second search across active workspace entities.
 - *Requirements:* Full-text search, multi-field filtering (Date, Category, Status, Owner), and export capabilities (CSV, PDF).

Module 3: Communications & Audit (CAA)

- **CAA-01: Notification Service**
 - *Description:* Event-driven notifications for state changes and user actions.
 - *Requirements:* In-app notification center + asynchronous email delivery (SendGrid/AWS SES integration).
- **CAA-02: Immutable Audit Logging**
 - *Description:* Compliance logging for high-security transactions.
 - *Requirements:* Records Timestamp, ActorID, Action, ResourceID, IP_Address, and Delta_Payload.

24.5 Development Backlog & Sprint Plan

Prioritization Key

- **Must Have (M):** Critical path for MVP launch.
- **Should Have (S):** High value, targeted for MVP if velocity permits.
- **Could Have (C):** Non-blocking enhancements.

Story ID	Epic	User Story Summary	Acceptance Criteria	Priority	Est. (Pts)
KC-101	ISE	As a new user, I want to create an account using my email or SSO so I can access the platform.	Successful account creation sends verification link; SSO redirects correctly.	M	5
KC-102	ISE	As an Admin, I want to assign roles to team members so access is restricted appropriately.	Admin can change roles in Settings; user permissions update immediately without session drop.	M	3
KC-103	ISE	As a user, I want to enable 2FA on my account to increase security.	System generates QR code for TOTP apps; login requires 6-digit code validation.	S	5
KC-201	CWE	As a Standard User, I want to create and submit a new core entity record.	Form validates required fields, updates status to "Submitted", and logs creator ID.	M	8
KC-202	CWE	As a Standard User, I want to filter entities by status and date range.	Data table updates live; active filters can be cleared with one click.	M	5
KC-203	CWE	As a user, I want to export filtered query results to CSV or PDF.	Generated file contains exact matching rows and accurate field formatting.	S	3
KC-301	CAA	As a user, I want to receive in-app notifications when my record status changes.	Bell icon badges unread count; clicking opens drawer with direct link to record.	M	5
KC-302	CAA	As an Auditor, I want to view system audit logs to verify compliance.	Admin/Auditor view presents paginated, searchable, tamper-proof logs.	M	5
KC-401	Admin	As a SuperAdmin, I want a operational dashboard showing active users and error rates.	Dashboard renders real-time counts, system health indicators, and latency graphs.	S	8

24.6 Non-Functional Requirements (NFRs) & Quality Gates

6.1 Performance & Scalability

- **API Response Time:** P95 response times under 300 ms for standard read/write requests.
- **UI Performance:** First Contentful Paint (FCP) $\leq 1.2\text{s}$, Cumulative Layout Shift (CLS) < 0.1 .
- **Scalability:** Support 1,000 concurrent active users without degradation.

6.2 Security & Compliance

- **Data Encryption:** TLS 1.3 in transit; AES-256 at rest for database and data assets.
- **Session Safety:** JWT tokens with short expiry (15 mins) and HTTP-only secure refresh tokens.
- **Penetration Testing:** Zero Critical or High OWASP Top 10 vulnerabilities prior to sign-off.

6.3 Release Readiness Definition of Done (DoD)

1. Unit test coverage exceeds **80%** across core service modules.
2. All **Must Have (M)** user stories pass QA acceptance tests.
3. CI/CD pipeline builds, tests, and deploys cleanly to Staging without manual intervention.
4. Product Management and Security sign-off completed.

24.7 MVP COMPLETION DEFINITION

KAFRIADA CORE v1.0 is complete when:

1. Athlete can register, get KUID, and share QR in < 3 min
2. Club can register, create team, and manage 30 players
3. Admin can approve Stage 2 verification in < 24 hrs
4. Athlete can see and apply to 1 opportunity
5. Platform is secure, stable, and ready to scale

KAFRIADA COMPETE

COMPETITION MANAGEMENT, SCOUTING, PERFORMANCE

ANALYTICS & AI

PHASE 2

See: [2.0 Business Overview](#)

KAFRIADA COMPETE: Phase 2 Feature Specification & Development Backlog

1. Executive Summary & Product Overview

KAFRIADA COMPETE is Software 2 of the KAFRIADA NET SportTech ecosystem developed by KowaGuru Technology Limited. Building directly upon the verified athlete identity, club profiles, and medical foundations established in **KAFRIADA CORE (Phase 1)**, KAFRIADA COMPETE digitizes tournament administration, scouting workflows, and performance analytics.

By unifying competition management with AI-assisted scouting and live match operations, KAFRIADA COMPETE addresses key regional challenges: manual fixture creation, paper-based tournament scoring, talent invisibility, age/identity fraud in events, and lack of verified athletic performance metrics.

1.1 Core Value Proposition & Objectives

- **End-to-End Tournament Lifecycle:** Streamline registration, automated fixture generation, match official assignments, live scoring, and instant standings updates.
- **Verified Talent Discovery Engine:** Connect verified athletes with 500+ verified scouts and agents through advanced search algorithms, automated trial invitations, and recruitment pipelines.
- **Data-Driven Performance Analytics:** Transition regional sports from anecdotal evaluation to objective metrics using AI match analysis, physical test tracking, and standardized talent scoring.

2. System Scope & Boundaries (Phase 2)

In-Scope (Phase 2: KAFRIADA COMPETE)	Out-of-Scope (Deferred to Phase 3 & 4)
Competition & Tournament Lifecycle Management	School Academic Management & LMS (KAFRIADA EDU)
Automated Fixture, Bracket, and Schedule Generation	Coach & Referee Academic Certification Courses
Live Match Event Logging & Official Match Sheets	National Sports Policy & Ministry Intelligence (KAFRIADA ENTERPRISE)
Scout Portal, Advanced Search & Trial Invitations	Direct In-App P2P Transfer Fee Escrow Processing
AI Performance Analytics & Athlete Trend Scoring	Custom Third-Party Hardware Camera Integration

Public Tournament Portals, Standings, & Leaderboards	Offline P2P Local Mesh Synchronization
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3. Target User Personas & System Roles

1. **Tournament Organizer (STK-07):** Creates competitions, enforces entry requirements, schedules fixtures, assigns match officials, and approves official match sheets.
2. **Verified Scout / Agent (STK-08, STK-10):** Searches verified athlete databases, views match performance histories, tracks talent pipelines, and issues formal trial invitations.
3. **Scout Agency Manager (STK-09):** Oversees team scout activity, manages agency talent lists, and accesses regional performance analytics.
4. **Match Official / Referee (Primary User):** Submits digital match reports, logs live events (goals, cards, substitutions), and signs off on match results.
5. **Media & Public Fans (STK-18, STK-21):** Consumes live scores, tournament statistics, leaderboards, and verified athlete match highlights via public APIs and portals.

4. Phase 2 Functional Feature Specifications

Module 1: Competition Management Engine (CME)

- **CME-01: Tournament Setup & Entry Management**
 - *Description:* Allows Tournament Organizers to create single-elimination, double-elimination, round-robin, hybrid group/knockout, or custom league formats.
 - *Requirements:* Integration with KAFRIADA CORE for team/athlete eligibility checks (verifying Stage 2/Stage 3 ID status and medical clearances). Automated 5% hosting fee collection.
- **CME-02: Automated Fixture & Schedule Generator**
 - *Description:* Algorithmic generation of match schedules avoiding team venue conflicts and rest-day violations.
 - *Requirements:* Support for manual schedule adjustments, venue mapping, and instant notification dispatch to coaches (STK-02) and managers (STK-04).
- **CME-03: Digital Match Sheet & Live Scoring**
 - *Description:* Mobile-optimized digital match protocol for match officials.
 - *Requirements:* Capture match events (goals, assists, fouls, yellow/red cards, player substitutions, match duration). Generate instant digital match sheets signed off by referees. Automatically update standings, goal-difference tables, and top-scorer leaderboards.

Module 2: Scouting & Talent Discovery Engine (STE)

- **STE-01: Advanced Scout Search & Filter Matrix**
 - *Description:* Multi-parametric query engine tailored for scouts and agents holding paid subscriptions (₦100,000/yr).
 - *Requirements:* Filter by age, verified position, physical attributes (height, speed, dominant foot), competition tier, performance stats, and KAFRIADA verification stage.
- **STE-02: Trial Invitation & Recruitment Workflow**
 - *Description:* Standardized communication bridge between scouts, athletes, parents (for minors), and current clubs.
 - *Requirements:* Issue formal trial invitations, track response states (Pending $\rightarrow$ Accepted $\rightarrow$ Declined $\rightarrow$ Completed), log transfer requests, and maintain compliance with parental consent workflows (STK-12).

Module 3: Performance Analytics & AI Engine (PAE)

- **PAE-01: Performance Index (KPI) Calculation**
 - *Description:* Automated aggregation of match data into objective athlete performance ratings.
 - *Requirements:* Calculate metrics per sport (e.g., pass completion %, minutes per goal, defensive clearances, sprint totals). Generate dynamic Radar Charts on athlete profiles.
- **PAE-02: AI Talent Scoring & Predictive Trend Modeling**
 - *Description:* Machine learning model evaluating performance trajectories against historical regional benchmarks.
 - *Requirements:* Flag standout grassroots talent ("High Potential Identification"), detect anomalous performance spikes, and generate downloadable Scout Intelligence Reports (PDF).

5. Phase 2 Development Backlog & Sprint Plan

Prioritization Key

- **Must Have (M):** Essential for Phase 2 competition launch.
- **Should Have (S):** High value, targeted for Phase 2 completion.
- **Could Have (C):** Secondary enhancements for future iterations.

Story ID	Epic	User Story Summary	Acceptance Criteria	Priority	Est. (Pts)
KC2-101	CME	As an Organizer, I want to create a new tournament with specific age and verification rules so that only eligible teams register.	System validates team eligibility using CORE IDs and charges the 5% hosting fee.	M	8
KC2-102	CME	As an Organizer, I want to auto-generate a round-robin schedule for 16 teams.	Fixture tree is built without venue/time overlaps and allows manual edits.	M	5
KC2-103	CME	As a Match Official, I want to record match events live on a tablet/mobile interface.	Events update the live match engine; match summary updates league standings immediately upon final whistle.	M	8
KC2-104	CME	As a Fan/Media user, I want to view live tournament standings and statistics.	Standings recalculate points, goal differentials, and head-to-head metrics in under 2 seconds.	M	3
KC2-201	STE	As a Scout, I want to search for verified midfielders aged 16-19 with >3 goals in active tournaments.	Search results return verified profiles only, sorting by performance index.	M	5
KC2-202	STE	As a Scout, I want to send a trial invitation to an athlete through their club.	System routes notification to Club Manager; parent consent flow triggers if athlete is under 18.	M	5
KC2-203	STE	As an Agent, I want to export a full scouting dossier for a shortlisted player.	Exported PDF contains aggregated stats, match history, verified ID badge, and medical clearance status.	S	3
KC2-301	PAE	As a Club Coach, I want to see radar performance charts for my squad post-match.	System calculates match metrics and renders visual performance indicators on the coach dashboard.	M	5
KC2-302	PAE	As a Scout, I want AI recommendations for top-performing players in a specific region.	AI model identifies top 5% performers based on standardized performance metrics and display flags.	S	8

KC2-303	PAE	As an Admin, I want to track overall competition metrics across regions.	System aggregates active tournaments, matches played, total goals, and card stats for reporting.	S	3
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6. Phase 2 Target KPIs & Success Metrics

Metric	Business Target (Phase 2 Scope)	Technical / Operational KPI
Competitions Digitized	500+ Tournaments managed	100% automated standings generation accuracy
Verified Scouts Active	500+ Paying Scout/Agent Subscriptions	\$< 1.0\text{s}\$ response time on complex multi-filter searches
Performance Records Logged	100,000+ Individual Match Records	Zero lost match sheets during offline-to-online sync
Trial Invitations Generated	2,500+ Formal Trial Requests	Automated notification delivery within 30 seconds

7. Non-Functional Requirements & Security

1. **High Concurrency for Live Scoring:** The live match engine must support up to 50,000 concurrent public updates without API response latency exceeding 500ms
2. **Data Integrity & Fraud Prevention:** Match officials must be verified users with role-gated access. Official match sheets lock permanently after dual sign-off (referee and organizer).
3. **Privacy Compliance (NDPR / GDPR):** Athlete performance data and minor scout communications strictly adhere to parental consent gates established in CORE (STK-12).

TOURNAMENT & COMPETITION MANAGEMENT MODULE

9.1 OVERVIEW

The Tournament & Competition Management Module is a comprehensive competition administration system that enables KAFRIADA, Sports Federations, Ministries of Sports, Leagues, Associations, Schools, Academies, Clubs, Universities, NGOs, Community Sports Organizations, and Approved Tournament Organizers to digitally create, approve, manage, monitor, and report sporting competitions.

The module supports the complete competition lifecycle, including planning, registration, approvals, fixtures, officiating, live scoring, disciplinary management, financial management, awards, analytics, and post-event reporting.

The platform is designed as a **sport-agnostic competition engine**, allowing new sports to be introduced through configurable templates without requiring software redevelopment.

9.2 OBJECTIVES

The Tournament & Competition Management Module shall:

- Digitize competition administration.
- Standardize tournament governance.
- Support both grassroots and elite competitions.
- Automate registration and fixture generation.
- Support online and offline competition management.
- Manage officials, referees, judges, and technical delegates.
- Process competition registration fees.
- Automate revenue distribution.
- Track disciplinary actions and sanctions.
- Produce live competition statistics.
- Generate rankings and awards.
- Support unlimited sporting disciplines.

9.3 COMPETITION CATEGORIES

The system shall support competitions organized by:

- KAFRIADA
- National Sports Federations
- State Sports Associations
- League Management Bodies
- Ministries of Sports
- Schools
- Universities
- Academies
- Clubs
- Community Organizations
- NGOs
- International Organizations

Supported competition formats include:

- League
- Knockout
- Group Stage
- Round Robin
- Swiss Format
- Double Elimination
- Single Elimination
- Championship
- Invitational
- Festival Games
- Multi-Sport Games
- Friendly Competition

9.4 COMPETITION REGISTRATION

Every organizer shall submit:

- Competition Name
- Competition Logo
- Organizer Logo
- Competition Banner
- Sport
- Competition Type
- Competition Level
- Host Organization
- Country
- State
- Local Government Area
- Venue(s)
- Registration Opening Date
- Registration Closing Date
- Competition Start Date
- Competition End Date
- Rule Book
- Sponsorship Information
- Prize Details
- Medical Requirements
- Insurance Requirements

Only verified organizations shall register competitions.

9.5 COMPETITION APPROVAL WORKFLOW

KAFRIADA shall implement a **dual-level approval system**.

9.5.1 STAGE ONE — ORGANIZATIONAL APPROVAL

The competition may first be reviewed by:

- Sports Federation
- League Management Body
- State Association
- School Board
- Academy Management
- Ministry of Sports
- Other recognized sports organizations

Once approved, the tournament status becomes:

Organization Approved

9.5.2 STAGE TWO — KAFRIADA APPROVAL

KAFRIADA Administrators shall perform the final platform review.

When approved, the status becomes:

KAFRIADA Approved

Only tournaments with **KAFRIADA Approved** status shall be published.

9.5.3 ADMINISTRATIVE OVERRIDE

KAFRIADA Administrators possess overriding approval authority.

This means:

- KAFRIADA may approve competitions even when no external organization has approved them.
- KAFRIADA may reject competitions already approved by another organization if they violate platform policies.
- KAFRIADA may suspend or withdraw approval at any time.

9.5.4 Competition Status

- Draft
- Submitted
- Registration Closed
- Live

- Pending Review
- Organization Approved
- KAFRIADA Approved
- Published
- Registration Open
- Suspended
- Cancelled
- Completed
- Archived

9.6 COMPETITION REGISTRATION FEES

Competitions shall support:

- Free Registration
- Paid Registration
- Invitation Only
- Hybrid Registration

Organizers may configure:

- Registration Fee
- Currency
- Maximum Participants
- Refund Policy
- Registration Deadline
- Late Registration Fee

9.7 COMPETITION REVENUE DISTRIBUTION

Every paid competition shall automatically distribute revenue.

Revenue distribution shall be configurable.

Revenue Components

- Hosting Organization
- Referee Fees
- Match Officials
- Judges
- Medical Officials
- Players Welfare Fund
- Competition Emergency Fund
- Governing Sports Organization
- KAFRIADA Platform Service Fee

The total allocation shall always equal **100%**.

Every transaction shall generate:

- Digital Receipt
- Financial Ledger
- Audit Log
- Payment Statement

9.8 OFFICIALS PAYMENT SYSTEM

Every official shall maintain a personal financial dashboard.

Supported officials include:

- Referees
- Assistant Referees
- Judges
- Timekeepers
- Match Commissioners
- Medical Officers

After each completed match:

- Match fees shall automatically accumulate.
- Officials shall view earnings.
- Payment history shall be available.
- Withdrawals shall be processed through approved payment methods.

- Technical Delegates

9.9 PLAYER WELFARE FUND

Competition organizers may configure welfare contributions.

Funds may support:

- Injury Treatment
- Medical Assistance
- Insurance
- Emergency Relief
- Performance Incentives

The welfare dashboard shall display:

- Total Fund
- Contributions
- Beneficiaries
- Payments
- Remaining Balance

9.10 DISCIPLINARY CHARGES & SANCTIONS

The platform shall support configurable disciplinary penalties.

Examples include:

- Yellow Card Fine
- Red Card Fine
- Violent Conduct
- Match Abandonment
- Walkover
- Team Withdrawal
- Appeal Fee
- Protest Fee
- Property Damage
- Match Manipulation
- Doping Violations
- Unsporting Behaviour

Every disciplinary payment shall generate:

- Invoice
- Receipt
- Financial Record
- Audit Trail

NOTE: Outstanding penalties may automatically prevent participation in future competitions.

9.11 TEAM & ATHLETE REGISTRATION

Eligibility validation shall include:

- Valid KAFRIADA ID
- Medical Clearance
- Competition Registration
- Club Verification
- Age Eligibility

- Transfer Status
- Disciplinary Status
- Insurance Status

Only eligible participants shall be registered.

9.12 FIXTURE & MATCH MANAGEMENT

The system shall support:

Automatic generation of:

- League Fixtures
- Group Fixtures
- Knockout Brackets
- Round Robin Fixtures

Manual editing by authorized administrators.

Each fixture shall include:

- Match Number
- Teams
- Officials
- Venue
- Time
- Medical Team
- Security Officer
- Match Commissioner

9.13 LIVE MATCH MANAGEMENT

Officials may record:

- Goals
- Fouls
- Cards
- Assists
- Penalties
- Substitutions
- Match Events
- Injury Time
- Possession
- Statistics

Real-time scoreboards shall update automatically.

9.14 UNIVERSAL SPORT TEMPLATE ENGINE

The Tournament Module shall include a configurable **Universal Sport Template Engine**.

Every template shall support:

- Competition Logo
- Organizer Logo
- Sponsor Logos
- Trophy Images
- Medal Graphics
- Team Logos
- Athlete Photos
- Event Branding

- Editable Rules
- Editable Scoring
- Editable Statistics
- Editable Ranking Formula

Templates shall be configurable without software development.

9.15 SPORT TEMPLATES

The platform shall provide editable templates for:

Football

- League
- Knockout
- Group Stage
- Fixtures
- Match Statistics
- Player Statistics
- League Table
- Team Sheet
- Penalty Shootout

Basketball

- Quarter Scores
- Team Statistics
- Player Statistics
- Fouls
- Timeouts
- Standings
-

Athletics

Supported Events:

- 100m
- 200m
- 400m
- 800m
- 1500m
- Relay
- Marathon
- Long Jump
- High Jump
- Triple Jump
- Discus
- Javelin
- Shot Put
- Hammer Throw

Templates support:

- Heat Allocation
- Qualification
- Timing
- Distance
- Medal Ranking

Cycling

- Road Race
- Stage Race
- Sprint
- Time Trial
- Mountain Classification

Tennis

- Singles
- Doubles
- Sets
- Games
- Tie-break
- Brackets

Volleyball

- Sets
- Rotation
- Team Statistics
- Rankings

Swimming

- Individual Events
- Relay
- Heat Management
- Lane Assignment
- Timing

Handball

- Fixtures
- Statistics
- League Table

Combat Sports

Supports:

- Boxing
- Wrestling
- Judo
- Karate
- Taekwondo
- Kickboxing
- MMA

Includes:

- Weight Categories
- Judges' Scores
- Rounds
- Knockout
- Technical Knockout
- Decision

African Traditional Sports

DAMBE (African One-on-One Boxing)

The platform shall provide a dedicated competition template for Dambe.

Features include:

- Fighter Registration
- Traditional Weight Classes
- Fight Card
- Round Scoring
- Judge Decisions
- Championship Rankings
- Cultural Branding
- Event Posters
- Traditional Symbols

9.16 FUTURE SPORT EXPANSION

The platform shall remain completely **sport-agnostic**.

Authorized administrators shall:

- Create new sport templates.
- Modify rules.
- Configure scoring.
- Configure rankings.
- Upload sport logos.
- Upload competition symbols.
- Configure officials.
- Configure medals.
- Configure statistics.

No software redevelopment shall be required to introduce new sports.

9.17 FUNCTIONAL REQUIREMENTS

Requirement ID	Description
FR-COMP-1	The system shall support multi-level competition approvals.
FR-COMP-2	KAFRIADA Administrators shall possess overriding approval authority.
FR-COMP-3	Competitions shall support free and paid registration.
FR-COMP-4	Registration revenue shall be automatically distributed using configurable percentages.
FR-COMP-5	Officials shall automatically receive accumulated match fees in their personal dashboards.
FR-COMP-6	The system shall support configurable disciplinary fines and sanctions.

FR-COMP-7	Every competition shall use editable sport templates.
FR-COMP-8	The platform shall remain fully sport-agnostic.
FR-COMP-9	New sports shall be introduced through configuration only.
FR-COMP-10	Competition templates shall support logos, images, branding, trophies, medals, and federation symbols.

9.18 BUSINESS RULES

Rule ID	Business Rule
BR-COMP-1	Only verified organizations may organize competitions.
BR-COMP-2	KAFRIADA may approve competitions with or without prior organizational approval.
BR-COMP-3	Only KAFRIADA Approved competitions may be published.
BR-COMP-4	Every paid competition shall use automated revenue distribution.
BR-COMP-5	Officials shall automatically accumulate earnings after each completed match.
BR-COMP-6	Outstanding disciplinary fines may restrict future registrations until settled.
BR-COMP-7	Every sport shall use an editable competition template.
BR-COMP-8	The competition engine shall remain extensible for future sports without source code modification.
BR-COMP-9	Every financial transaction, approval, sanction, and competition activity shall generate a permanent audit record.

Summary

The Tournament & Competition Management Module provides a complete enterprise-grade competition ecosystem that supports multi-level governance, configurable approval workflows, free and paid competitions, automated financial distribution, referee earnings management, player welfare funding, disciplinary enforcement, live match operations, and a universal sport template engine. By remaining fully sport-agnostic and configuration-driven, KAFRIADA NET can support traditional African sports such as **Dambe** alongside international sports while enabling governments, federations, leagues, schools, academies, and clubs to manage competitions within a single scalable digital platform.

CHAPTER 10: SCOUTING & TALENT DISCOVERY MODULE

10.1 Overview

The Scouting & Talent Discovery Module is a centralized digital recruitment platform that enables verified scouts, clubs, academies, schools, national teams, sports federations, licensed agents, and KAFRIADA administrators to identify, evaluate, recruit, and monitor athletes across all supported sports.

The module eliminates fragmented scouting processes by providing verified athlete profiles, performance analytics, scouting reports, trial management, recruitment workflows, and talent intelligence within a single ecosystem.

The platform shall support grassroots talent identification through elite professional recruitment while maintaining transparency, athlete protection, and regulatory compliance.

10.2 Objectives

The module shall:

- Discover athletes from grassroots to elite level.
- Support digital scouting across all supported sports.
- Provide verified athlete profiles.
- Eliminate identity fraud during recruitment.
- Digitize trial invitations.
- Manage scouting reports.
- Rank athletes using configurable performance metrics.
- Connect scouts with clubs, academies, schools, and national teams.
- Improve athlete visibility across Africa and internationally.

10.3 Eligible Users

The following users may access the Scouting Module according to their permissions.

Individual Users

- Verified Scout
- Scout Manager
- Licensed Sports Agent
- Athlete

- Coach

Organizations

- Clubs
- Academies
- Schools
- Universities
- Sports Federations
- National Teams
- Ministries of Sports

Platform Administration

- KAFRIADA Administrator
 - Federation Administrator
-

10.4 Scout Registration

Only verified individuals and organizations may register as scouts.

Required Information

- Full Name
- KAFRIADA ID
- Nationality
- Organization
- Sports Specialization
- Years of Experience
- Professional Certification
- Identity Verification
- Photograph
- Contact Information

Applications shall be reviewed by KAFRIADA before activation.

Scout Status

- Draft
- Pending Review

- Verified
 - Suspended
 - Revoked
-

10.5 Athlete Discovery

Scouts shall search athletes using advanced filters.

Search Criteria

- KAFRIADA ID
- Athlete Name
- Sport
- Position
- Gender
- Age
- Country
- State
- Local Government Area
- Club
- School
- Academy
- Competition
- Performance Rating
- Verification Status
- Medical Clearance
- Availability Status
- Transfer Status

Only information permitted by athlete privacy settings shall be displayed.

10.6 Athlete Availability

Athletes may define their recruitment status.

Availability Options

- Available for Trials
- Available for Transfer
- Available for Scholarship
- Available for Sponsorship
- Available for National Team Selection
- Not Available

Transfer availability shall synchronize automatically with the Player Transfer Marketplace.

10.7 Scouting Reports

Verified scouts shall create structured scouting reports.

Each report shall contain:

- Athlete Information
- Competition Observed
- Match Date
- Venue
- Technical Ability
- Tactical Awareness
- Physical Performance
- Mental Attributes
- Leadership
- Discipline
- Communication
- Strengths
- Weaknesses
- Development Areas
- Overall Rating
- Recommendation

Recommendation Status

- Immediate Signing
- Trial Recommended

- Continue Monitoring
- Future Prospect
- Not Recommended

Reports shall remain confidential and accessible only to authorized users.

10.8 Trial Management

Organizations may invite athletes for trials.

Trial Workflow

1. Scout identifies athlete.
2. Trial invitation is created.
3. Athlete receives notification.
4. Athlete accepts or declines.
5. Trial schedule is confirmed.
6. Attendance is recorded.
7. Evaluation is completed.
8. Decision is issued.

Decision Outcomes

- Selected
 - Reserve
 - Further Assessment
 - Not Selected
-

10.9 Recruitment Workflow

Recruitment shall follow an auditable digital process.

1. Scout Recommendation.
2. Club Review.
3. Athlete Acceptance.
4. Medical Verification.
5. Contract Negotiation.
6. Transfer Validation (where applicable).

7. Digital Contract Execution.
8. Player Registration Update.

Every recruitment activity shall be permanently logged.

10.10 Athlete Performance Index

The platform shall calculate an Athlete Performance Index (API Score).

The score may include:

- Competition Performance
- Technical Rating
- Tactical Rating
- Physical Fitness
- Medical Status
- Discipline
- Attendance
- Awards
- Competition Level
- Recent Form

Weighting shall be configurable by sport and competition.

10.11 Scout Dashboard

Each verified scout shall have a personal dashboard displaying:

- Athletes Observed
- Active Reports
- Trial Invitations
- Successful Signings
- Recruitment History
- Pending Reviews
- Performance Analytics
- Favorite Athletes
- Upcoming Competitions

- Earnings (where applicable)
-

10.12 Notifications

The system shall notify users regarding:

- Scout Verification
- Trial Invitations
- Trial Responses
- Recruitment Decisions
- Report Approval
- Contract Offers
- Transfer Approval
- Athlete Availability Changes

Notification channels:

- In-App Notification
 - Email
 - SMS
 - Push Notification
-

10.13 Functional Requirements

Requirement ID	Description
FR-SCT-1	The system shall support verified digital scouting.
FR-SCT-2	Advanced athlete search shall be available using configurable filters.
FR-SCT-3	Scouts shall create structured scouting reports.
FR-SCT-4	Athlete privacy settings shall govern profile visibility.
FR-SCT-5	Trial invitations shall be managed digitally.
FR-SCT-6	Recruitment workflows shall integrate with Club Management and Player Transfer modules.
FR-SCT-7	Athlete Performance Index scores shall be configurable by sport.

Requirement ID	Description
FR-SCT-8	Every scouting activity shall generate a permanent audit log.

10.14 Business Rules

Rule ID	Business Rule
BR-SCT-1	Only verified scouts and approved organizations may access advanced scouting features.
BR-SCT-2	Athlete data shall only be visible according to privacy settings and verification level.
BR-SCT-3	Trial invitations shall be accepted or declined within the configured response period.
BR-SCT-4	Every scouting report shall be linked to the observing scout and competition.
BR-SCT-5	Recruitment processes involving transfers shall follow the Player Transfer approval workflow.
BR-SCT-6	Every scouting, trial, recruitment, and recommendation activity shall be permanently recorded for compliance and reporting.

Chapter Summary

The Scouting & Talent Discovery Module provides a trusted, data-driven recruitment ecosystem that connects athletes with verified scouts, clubs, federations, academies, schools, and national teams. By combining verified identities, structured scouting reports, digital trial management, configurable athlete performance analytics, and integrated recruitment workflows, the module strengthens talent identification and career development while ensuring transparency, athlete protection, and regulatory compliance across the KAFRIADA ecosystem.

KAFRIADA COMPETE MODULES SUMMARY

Module focus: Competition Management, Scouting Workflows, Performance Analytics & AI Intelligence

Target Timeline: Phase 2 Execution Cycle

1. Executive Summary & Objectives

KAFRIADA COMPETE expands the platform beyond core administrative identity and club management into live operational tools for tournaments, real-time match tracking, verified scouting metrics, and predictive AI modeling.

- **Primary Goal:** Provide leagues, academies, and tournament organizers with a digital command center while enabling scouts to evaluate verified athlete performance data with AI-assisted insights.
- **Key Pillars:**
 1. Automated Competition & Tournament Infrastructure
 2. Multi-Tiered Talent Scouting & Recommendation Engine
 3. Advanced Match Performance & Biometric Analytics
 4. Predictive AI Insights (Career Projection, Risk Assessment, Valuation)

2. Key Functional Modules

A. Competition & Tournament Management

- **Tournament Setup & Scheduling:** Fixture generation (round-robin, knockout, hybrid), venue mapping, referee assignments, and automated match-sheet generation.
- **Live Match Engine:** Real-time scoring, event logging (goals, assists, cards, substitutions, VAR/review notes), and official match report sign-offs.
- **Discipline & Standings Engine:** Automatic points calculation, tie-breaker handling, yellow/red card accumulation tracking, and automated suspension enforcement tied to **KUID** profiles.

B. Scouting & Talent Identification

- **Verified Scout Dashboards:** Filterable player databases by age, position, physical attributes, KUID verification level, region, and match rating.
- **Custom Watchlists & Scouting Reports:** Structured evaluation templates (Technical, Tactical, Physical, Psychological) with custom tagging and target lists.
- **Talent Discovery Marketplace:** Direct channel for verified scouts to signal interest, request verified video assets, or coordinate official trial invitations through host clubs.

C. Performance Analytics Core

- **Match & Event Metrics:** Deep-dive statistics tracking pass completion, duel success rates, distance covered, sprint frequency, and spatial heatmap generation.
- **Player Development Radar:** Visual spider charts comparing an athlete's metrics against age-group benchmarks, regional averages, or professional positional standards.
- **Video & Clipping Integration:** Timestamped match events linked directly to player profiles for quick video retrieval and review.

D. AI Engine Integration

- **Match Performance Scoring (K-Score):** Algorithmic performance index generated post-match based on position-specific impact vectors.
- **Talent Match & Similarity Engine:** AI-driven discovery matching young athletes to established professional player profiles based on physical and tactical statistical fingerprints.
- **Injury Risk & Workload Indicator:** Predictive health analytics highlighting fatigue thresholds based on match density, physical load data, and recovery intervals.

- **Scouting Valuation Estimator:** Data-driven valuation model calculating athlete market potential based on age, performance trajectory, league level, and contract status.

3. Data Architecture & Integration Points

Integration Point	Source / Dependency	Function
Identity & KUID	KAFRIADA CORE (Phase 1)	Ensures all stats, match logs, and scouting reports permanently attach to verified athlete records.
Club & Squads	KAFRIADA CORE (Phase 1)	Pulls eligible rosters directly into competition match sheets.
Live Event Stream	Mobile App / Web Scorer	Low-latency event streaming via WebSockets for real-time leaderboards and live updates.
AI Processing Pipeline	AWS Lambda / Python AI Core	Runs asynchronous post-match processing for player performance indexing and predictive models.

4. Phase 2 Key Deliverables & Milestones

Phase 2 Execution Roadmap

- Sprint 1–2: Competition Architecture & Dynamic Fixture Engine
- Sprint 3–4: Live Match Scoring Interface & Automated Disciplinary Logic
- Sprint 5–6: Scout Workstation, Advanced Search Filters & Watchlists
- Sprint 7–8: Performance Analytics Engine, Heatmaps & Benchmark Radars
- Sprint 9–10: K-Score AI Pipeline, Similarity Matching & Predictive Models
- Sprint 11–12: Beta Testing with Partner Leagues, Security Audit & Phase 2 Launch
 - **Deliverable 1:** Fully functional Competition & Tournament Management System.
 - **Deliverable 2:** Verified Scouting Suite with customized evaluation tools.
 - **Deliverable 3:** Interactive Analytics & Heatmap Visualization Module.
 - **Deliverable 4:** Phase 1 AI Integration (K-Score, Player Similarity, Workload Risk).

Execution Cycle: Phase 2 (Months 4–6 / Sprints 1–12)

Core Dependencies: KAFRIADA CORE (Phase 1 Identity, KUID System, and Roster Foundations)

1. Executive Summary & Strategic Objectives

KAFRIADA COMPETE transitions the platform from administrative management into live operational engagement. It establishes an end-to-end digital ecosystem for tournament organizers, talent scouts, club analysts, and athletes.

- **Primary Objective:** Deliver a centralized competition engine for regional and international leagues while providing data-backed scouting and predictive AI metrics tied directly to verified athlete **KUID** profiles.
- **Core Value Proposition:**

- **Automated Tournament Operations:** Multi-tier fixture generation, live match engines, and real-time disciplinary tracking.
- **Verified Scouting Workflows:** Structured evaluation models, talent pipelines, and direct trial signaling.
- **Deep Performance Analytics:** Spatial tracking, biometric data indexing, and benchmark radar visualizers.
- **Predictive AI Engine:** Algorithmic match ratings (**K-Score**), player similarity matching, workload/injury risk modeling, and market valuation estimates.

2. Core Operational Modules

A. Competition & Tournament Management System

- **Dynamic Fixture Engine:** Supports complex competition formats (Round Robin, Single/Double Elimination, Swiss System, Multi-stage Group + Playoff).
- **Live Match Scorer & Webhooks:** Real-time event logging (Goals, Assists, Cards, Submissions, VAR/Review notes, Substitutions) synced via WebSockets.
- **Automated Disciplinary Logic:** Dynamic point calculations, goal-difference tie-breakers, automatic card tracking, and cross-match suspension flags enforced at the KUID level.

B. Scout Workstation & Talent Marketplace

- **Advanced Multi-Filter Search:** Filter athlete databases by age, primary/secondary positions, dominant foot, physical metrics, verification status, and match impact scores.
- **Scouting Evaluation Models:** Standardized evaluation forms (Technical, Tactical, Physical, Psychological) with customizable scoring grids for club-specific requirements.
- **Watchlists & Talent Signaling:** Direct channels for accredited scouts to request verified match tape, trigger trial invitations, and initiate official club-to-club inquiries.

C. Performance Analytics Core

- **Match & Event Metrics:** Deep-dive statistical tracking covering pass completion percentages, duel success rates, distance covered, sprint intensity, and turnover location.
- **Spatial Heatmaps & Radar Plots:** Visual spider charts comparing individual performance metrics against age-group benchmarks, regional standards, or professional position baselines.
- **Timestamped Video Integration:** Automated clipping engine linking log events directly to full-match footage for rapid review.

D. AI Intelligence Engine (K-AI Core)

- **Match Impact Rating (K-Score):** Positional index weighting key performance indicators (KPIs) to produce a composite match rating (1.00 – 10.00).

- **Player Similarity Engine:** Machine learning model comparing physical and tactical statistics to match youth prospects with established professional profiles.
- **Workload & Fatigue Risk Indicator:** Algorithmic strain tracking based on match density, minutes played, and physical load vectors to flag potential injury risks.
- **Data-Driven Valuation Model:** Predictive framework calculating estimated player value based on age, trajectory, league strength, and contract status.

3. Data Architecture & Integration Points

Module / System	Source / Dependency	Functional Role
KUID Identity Engine	KAFRIADA CORE (Phase 1)	Binds all competition statistics, match logs, and scouting notes to the athlete's permanent identity record.
Squad & Roster Control	KAFRIADA CORE (Phase 1)	Feeds active rosters and transfer status directly into match-sheet generation.
Real-Time Data Stream	Web/Mobile Scorer App	Delivers low-latency event streams to public leaderboards and live match dashboards.
K-AI Microservices	AWS Lambda / Python AI Core	Runs asynchronous post-match processing for K-Scores, radar charts, and predictive models.

4. 12-Week Milestone Execution Roadmap

Timeline	Sprint	Technical & Operational Focus	Key Evaluation Gate
Weeks 01–02	Sprint 1	Competition Architecture & Dynamic Fixture Engine	Automated fixture generation functional across all tournament formats.
Weeks 03–04	Sprint 2	Live Match Scoring Interface & Real-time WebSockets	Live event streaming latency below 500ms; match report generation pass.
Weeks 05–06	Sprint 3	Automated Disciplinary & Standings Logic	Zero-error tie-breaker and automatic card suspension calculation.
Weeks 07–08	Sprint 4	Scout Workstation & Advanced Filter Engine	Multi-parameter search queries return results in <300ms across 100k records.
Weeks 09–10	Sprint 5	Scouting Evaluation Templates & Watchlist Systems	Custom report generation and scout notification workflows functional.

Weeks 11–12	Sprint 6	Performance Data Processing & Event Metric Storage	Post-match statistical sync completed within 60 seconds of full-time.
Weeks 13–14	Sprint 7	Spatial Heatmap & Radar Visualization Module	Dynamic chart rendering active on Web and Mobile apps.
Weeks 15–16	Sprint 8	Video Clipping & Match Timestamp Linkage	Match event clicks successfully trigger accurate video playback timestamps.
Weeks 17–18	Sprint 9	K-Score AI Engine & Position Impact Vectors	K-Score algorithm producing consistent post-match ratings.
Weeks 19–20	Sprint 10	Player Similarity & Workload Risk Models	Predictive engine producing valid comparative profiles and load alerts.
Weeks 21–22	Sprint 11	Closed Beta Testing with Partner Leagues	Live tournament dry-run executed with 20+ teams and active scouts.
Weeks 23–24	Sprint 12	System Hardening, Security Audits & Phase 2 Launch	Pen test approval, zero P1/P2 bugs, and formal Phase 2 deployment.

KAFRIADA EDU (PHASE 3)

Focus: School/Academy Management, Learning Management System (LMS), Dual-Career Tracking, and Guardian Governance

See [2.0 Business Overview](#)

1. Architectural & Domain Objective

KAFRIADA EDU bridges academic education and elite sports training across primary, secondary, and tertiary institutions as well as private academies. It addresses **Problem Domain #6 (Education)** by ensuring that young athletes maintain academic standing alongside athletic growth, while giving coaches and officials access to accredited training programs.

2. Functional Feature Specifications (Deep Dive)

Module 1: Institution & Academy Management System (IAMS)

- **IAMS-01: Dual-Career Tracking Engine**
 - **Detailed Functionality:** Tracks GPA, attendance, classroom behavior, and academic eligibility alongside physical development indicators (e.g., growth spurts, speed metrics, fatigue recovery).
 - **Business Logic:** If an athlete's cumulative GPA falls below institutionally set thresholds (e.g., 2.0/4.0 or 60%), the engine automatically flags the profile in **KAFRIADA COMPETE**, preventing tournament match-sheet (for university sport courses that are strictly under academic concern) inclusion until academic clearance is logged by a verified School Admin (STK-05).
 - **Data Model Attributes:** StudentAthleteID (KAFRIADA ID), InstitutionID, CurrentGPA, AttendanceRate, AcademicEligibilityStatus (Cleared, Warning, Ineligible), LastUpdatedBy.
- **IAMS-02: Guardian Governance & Parental Consent Portal (CONSIDER KAFRIADA CORE/ ID PHASE 1)**
 - **Detailed Functionality:** A dedicated portal for parents/guardians (STK-12) to manage legal permissions for minor athletes (<18 years old).
 - **Business Logic:** Enforces explicit digital opt-in/opt-out for:
 1. Public profile visibility to third-party scouts (STK-08).
 2. Direct messaging or trial invitations from agents (STK-10).
 3. Academy contract transfers and medical record sharing.

- **Compliance Gate:** Adheres strictly to NDPR and GDPR child privacy mandates. Any scout trial request issued in COMPETE triggers a mandatory SMS/In-App authorization flow to the guardian before releasing contact info.

Module 2: Sports Learning Management System & Certification (LMS)

- **LMS-01: Interactive Course Delivery & Assessment Infrastructure**

- **Detailed Functionality:** Delivers structured online coursework for coaches (STK-02), referees/match officials, athletes, and club administrators (STK-04). Course types include Tactical Training, Sports Nutrition, Anti-Doping Regulations, First Aid, and Tournament Officiating.
- **E-Commerce & Monetization Logic:** Integrates direct payment processing supporting course fees ranging from ₦10,000 to ₦50,000. Supports modular video streaming, SCORM-compliant interactive materials, and automated quiz evaluations.
- **Prerequisite Enforcement:** High-level certification tracks require verified prerequisite credentials (e.g., *Level 1 Grassroots Coaching Certificate* required before unlocking *Level 2 Advanced Tactics*).

- **LMS-02: Cryptographically Verified Badge & Certification Issuer**

- **Detailed Functionality:** Automatically issues digital certificates and badges upon course completion and passing score verification ($\geq 80\%$).
- **Integration Logic:** Issued credentials immediately write to the user's universal **KAFRIADA ID** across CORE, COMPETE, and ENTERPRISE. Scouts and clubs can instantly verify whether a coach or match official holds active, official certifications.

3. Comprehensive Phase 3 Development Backlog

Story ID	Epic	User Story	Acceptance Criteria	Priority	Complexity (Pts)
EDU-101	IAMS	As a School Admin, I want to batch-onboard student-athletes via CSV or manual form and link them to their KAFRIADA CORE ID.	System validates existing CORE IDs, creates student-athlete records, and maps academic metrics to profiles.	Must	8
EDU-102	IAMS	As a Parent/Guardian, I want to receive real-time push/SMS alerts when a scout requests access to my minor child's profile.	System sends notification with approval toggle; profile access remains blocked	Must	5

			until guardian approves.		
EDU-103	IAMS	As an Academy Manager, I want to record periodic physical assessment data (height, weight, 40m sprint times) for my players.	System saves metrics, logs historical trends, and renders visual growth charts on the player dashboard.	Should	5
EDU-104	IAMS	As a School Registrar, I want to input end-of-term GPAs so that academic eligibility is enforced automatically.	System updates status; if GPA < threshold, COMPETE module locks player from fixture generation.	Must	5
EDU-201	LMS	As a Coach, I want to browse available certified courses, filter by price, and pay via local payment channels.	Payment gateway processes transaction (₦10k-₦50k), issues receipt, and unlocks course content.	Must	8
EDU-202	LMS	As a Course Participant, I want to watch video modules and complete end-of-unit quizzes.	Video progress is tracked; quiz attempts calculate scores instantly and store results.	Must	5
EDU-203	LMS	As a certified user, I want my digital certificate to append to my public profile with a verification badge.	Profile displays active badge with a unique verification code and issuance date.	Must	3
EDU-204	LMS	As an LMS Admin, I want to upload new course materials, create quiz question banks, and set passing criteria.	Admin panel allows rich text/video upload, quiz creation, and fee configuration per course.	Should	8

KAFRIADA ENTERPRISE (PHASE 4)

Focus: National Sports Intelligence, Government Policy Dashboards, B2B Tiered API Gateway, and Pan-African Expansion

See: [2.0 Business Overview](#)

1. Architectural & Domain Objective

KAFRIADA ENTERPRISE represents the macro-governance layer of the ecosystem. It addresses **Problem Domain #7 (Government & Federations)** by converting multi-year grassroots data across CORE, COMPETE, and EDU into high-level intelligence, enabling Ministries of Sports (STK-15), National Federations (STK-14), and Enterprise Sponsors (STK-13) to drive data-led investments, policy changes, and commercial campaigns.

2. Functional Feature Specifications (Deep Dive)

Module 1: Governance & National Sports Intelligence (GNSI)

- **GNSI-01: National Executive Sports Dashboard**
 - **Detailed Functionality:** Provides macro-level visual intelligence displaying active athlete density, sport-by-sport participation, facility utilization rates, gender distribution, and geographic talent concentration across states/provinces.
 - **Business Value:** Replaces intuition-based decision-making with verified statistical evidence. Allows government entities to identify underserved grassroots regions and allocate infrastructure budgets effectively.
 - **Data Aggregation Engine:** Aggregates anonymized, real-time telemetry from CORE (registrations), COMPETE (matches played, tournament density), and EDU (school participation).
- **GNSI-02: Federation Regulatory & Compliance Engine**
 - **Detailed Functionality:** Provides sports governing bodies with centralized control to monitor club licensing, referee active status, transfer compliance, and age-verification audit trails.
 - **Fraud Audit Mechanism:** Scans national databases for duplicate identity flags, unverified Stage 1 profiles attempting elite competition entry, or age anomalies across multi-season records.

Module 2: Enterprise API Gateway & Commercialization Engine (EAGE)

- **EAGE-01: Multi-Tiered Partner API Gateway**
 - **Detailed Functionality:** Secure, monitored REST API infrastructure allowing commercial sponsors (STK-13), media partners (STK-18), and research universities (STK-16) to query structured sports data.

- **Monetization & Technical Controls:**
 1. *Free / Public Tier*: Basic public fixture lists and tournament standings.
 2. *Standard Tier (Paid)*: Aggregated match statistics, anonymized talent demographic reports.
 3. *Enterprise Tier (₦10M - ₦50M/yr)*: Real-time live match telemetry data streams, deep performance analytics feeds, and custom national reporting pipelines.
- **Technical Specs**: Enforces OAuth 2.0 authentication, strict rate-limiting (e.g., 10,000 calls/min for Enterprise), payload encryption, and automated billing integration based on API usage metrics.
- **EAGE-02: Pan-African Cross-Border Data & Talent Migration Engine**
 - **Detailed Functionality**: Multi-currency, multi-language (English, French, Arabic) synchronization infrastructure facilitating international scouting, cross-border transfers, and continental tournament administration across Africa and the Middle East.
 - **Compliance & Currency Logic**: Handles dynamic multi-currency conversions (NGN, KES, GHS, USD, EUR, AED), dynamic time-zone localization, and legal compliance routing for cross-border player transfers compliant with FIFA/CAF regulatory guidelines.

3. Comprehensive Phase 4 Development Backlog

Story ID	Epic	User Story	Acceptance Criteria	Priority	Complexity (Pts)
ENT-101	GNSI	As a Ministry Official, I want an interactive heat map showing active registered athletes across all states/regions.	Dashboard renders dynamic map with filter controls (sport, age bracket, gender) within 1.5s.	Must	8
ENT-102	GNSI	As a Federation Admin, I want to audit player transfers across regional leagues to flag unverified or duplicate identities.	System runs automated integrity checks and flags suspicious contracts or identity duplicate records.	Must	8
ENT-103	GNSI	As a Policy Maker, I want to generate downloadable PDF reports on youth	Report engine generates formatted PDF summarizing participation trends,	Should	5

1. **System Uptime & Reliability:** 99.9% availability across all API gateways and end-user web application routes.
2. **Data Privacy & Encryption:**
 - **In Transit:** TLS 1.3 protocol for all client-server communications.
 - **At Rest:** AES-256 encryption applied to database records containing PII (Personally Identifiable Information), academic GPAs, and medical charts.
3. **API Performance Thresholds:** API Gateway P99 latency guaranteed under 250 ms under concurrent load up to 100,000 active sessions.
4. **Localization & Accessibility:** Responsive design adhering to WCAG 2.1 AA accessibility guidelines, supporting localized English, French, and Arabic interfaces.

OTHERS

CHAPTER 01: SPONSORSHIP, PARTNERSHIP, DONATIONS & FUNDRAISING MANAGEMENT

MODULE

01.1 Overview

The Sponsorship, Partnership, Donations & Fundraising Management Module provides a comprehensive digital platform for attracting, managing, monitoring, and reporting sponsorships, partnerships, grants, donations, crowdfunding campaigns, scholarships, corporate social responsibility (CSR) initiatives, and sports development funding across the KAFRIADA ecosystem. The module enables governments, ministries, federations, clubs, academies, schools, universities, NGOs, development partners, foundations, private organizations, corporate sponsors, philanthropists, and individuals to invest in sports development through a transparent and accountable financial management framework.

The module integrates seamlessly with Competition Management, Athlete Management, Club Management, Marketplace, Financial Management, Education, Athlete Care, and Reporting modules to ensure that every sponsorship, donation, or partnership is traceable from initiation to utilization.

01.2 Objectives

The system shall:

- Digitize sponsorship management.
- Support corporate partnerships.
- Manage donations and grants.
- Enable crowdfunding campaigns.
- Promote athlete sponsorship.
- Support club and competition sponsorship.
- Manage scholarship funding.
- Ensure transparency and accountability.
- Automate revenue allocation.
- Produce sponsorship performance analytics.

01.3 Sponsorship Categories

The platform shall support sponsorship for:

Athlete Sponsorship

- Individual Athletes
- Youth Athletes
- Elite Athletes
- Paralympic Athletes

Club Sponsorship

- Professional Clubs
- Amateur Clubs
- Community Clubs

Organization Sponsorship

- Sports Federations
- Academies
- Schools
- NGOs
- Ministries of Sports
- Sports Associations

Event Sponsorship

- Conferences

- School Teams
- University Teams

Competition Sponsorship

- League Competitions
- Cups
- Championships
- School Sports Festivals
- Community Games
- Continental Championships

- Coaching Clinics
- Referee Courses
- Sports Seminars
- Awards Ceremonies
- Sports Festivals

O1.4 Partnership Management

The platform shall support strategic partnerships with:

- Governments
- International Sports Organizations
- Olympic Committees
- Educational Institutions
- Development Agencies
- NGOs
- Corporate Organizations
- Financial Institutions
- Insurance Companies
- Technology Companies
- Media Organizations

Each partnership shall include:

- Partner Name
- Partnership Type
- Objectives
- Start Date
- End Date
- Deliverables
- Financial Value
- Contact Persons
- Documents
- Approval Status

O1.5 Sponsorship Workflow

Every sponsorship shall follow an auditable workflow.

Workflow

- | | |
|--|---|
| <ol style="list-style-type: none"> 1. Sponsor Registration 2. Sponsor Verification 3. Sponsorship Proposal 4. Proposal Review 5. Negotiation 6. Agreement Approval | <ol style="list-style-type: none"> 7. Digital Contract Execution 8. Payment Processing 9. Fund Allocation 10. Performance Monitoring 11. Sponsorship Completion 12. Final Reporting |
|--|---|

O1.6 Donations Management

The system shall support:

- Individual Donations
- Corporate Donations
- Anonymous Donations (subject to applicable legal and compliance requirements)
- Emergency Relief Donations
- Equipment Donations

Donors may specify:

- Beneficiary
- Purpose
- Amount
- Currency

- | | |
|---|---|
| <ul style="list-style-type: none"> • Facility Development Donations • Scholarship Donations <p>Donation options include:</p> <ul style="list-style-type: none"> • One-Time Donation • Monthly Donation • Annual Donation • Recurring Donation | <ul style="list-style-type: none"> • Anonymous Preference (where permitted) • Public Recognition Preference |
|---|---|

O1.7 Crowdfunding Campaigns

Authorized organizations may create fundraising campaigns.

Campaign information shall include:

- Campaign Title
- Campaign Description
- Campaign Banner
- Campaign Video
- Financial Goal
- Amount Raised
- Closing Date
- Beneficiary
- Progress Indicator
- Updates
- Donor List (subject to privacy settings)

Campaign types include:

- Medical Assistance
- Athlete Support
- Competition Funding
- Club Development
- Equipment Purchase
- Stadium Development
- Community Sports Projects
- Disaster Relief

O1.8 Scholarship & Grant Management

The module shall support:

- Educational Scholarships
- Sports Scholarships
- Research Grants
- Infrastructure Grants
- Youth Development Grants
- Women's Sports Grants
- Disability Sports Grants

Application workflow:

1. Scholarship Publication
2. Online Application
3. Eligibility Verification
4. Document Review
5. Interview (Optional)
6. Selection
7. Award Notification
8. Fund Disbursement
9. Progress Monitoring

O1.9 Financial Allocation & Distribution

The platform shall automatically allocate sponsorship funds according to configurable distribution rules.

Beneficiaries may include:

- Sponsored Athlete
- Club

Distribution percentages shall be configurable by authorized administrators.

- | | |
|--|--|
| <ul style="list-style-type: none"> • Competition Organizer • Sports Federation • Educational Institution • Service Provider • KAFRIADA Platform Service Fee | <p>Every allocation shall generate:</p> <ul style="list-style-type: none"> • Financial Ledger Entry • Audit Record • Digital Receipt • Payment Voucher |
|--|--|

O1.10 Sponsor Dashboard

Sponsors shall have access to a dedicated dashboard displaying:

- Active Sponsorships
- Completed Sponsorships
- Donation History
- Campaign Performance
- Fund Utilization Reports
- Beneficiary Reports
- Contracts
- Financial Statements
- Tax-Related Reports (where applicable)

O1.11 Beneficiary Dashboard

Athletes, clubs, organizations, and competition organizers shall view:

- Active Sponsorships
- Grants Received
- Donations Received
- Funding Status
- Spending Reports
- Compliance Requirements
- Reporting Deadlines

O1.12 Branding & Marketing Assets

Sponsors and partners may upload and manage:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Company Logo • Brand Colours • Promotional Banners • Advertisements • Videos • Event Branding Materials • Social Media Assets | <p>The system shall support configurable placement of branding assets across:</p> <ul style="list-style-type: none"> • Competition Pages • Marketplace Listings • Event Pages • Athlete Profiles (where approved) • Club Profiles (where approved) • Educational Courses • Digital Certificates (where applicable) <p>Branding shall comply with KAFRIADA policies and sponsorship agreements.</p> |
|---|---|

O1.13 Reporting & Compliance

The platform shall generate reports including:

- Sponsorship Revenue
- Donation Reports
- Campaign Performance
- Beneficiary Reports
- Financial Statements
- ROI Analysis
- Impact Assessment
- Grant Utilization
- Partnership Performance

Reports shall be exportable in multiple formats.

O1.14 Notifications

The system shall notify stakeholders regarding:

- Sponsorship Approval
- Partnership Requests
- Donation Confirmation
- Campaign Milestones
- Grant Awards
- Payment Processing
- Reporting Deadlines
- Contract Expiry
- Renewal Opportunities

Notification channels include:

- In-App Notifications
- Email
- SMS
- Push Notifications

O1.15 Functional Requirements

Requirement ID	Description
FR-SPON-1.1	The system shall manage sponsorships, partnerships, donations, grants, and crowdfunding campaigns.
FR-SPON-1.2	Authorized organizations shall create and manage fundraising campaigns.
FR-SPON-1.3	Sponsorship and grant workflows shall support configurable approval processes.
FR-SPON-1.4	Financial allocations shall be automatically distributed according to configurable percentage rules.
FR-SPON-1.5	Sponsors and beneficiaries shall have dedicated dashboards with financial and performance analytics.
FR-SPON-1.6	The system shall support branding and sponsor asset management.
FR-SPON-1.7	Reports shall provide financial transparency, utilization tracking, and impact analysis.
FR-SPON-1.8	Every sponsorship, donation, payment, agreement, and allocation shall generate a permanent audit record.

O1.16 Business Rules

Rule ID	Business Rule
BR-SPON-1	Only verified sponsors, partners, and organizations may initiate sponsorship agreements or fundraising campaigns.
BR-SPON-2	Every sponsorship and partnership agreement shall undergo the configured approval workflow before activation.
BR-SPON-3	Financial distributions shall follow approved percentage allocations and generate corresponding audit records.
BR-SPON-4	Beneficiaries shall submit utilization reports for grants or sponsorships when required by the sponsoring organization.
BR-SPON-5	Crowdfunding campaigns shall automatically close upon reaching their end date or funding goal, according to campaign configuration.
BR-SPON-6	Sponsor branding shall only appear in locations authorized by the applicable sponsorship agreement and KAFRIADA branding policies.
BR-SPON-7	All donations, grants, sponsorships, and partnerships shall comply with applicable financial regulations, anti-money laundering (AML), and know-your-customer (KYC) requirements.
BR-SPON-8	Every financial transaction, agreement, approval, allocation, and report shall be permanently stored for compliance, transparency, and auditing purposes.

Summary

The Sponsorship, Partnership, Donations & Fundraising Management Module establishes a transparent, accountable, and scalable funding ecosystem for African sports. It enables governments, federations, clubs, educational institutions, corporate organizations, development partners, and individual supporters to invest confidently in athletes, competitions, infrastructure, education, and community development. Through configurable workflows, automated financial allocation, integrated reporting, branding management, and compliance controls, the module strengthens sustainable sports financing and long-term growth across the entire KAFRIADA Network.

CHAPTER 02: REPORTING, BUSINESS INTELLIGENCE, DATA ANALYTICS & DECISION SUPPORT MODULE

O2.1 Overview

The Reporting, Business Intelligence (BI), Data Analytics & Decision Support Module is the centralized data intelligence platform of KAFRIADA NET. It collects, consolidates, analyzes, visualizes, and reports data generated across all modules to support strategic planning, operational monitoring, governance, policy formulation, performance evaluation, and evidence-based decision-making.

The module transforms raw operational data into actionable insights through configurable dashboards, reports, scorecards, predictive analytics, artificial intelligence (AI), machine learning (ML), and statistical analysis.

The module shall integrate with all KAFRIADA NET modules including Athlete Management, Club Management, Competition Management, Marketplace, Finance, Education, Sponsorship, Communication, Athlete Care, Scouting, Technical Personnel, and future modules.

O2.2 Objectives

The module shall:

- Provide enterprise-wide reporting.
- Deliver real-time dashboards.
- Support executive decision-making.
- Generate operational and strategic reports.
- Measure organizational performance.
- Provide sports development analytics.
- Monitor financial performance.
- Track athlete development.
- Support predictive analytics.
- Enable AI-powered recommendations.
- Support data export and integration.

O2.3 Reporting Categories

The platform shall provide reports for:

Athlete Reports

- Athlete Registration
- Athlete Growth
- Athlete Performance
- Medical Status
- Injury Statistics
- Scholarship Reports
- Career Progression
- Retirement Statistics

Marketplace Reports

- Sales Performance
- Product Performance
- Vendor Performance
- Transfers
- Sponsorship
- Marketplace
- Scholarships

Club Reports

- Club Registration
- Club Performance
- Membership
- Revenue
- Player Transfers
- Club Rankings

Competition Reports

- Competition Registration
- Match Results
- League Tables
- Attendance
- Officials Performance
- Discipline Reports
- Competition Revenue

Financial Reports

- Income
- Expenses
- Revenue Distribution
- Wallet Transactions
- Sponsorship Revenue
- Marketplace Revenue
- Course Revenue
- Tax Reports
- Audit Reports

Education Reports

- Student Enrolment
- Course Completion
- Examination Performance
- Certification Statistics
- CPD Statistics

Sponsorship Reports

- Sponsorship Performance
- Donation Analysis
- Campaign Performance
- ROI Reports
- Partner Contributions

Technical Personnel Reports

- Active Licences
- Referee Performance
- Coaching Statistics
- Assignments
- CPD Compliance

Communication Reports

- Broadcast Statistics
- Audience Reach
- Social Media Engagement
- Communication Performance
- Notification Delivery

O2.4 Executive Dashboard

The Executive Dashboard shall provide a real-time overview of the entire KAFRIADA ecosystem.

Key Performance Indicators (KPIs) may include:

- | | |
|---|--|
| <ul style="list-style-type: none">• Total Registered Athletes• Active Clubs• Active Competitions• Registered Coaches• Registered Officials• Total Sponsors• Marketplace Revenue• Competition Revenue• Total Wallet Transactions• Active Scholarships | <ul style="list-style-type: none">• Total Donations• Athlete Transfers• Total Courses• Active Users• Country Participation• Sport Participation• Gender Distribution• Youth Participation• Disability Sports Participation |
|---|--|

Widgets shall be configurable by role.

O2.5 Interactive Dashboards

Dashboards shall support:

- Charts
- Graphs
- Maps
- Tables
- Scorecards
- Heat Maps
- Trend Lines
- Drill-down Analysis
- KPI Cards

Users may:

- Rearrange widgets
- Save dashboard layouts
- Filter reports
- Export visualizations

O2.6 Geographic Analytics

The system shall visualize sports development geographically.

Supported views include:

- Africa
- Regional Communities
- Countries
- States/Regions
- Local Government Areas
- Cities
- Communities

Maps shall display:

- Athlete Distribution
- Club Distribution
- Competition Locations
- Sports Facilities
- Sponsorship Coverage
- Education Centres
- Participation Levels

O2.7 Artificial Intelligence & Predictive Analytics

The platform shall support AI-powered analytics including:

Athlete Intelligence

- Talent Prediction
- Injury Risk Analysis
- Performance Forecasting
- Career Development Recommendations

Competition Intelligence

- Attendance Forecasting
- Scheduling Optimization
- Referee Assignment Recommendations

Financial Intelligence

- Revenue Forecasting
- Fraud Detection
- Budget Forecasting
- Spending Trends

Sponsorship Intelligence

- Sponsorship Matching
- Campaign Success Prediction
- Donor Behaviour Analysis

AI models shall be configurable and continuously improved using authorized datasets.

O2.8 Custom Report Builder

Authorized users shall create custom reports without software development.

Users may select:

- Data Sources
- Fields
- Filters
- Calculations
- Charts
- Grouping
- Sorting
- Output Format

Reports may be saved as reusable templates.

O2.9 Scheduled Reports

Users shall schedule reports for automatic generation.

Supported schedules include:

- Hourly
- Daily
- Weekly
- Monthly
- Quarterly
- Yearly

Reports may be delivered via:

- Email
- Secure Download
- Dashboard
- API

O2.10 Data Export & Integration

Reports shall support export to:

- | | |
|---|---|
| <ul style="list-style-type: none"> • PDF • Excel • CSV | <ul style="list-style-type: none"> • Word • JSON • XML |
|---|---|

Authorized external systems may access approved reports through secure APIs.

O2.11 Benchmarking

Organizations shall compare performance against:

- National Average
- State Average
- Federation Average
- League Average
- Historical Performance
- Organizational Goals

Benchmark indicators shall be configurable.

O2.12 Audit & Compliance Reporting

The module shall provide:

- User Activity Logs

- Security Reports
- Financial Audit Reports
- Compliance Reports
- Data Access Logs
- Change History
- Regulatory Reports

Reports shall support investigations and governance reviews.

O2.13 Data Warehouse

The platform shall maintain a centralized data warehouse for analytical processing.

The data warehouse shall:

- Consolidate information from all modules.
- Preserve historical records.
- Support large-scale analytics.
- Enable trend analysis.
- Improve reporting performance.

Data refresh schedules shall be configurable.

O2.14 Performance Monitoring

The platform shall monitor:

- System Performance
- API Performance
- User Activity
- Transaction Volume
- Storage Usage
- Response Time
- Error Rates

Administrators shall receive alerts when thresholds are exceeded.

O2.15 Notifications

The system shall notify administrators regarding:

- Scheduled Report Completion
- Failed Reports
- KPI Threshold Breaches
- Fraud Alerts
- Revenue Milestones
- System Performance Issues
- Compliance Exceptions

Notification channels include:

- In-App Notifications
- Email
- SMS
- Push Notifications

O2.16 Functional Requirements

Requirement ID	Description
FR-BI-1.1	The system shall consolidate data from all KAFRIADA modules into a centralized reporting environment.
FR-BI-1.2	The platform shall provide configurable dashboards with real-time and historical analytics.
FR-BI-1.3	Authorized users shall create custom reports without software development.
FR-BI-1.4	Reports shall support scheduling, export, and secure sharing.
FR-BI-1.5	The platform shall provide geographic analytics and interactive visualizations.
FR-BI-1.6	AI-powered predictive analytics shall support performance, financial, and operational decision-making.
FR-BI-1.7	The module shall maintain a centralized analytical data warehouse with configurable refresh schedules.
FR-BI-1.8	Every report generation, export, dashboard modification, and analytical query shall generate an audit record.

O2.17 Business Rules

Rule ID	Business Rule
BR-BI-1	Users shall only access reports and dashboards permitted by their assigned roles and permissions.
BR-BI-2	Sensitive financial, medical, and personal data shall be protected through role-based access control and data privacy policies.
BR-BI-3	Scheduled reports shall execute according to configured schedules and notify users upon completion or failure.
BR-BI-4	AI-generated insights shall support decision-making but shall not automatically override authorized human decisions.
BR-BI-5	Historical analytical data shall be retained according to configurable data retention policies and applicable regulations.
BR-BI-6	Every exported report shall include generation metadata, report owner, generation date, and version information where applicable.
BR-BI-7	Dashboard widgets, KPIs, benchmarks, and calculations shall be configurable without software redevelopment.
BR-BI-8	All reporting, analytics, exports, and audit activities shall be permanently logged for governance, compliance, and traceability.

Summary

The Reporting, Business Intelligence, Data Analytics & Decision Support Module transforms KAFRIADA NET into a data-driven sports governance platform. By integrating operational data from every module, providing configurable dashboards, AI-assisted analytics, predictive insights, custom reporting, geographic visualization, benchmarking, and comprehensive audit capabilities, the module empowers administrators, governments, federations, clubs, and other stakeholders to make informed decisions, measure impact, and drive the sustainable development of sports across Africa.

CHAPTER 03: COACHING, OFFICIATING & TECHNICAL PERSONNEL MANAGEMENT

MODULE

03.1 Overview

The Coaching, Officiating & Technical Personnel Management Module provides a comprehensive digital ecosystem for registering, verifying, certifying, licensing, assigning, evaluating, and managing all technical personnel participating in sporting activities within the KAFRIADA Network.

The module supports coaches, assistant coaches, referees, umpires, judges, instructors, trainers, match commissioners, technical delegates, assessors, medical personnel, physiotherapists, fitness trainers, sports psychologists, nutritionists, statisticians, analysts, equipment managers, team managers, and other sports professionals.

The module integrates with KAFRIADA ID, Club Management, Competition Management, Athlete Care, Marketplace, Financial Management, and Reporting modules to ensure that only qualified and verified personnel participate in competitions and sports development programmes.

03.2 Objectives

The system shall:

- Register all sports technical personnel.
- Verify professional qualifications.
- Manage licensing and renewals.
- Maintain personnel career histories.
- Assign officials to competitions.
- Evaluate technical performance.
- Manage referee and official payments.
- Track disciplinary actions.
- Maintain continuing professional development (CPD) records.
- Provide searchable technical personnel directories.

03.3 Supported Personnel Categories

The platform shall support registration of:

Coaching Personnel

- Head Coach
- Assistant Coach
- Goalkeeper Coach
- Fitness Coach
- Strength & Conditioning Coach
- Skills Development Coach

Match Officials

- Referee
- Assistant Referee

Medical Personnel

- Sports Doctor
- Physiotherapist
- Sports Nurse
- Nutritionist
- Sports Psychologist
- First Aid Officer

Team Officials

- Team Manager
- Equipment Manager

- Fourth Official
- Umpire
- Judge
- Timekeeper
- Scorer

Technical Officials

- Match Commissioner
- Technical Delegate
- Competition Coordinator
- Referee Assessor
- Referee Instructor

- Welfare Officer
- Media Officer
- Performance Analyst
- Video Analyst

The Super Administrator may create additional personnel categories without modifying the software.

O3.4 Registration & Verification

Every technical personnel profile shall include:

- KAFRIADA ID
- Full Name
- Photograph
- Date of Birth
- Gender
- Nationality
- Contact Information
- Professional Category
- Sport(s)
- Qualifications
- Certifications
- Issuing Authority
- Years of Experience
- Employer or Affiliated Organization
- Government Identification
- Emergency Contact

Verification documents may include:

- Coaching Licence
- Referee Licence
- Professional Certificates
- Academic Qualifications
- National Identification
- Passport
- Police Clearance (Optional)

O3.5 Certification & Licensing

The platform shall maintain digital records of certifications and licences.

Each licence shall include:

- Licence Number
- Licence Level
- Issuing Organization
- Date Issued
- Expiry Date
- Renewal Status
- Verification Status

Licence Status

- Pending Verification
- Active
- Suspended
- Expired
- Revoked

The system shall notify users before licence expiry and support online renewal requests.

O3.6 Appointment & Assignment Management

Authorized competition administrators may assign personnel to competitions, matches, or events.

Assignments may include:

- Competition
- Match
- Venue
- Date
- Time
- Assigned Role
- Reporting Time
- Travel Details (Optional)

The system shall prevent scheduling conflicts and double-booking.

O3.7 Performance Evaluation

After each assignment, authorized evaluators may assess personnel using configurable criteria.

Example Evaluation Criteria

- Knowledge of Rules
- Decision Accuracy
- Communication
- Fitness
- Professional Conduct
- Leadership
- Match Control
- Punctuality
- Teamwork

Evaluation results shall contribute to the individual's performance history and ranking.

O3.8 Referee & Official Payments

The module shall integrate with the Competition Financial Management system.

For every completed assignment:

- Match fees shall be calculated automatically.
- Approved allowances shall be added.
- Travel reimbursements may be included.
- Deductions (if applicable) shall be applied.
- Net earnings shall be credited to the individual's dashboard.

Payment records shall include:

- Assignment Reference
- Gross Amount
- Deductions
- Net Amount
- Payment Status
- Payment Date
- Receipt Number

O3.9 Continuing Professional Development (CPD)

The platform shall support professional development programmes.

Activities may include:

- Courses
- Workshops
- Seminars
- Certifications
- Webinars
- Practical Assessments

Each activity shall record:

- Course Title
- Provider
- Date Completed
- Credit Hours
- Certificate
- CPD Points

Organizations may define minimum CPD requirements for licence renewal.

O3.10 Discipline & Compliance

The system shall record disciplinary actions against technical personnel.

Examples include:

- Warning
- Suspension
- Fine
- Licence Suspension
- Licence Revocation
- Misconduct
- Match Fixing Investigation
- Code of Conduct Violations

Every disciplinary action shall be linked to evidence, investigation records, and appeal outcomes.

O3.11 Rankings & Recognition

The platform shall maintain performance-based rankings.

Examples include:

- Top Referees
- Top Coaches
- Top Assessors
- Top Technical Delegates

Recognition awards may include:

- Official of the Year
- Coach of the Year
- Best Emerging Coach
- Best Referee
- Best Technical Delegate

Ranking formulas shall be configurable by governing organizations.

O3.12 Dashboard

Each technical personnel shall have a personalized dashboard displaying:

- Active Licence
- Upcoming Assignments
- Completed Assignments
- Earnings

Administrators shall have dashboards showing:

- Performance Rating
- CPD Points
- Disciplinary Status
- Certificates
- Notifications
- Career Statistics

- Total Registered Personnel
- Verification Status
- Licence Expiry Alerts
- Assignment Statistics
- Evaluation Reports
- Payment Summaries
- Discipline Cases

O3.13 Notifications

The platform shall automatically notify users about:

- Registration Approval
- Verification Results
- Licence Expiry
- Assignment Invitations
- Assignment Changes
- Match Reminders
- Evaluation Results
- Payment Confirmation
- CPD Opportunities
- Disciplinary Decisions

Notifications shall be delivered through:

- In-App Notifications
- Email
- SMS
- Push Notifications

O3.14 Functional Requirements

Requirement ID	Description
FR-TECH-1.1	The system shall register and verify technical personnel across all supported sports.
FR-TECH-1.2	The system shall maintain digital licences and certifications with renewal workflows.
FR-TECH-1.3	Competition organizers shall assign verified personnel to events and matches.
FR-TECH-1.4	The system shall prevent assignment conflicts and duplicate bookings.
FR-TECH-1.5	Performance evaluations shall contribute to personnel rankings and career history.
FR-TECH-1.6	Match fees and approved allowances shall be calculated and credited automatically.
FR-TECH-1.7	The platform shall track CPD activities and support licence renewal requirements.
FR-TECH-1.8	Every registration, assignment, evaluation, payment, and disciplinary action shall generate a permanent audit record.

O3.15 Business Rules

Rule ID	Business Rule
BR-TECH-1	Only verified personnel with active licences may be assigned to official competitions where licences are required.

BR-TECH-2	Expired or suspended licences shall prevent assignment until reinstated.
BR-TECH-3	The system shall not assign an individual to overlapping events or matches.
BR-TECH-4	Performance evaluations shall only be submitted by authorized assessors or administrators.
BR-TECH-5	Referee and official payments shall only be processed after assignment completion and approval.
BR-TECH-6	CPD requirements for licence renewal shall be configurable by the governing organization.
BR-TECH-7	Every disciplinary action shall be recorded with supporting evidence and an audit trail.
BR-TECH-8	The module shall support additional personnel categories, licence types, and evaluation criteria through configuration without requiring software redevelopment.

Summary

The Coaching, Officiating & Technical Personnel Management Module establishes a unified framework for managing the registration, verification, licensing, appointments, evaluations, payments, professional development, and compliance of sports technical personnel. Integrated with the broader KAFRIADA ecosystem, it ensures that only qualified and verified professionals participate in sporting activities while supporting transparent governance, career development, financial accountability, and configurable management across all current and future sports disciplines.

CHAPTER 05: SPONSORSHIP, PARTNERSHIP, DONATIONS & FUNDRAISING MANAGEMENT MODULE

O5.1 Overview

The Sponsorship, Partnership, Donations & Fundraising Management Module provides a comprehensive digital platform for attracting, managing, monitoring, and reporting sponsorships, partnerships, grants, donations, crowdfunding campaigns, scholarships, corporate social responsibility (CSR) initiatives, and sports development funding across the KAFRIADA ecosystem. The module enables governments, ministries, federations, clubs, academies, schools, universities, NGOs, development partners, foundations, private organizations, corporate sponsors, philanthropists, and individuals to invest in sports development through a transparent and accountable financial management framework.

The module integrates seamlessly with Competition Management, Athlete Management, Club Management, Marketplace, Financial Management, Education, Athlete Care, and Reporting modules to ensure that every sponsorship, donation, or partnership is traceable from initiation to utilization.

O5.2 Objectives

The system shall:

- Digitize sponsorship management.
- Support corporate partnerships.
- Manage donations and grants.
- Enable crowdfunding campaigns.
- Promote athlete sponsorship.
- Support club and competition sponsorship.
- Manage scholarship funding.
- Ensure transparency and accountability.
- Automate revenue allocation.
- Produce sponsorship performance analytics.

O5.3 Sponsorship Categories

The platform shall support sponsorship for:

Athlete Sponsorship

- Individual Athletes
- Youth Athletes
- Elite Athletes
- Paralympic Athletes

Club Sponsorship

- Professional Clubs
- Amateur Clubs
- Community Clubs
- School Teams
- University Teams

Organization Sponsorship

- Sports Federations
- Academies
- Schools
- NGOs
- Ministries of Sports
- Sports Associations

Event Sponsorship

- Conferences
- Coaching Clinics
- Referee Courses

Competition Sponsorship

- League Competitions
- Cups
- Championships
- School Sports Festivals
- Community Games
- Continental Championships

- Sports Seminars
- Awards Ceremonies
- Sports Festivals

O5.4 Partnership Management

The platform shall support strategic partnerships with:

- Governments
- International Sports Organizations
- Olympic Committees
- Educational Institutions
- Development Agencies
- NGOs
- Corporate Organizations
- Financial Institutions
- Insurance Companies
- Technology Companies
- Media Organizations

Each partnership shall include:

- Partner Name
- Partnership Type
- Objectives
- Start Date
- End Date
- Deliverables
- Financial Value
- Contact Persons
- Documents
- Approval Status

O5.5 Sponsorship Workflow

Every sponsorship shall follow an auditable workflow.

Workflow

1. Sponsor Registration
2. Sponsor Verification
3. Sponsorship Proposal
4. Proposal Review
5. Negotiation
6. Agreement Approval
7. Digital Contract Execution
8. Payment Processing
9. Fund Allocation
10. Performance Monitoring
11. Sponsorship Completion
12. Final Reporting

O5.6 Donations Management

The system shall support:

- Individual Donations
- Corporate Donations
- Anonymous Donations (subject to applicable legal and compliance requirements)
- Emergency Relief Donations
- Equipment Donations
- Facility Development Donations
- Scholarship Donations

Donation options include:

- One-Time Donation
- Monthly Donation
- Annual Donation
- Recurring Donation

Donors may specify:

- Beneficiary
- Purpose
- Amount
- Currency
- Anonymous Preference (where permitted)
- Public Recognition Preference

O5.7 Crowdfunding Campaigns

Authorized organizations may create fundraising campaigns.

Campaign information shall include:

- Campaign Title
- Campaign Description
- Campaign Banner
- Campaign Video
- Financial Goal
- Amount Raised
- Closing Date
- Beneficiary
- Progress Indicator
- Updates
- Donor List (subject to privacy settings)

Campaign types include:

- Medical Assistance
- Athlete Support
- Competition Funding
- Club Development
- Equipment Purchase
- Stadium Development
- Community Sports Projects
- Disaster Relief

O5.8 Scholarship & Grant Management

The module shall support:

- Educational Scholarships
- Sports Scholarships
- Research Grants
- Infrastructure Grants
- Youth Development Grants
- Women's Sports Grants
- Disability Sports Grants

Application workflow:

1. Scholarship Publication
2. Online Application
3. Eligibility Verification
4. Document Review

5. Interview (Optional)
6. Selection
7. Award Notification
8. Fund Disbursement
9. Progress Monitoring

O5.9 Financial Allocation & Distribution

The platform shall automatically allocate sponsorship funds according to configurable distribution rules.

Beneficiaries may include:

- Sponsored Athlete
- Club
- Competition Organizer
- Sports Federation
- Educational Institution
- Service Provider
- KAFRIADA Platform Service Fee

Distribution percentages shall be configurable by authorized administrators.

Every allocation shall generate:

- Financial Ledger Entry
- Audit Record
- Digital Receipt
- Payment Voucher

O5.10 Sponsor Dashboard

Sponsors shall have access to a dedicated dashboard displaying:

- Active Sponsorships
- Completed Sponsorships
- Donation History
- Campaign Performance
- Fund Utilization Reports
- Beneficiary Reports
- Contracts
- Financial Statements
- Tax-Related Reports (where applicable)

O5.11 Beneficiary Dashboard

Athletes, clubs, organizations, and competition organizers shall view:

- Active Sponsorships
- Grants Received
- Donations Received
- Funding Status

- Spending Reports
- Compliance Requirements
- Reporting Deadlines

O5.12 Branding & Marketing Assets

Sponsors and partners may upload and manage:

- Company Logo
- Brand Colours
- Promotional Banners
- Advertisements
- Videos
- Event Branding Materials
- Social Media Assets

The system shall support configurable placement of branding assets across:

- Competition Pages
- Marketplace Listings
- Event Pages
- Athlete Profiles (where approved)
- Club Profiles (where approved)
- Educational Courses
- Digital Certificates (where applicable)

Branding shall comply with KAFRIADA policies and sponsorship agreements.

O5.13 Reporting & Compliance

The platform shall generate reports including:

- Sponsorship Revenue
- Donation Reports
- Campaign Performance
- Beneficiary Reports
- Financial Statements
- ROI Analysis
- Impact Assessment
- Grant Utilization
- Partnership Performance

Reports shall be exportable in multiple formats.

O5.14 Notifications

The system shall notify stakeholders regarding:

- Sponsorship Approval
- Partnership Requests
- Donation Confirmation
- Campaign Milestones
- Grant Awards

Notification channels include:

- In-App Notifications
- Email
- SMS
- Push Notifications

- Payment Processing
- Reporting Deadlines
- Contract Expiry
- Renewal Opportunities

O5.15 Functional Requirements

Requirement ID	Description
FR-SPON-1.1	The system shall manage sponsorships, partnerships, donations, grants, and crowdfunding campaigns.
FR-SPON-1.2	Authorized organizations shall create and manage fundraising campaigns.
FR-SPON-1.3	Sponsorship and grant workflows shall support configurable approval processes.
FR-SPON-1.4	Financial allocations shall be automatically distributed according to configurable percentage rules.
FR-SPON-1.5	Sponsors and beneficiaries shall have dedicated dashboards with financial and performance analytics.
FR-SPON-1.6	The system shall support branding and sponsor asset management.
FR-SPON-1.7	Reports shall provide financial transparency, utilization tracking, and impact analysis.
FR-SPON-1.8	Every sponsorship, donation, payment, agreement, and allocation shall generate a permanent audit record.

O5.16 Business Rules

Rule ID	Business Rule
BR-SPON-1	Only verified sponsors, partners, and organizations may initiate sponsorship agreements or fundraising campaigns.
BR-SPON-2	Every sponsorship and partnership agreement shall undergo the configured approval workflow before activation.
BR-SPON-3	Financial distributions shall follow approved percentage allocations and generate corresponding audit records.
BR-SPON-4	Beneficiaries shall submit utilization reports for grants or sponsorships when required by the sponsoring organization.
BR-SPON-5	Crowdfunding campaigns shall automatically close upon reaching their end date or funding goal, according to campaign configuration.
BR-SPON-6	Sponsor branding shall only appear in locations authorized by the applicable sponsorship agreement and KAFRIADA branding policies.
BR-SPON-7	All donations, grants, sponsorships, and partnerships shall comply with applicable financial regulations, anti-money laundering (AML), and know-your-customer (KYC) requirements.
BR-SPON-8	Every financial transaction, agreement, approval, allocation, and report shall be permanently stored for compliance, transparency, and auditing purposes.

Summary

The Sponsorship, Partnership, Donations & Fundraising Management Module establishes a transparent, accountable, and scalable funding ecosystem for African sports. It enables governments,

federations, clubs, educational institutions, corporate organizations, development partners, and individual supporters to invest confidently in athletes, competitions, infrastructure, education, and community development. Through configurable workflows, automated financial allocation, integrated reporting, branding management, and compliance controls, the module strengthens sustainable sports financing and long-term growth across the entire KAFRIADA Network.

CHAPTER 06: COMMUNICATION, MEDIA, BROADCASTING & DIGITAL ENGAGEMENT

MODULE

O6.1 Overview

The Communication, Media, Broadcasting & Digital Engagement Module provides a unified communication ecosystem for KAFRIADA NET, enabling secure and real-time communication among athletes, clubs, coaches, officials, federations, sponsors, media organizations, supporters, and the general public.

The module combines internal messaging, announcements, notifications, email campaigns, SMS services, live broadcasting, press management, social media integration, media asset management, digital publications, podcasts, newsletters, fan engagement, and content monetization into a single platform.

It supports both operational communications and public engagement while ensuring compliance with data privacy regulations, media rights agreements, and organizational communication policies.

The module integrates seamlessly with Competition Management, Marketplace, Sponsorship, Athlete Management, Club Management, Financial Management, Education, and Reporting modules.

O6.2 Objectives

The module shall:

- Facilitate secure communication among all stakeholders.
- Deliver real-time notifications.
- Support internal and external messaging.
- Broadcast competitions and sporting events.
- Manage digital media assets.
- Integrate social media platforms.
- Support fan engagement.
- Publish official news and announcements.
- Manage press accreditation.
- Enable content monetization.
- Provide communication analytics.

O6.3 Communication Channels

The platform shall support multiple communication channels.

Internal Communication

- Direct Messaging
- Group Chats
- Organization Chats
- Team Chats
- Competition Chats

- Committee Discussions

External Communication

- Email
- SMS
- Push Notifications
- WhatsApp Integration (where supported)
- Telegram Integration (Optional)

Public Communication

- News Articles
- Press Releases
- Blogs
- Newsletters
- Public Announcements
- Event Updates

O6.4 Messaging System

Users shall communicate securely through the platform.

Supported features include:

- One-to-One Messaging
- Group Messaging
- Broadcast Messages
- File Sharing
- Image Sharing
- Video Sharing
- Voice Notes
- Document Sharing
- Message Search
- Message Archive
- Read Receipts
- Message Reactions

Organizations may restrict messaging according to configurable communication policies.

O6.5 Announcements

Authorized users may publish announcements.

Announcement categories include:

- Competition Notices
- Transfer Announcements
- Registration Deadlines
- Fixture Updates
- Medical Alerts
- Sponsorship News

- Educational Opportunities
- Employment Opportunities
- Emergency Notices
- Administrative Circulars

Announcements may target:

- Individual Users
- Teams
- Clubs
- Organizations
- Countries
- States
- Sports
- User Roles
- Entire Platform

O6.6 Media Centre

The platform shall provide a centralized Digital Media Centre.

Supported media types include:

- Images
- Videos
- Audio
- Podcasts
- Press Kits
- Competition Posters
- Event Banners
- Sponsor Advertisements
- Promotional Materials
- Digital Magazines

Media assets shall support:

- Categories
- Tags
- Metadata
- Copyright Information
- Licensing Information
- Download Permissions

O6.7 Live Broadcasting

The platform shall support live event broadcasting.

Supported broadcasting options include:

- Live Video Streaming
- Live Audio Commentary
- Live Match Statistics
- Live Score Updates

- Multi-Camera Streaming
- Event Highlights
- Instant Replay (where supported)

The system may integrate with external streaming providers and Content Delivery Networks (CDNs) through configurable APIs.

Broadcast administrators shall configure:

- Stream Quality
- Viewer Access
- Broadcast Schedule
- Language
- Commentary Team
- Sponsor Advertisements

O6.8 Press & Media Accreditation

Media organizations may apply for accreditation.

Applications shall include:

- Organization Name
- Journalist Name
- Photographer Name
- Identification
- Assignment Letter
- Event Selection
- Equipment Information

Application status:

- Submitted
- Under Review
- Approved
- Rejected
- Revoked

Approved applicants shall receive:

- Digital Media Pass
- QR Code Accreditation
- Access Permissions

O6.9 Social Media Integration

The platform shall support integration with official social media accounts.

Supported functions include:

- Publish Competition Updates
- Publish News
- Publish Match Results
- Publish Videos
- Publish Images
- Share Certificates

- Share Awards
- Share Campaigns

Administrators shall configure approval workflows before automatic publication.

O6.10 Fan Engagement

The platform shall provide interactive fan engagement features.

Features include:

- Fan Registration
- Favourite Teams
- Favourite Athletes
- Live Polls
- Match Predictions
- Fan Voting
- Comments
- Reactions
- Digital Fan Badges
- Loyalty Programmes
- Membership Programmes

Future releases may include fantasy sports and prediction games.

O6.11 Content Monetization

Authorized organizations may monetize digital content.

Supported monetization options include:

- Pay-Per-View Events
- Subscription Packages
- Premium Articles
- Premium Videos
- Sponsored Content
- Digital Advertisements
- Merchandise Promotions

Revenue distribution shall support configurable percentage allocations among content owners, event organizers, broadcasters, and KAFRIADA.

O6.12 Communication Dashboard

Users shall access dashboards displaying:

Personal Dashboard

- Messages
- Notifications
- Invitations
- Announcements
- Upcoming Events
- Saved Media

Organization Dashboard

- Communication Statistics
- Broadcast Performance
- Audience Reach
- Announcement Engagement
- Social Media Performance
- Media Library Usage
- Revenue Analytics

O6.13 Notification Management

The Notification Engine shall support:

- System Notifications
- Competition Notifications
- Transfer Notifications
- Medical Alerts
- Payment Notifications
- Sponsorship Notifications
- Learning Notifications
- Assignment Notifications
- Security Alerts

Users shall configure:

- Preferred Channels
- Quiet Hours
- Language
- Notification Categories

O6.14 Reports & Analytics

The module shall generate reports on:

- Message Delivery
- Broadcast Viewership
- Audience Demographics
- Social Media Reach
- Communication Performance
- Media Downloads
- Fan Engagement
- Advertisement Performance
- Content Revenue

Reports shall support graphical dashboards and export in multiple formats.

O6.15 Functional Requirements

Requirement ID	Description
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FR-COMM-1.1	The system shall provide secure messaging for individuals, teams, and organizations.
FR-COMM-1.2	Authorized users shall publish announcements, news, and digital content.
FR-COMM-1.3	The platform shall support live broadcasting of competitions and events.
FR-COMM-1.4	The system shall manage media accreditation with digital passes and QR code verification.
FR-COMM-1.5	The platform shall integrate with external broadcasting and social media services through configurable APIs.
FR-COMM-1.6	Users shall configure communication preferences and notification settings.
FR-COMM-1.7	Authorized organizations shall monetize eligible digital content using configurable revenue-sharing rules.
FR-COMM-1.8	Every communication, publication, broadcast, notification, and media transaction shall generate a permanent audit record.

O6.16 Business Rules

Rule ID	Business Rule
BR-COMM-1	Only authorized users may publish official announcements or media content.
BR-COMM-2	Media accreditation shall be approved before access credentials are issued.
BR-COMM-3	Broadcast rights and content distribution shall comply with applicable licensing agreements and organizational policies.
BR-COMM-4	Communication records shall be retained according to configurable retention and privacy policies.
BR-COMM-5	Users shall control their notification preferences, subject to mandatory system and security alerts.
BR-COMM-6	Content monetization revenue shall follow configurable distribution rules defined by KAFRIADA and participating organizations.
BR-COMM-7	All media assets shall include ownership and copyright metadata where applicable.
BR-COMM-8	Every communication, publication, broadcast, accreditation, and monetization activity shall be permanently logged for auditing and compliance.

Summary

The Communication, Media, Broadcasting & Digital Engagement Module establishes a modern digital communication and media ecosystem for KAFRIADA NET. It enables secure stakeholder communication, official announcements, live event broadcasting, digital media management, press accreditation, fan engagement, social media integration, and content monetization. Through configurable workflows, comprehensive analytics, and integration with all core platform modules, it enhances collaboration, public engagement, transparency, and the global visibility of African sports.